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Screening and social prescribing in healthcare and social services to address housing issues among children and families: a systematic review

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Title

Screening and social prescribing in healthcare and social services to address housing issues among children and families: a systematic review

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Abstract

Objectives: Housing is a social determinant of health that impacts the health and wellbeing of children and families. Screening and referral to address social determinants of health in clinical and social service settings has been proposed to support families with housing problems. This study aims to identify housing screening questions asked of families in healthcare and social services, determine validated screening tools and extract information about recommendations for action after screening for housing issues. Methods: The electronic databases MEDLINE, PsycINFO, EMBASE, Ovid Emcare, Scopus and CINAHL were searched from 2009 to 2019. Inclusion criteria were peer-reviewed literature that included questions about housing being asked of children or young people aged 0-18 years old and their families accessing any healthcare or social service. We extracted data on the housing questions asked, source of housing questions, validity, and descriptions of actions to address housing issues. Results: Twenty-three peer-reviewed papers met the inclusion criteria. The housing questions in social screening tools vary widely. There are no standard housing-related questions that clinical and social service providers ask families. Nine screening tools were validated. An action was included as part of 10 social screening tools. Actions for identified housing problems included provision of a community resource guide, referral to a social worker or social service agency, or referral to a housing and child welfare demonstration project.

Conclusion: This review provides a catalogue of housing questions and subsequent referral and actions that clinical and/or social service providers can ask families experiencing housing problems.

Strengths and Limitations

- The first systematic review to catalogue housing questions asked of families in clinical and social service settings.

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- Independent review of study selection, quality assessment and data extraction.
- Search terms may have been too narrow and may not have captured relevant papers that included information on housing questions in broader social screening tools.
- The quality of studies varied with relatively small sample sizes in many of the studies.
- Heterogeneity in study type, sample size, participant age, housing questions asked across studies, limiting ability to compare studies.

Introduction

Social determinants of health are the environments and conditions in which people are born, grow, work, live and age (1). The role of housing as a social determinant of health is well-established (2, 3). Healthy housing is described by WHO as "shelter that supports a state of complete physical, mental and social well-being" and "provides a feeling of home, including a sense of belonging, security and privacy" (3). Adequate housing is more than shelter and provision of quality physical dwelling conditions (4-6). Housing tenure and housing affordability are important factors in supporting psychological and developmental wellbeing (6-11). Similar to other Western countries, in Australia, where this review was conducted, almost one million people reside in housing considered to be in poor condition and there is an overrepresentation of young people, people in low-income households, Aboriginal and Torres Strait Islander people living in poor housing conditions (12).

There is growing evidence that housing aspects such as housing quality and environments materially affect health, wellbeing and developmental trajectories well into adulthood (13-16). Housing issues in childhood have a particularly detrimental effect on health and wellbeing. Poor housing conditions in childhood have been predicted to lead to poorer health and increased mortality later in life, after controlling for the confounding effect of socioeconomic conditions (13, 14). Material and social factors that impact on children's wellbeing include poor physical housing quality (mould and damp, utilities, toxicants, disrepair, lack of heating, injury hazards), crowding, lack of outdoor space for children to play, accessibility within the home, housing affordability, tenure type, frequent

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residential moves, homelessness, lack of cultural appropriateness and complex neighbourhood quality (6, 16-23). Psychological and emotional distress, as well as behavioural problems, have been independently associated with children living in poor housing conditions and experiencing housing instability (20, 24-29). Housing instability, frequent moving and homelessness also negatively affect children's educational outcomes (30). A range of physical health issues such as asthma, acute respiratory symptoms and low birthweight have been independently associated with damp and mouldy housing (22, 31, 32), and living in crowded conditions has been found to lead to infectious diseases such as tuberculosis, meningococcal disease, acute rheumatic fever and otitis media and gastrointestinal infections in New Zealand and Australia (33-40). Multiple forms of housing problems can coexist and have a compounded, cumulative impact on physical and mental health, particularly after prolonged exposure during childhood (11, 41).

Improving the quality of housing and the immediate environment around the home "can save lives, prevent disease, and increase quality of life" (3). There is evidence to support screening and social prescribing for social determinants in healthcare, including housing issues, to address inequity and improve child health (29, 42-48). Screening involves incorporating standardized tools, such as surveys or questionnaires, completed with families in paediatric or primary care practices (49, 50). Social prescribing is a mechanism for clinicians to link or refer patients to non-medical sources of support to address social determinants of health (48, 51). Screening for housing issues has been demonstrated to increase the occurrence of social prescribing, for example referring families to appropriate professionals or providing families with resources (52, 53), and can lead to improved child health (43). While many social screening tools addressing housing issues do exist (54, 55), to date there are no widely accepted guidelines for health and social service professionals to systematically identify specific housing problems for children and address them (29, 53, 56). This systematic review was conducted to catalogue housing screening questions asked of families in healthcare and social service settings, as described in the international literature; determine if any validated screening tools exist; and extract information about any recommendations for action after

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screening described in the literature, including referral though health, housing and social care pathways.

Findings from this review will inform the development of an integrated detection, referral and action pathway to be used in healthcare and social service settings with Aboriginal and Torres Strait Islander children and families who may be experiencing poor health due to housing problems in Australia. The detection, referral and action pathway will also be used with non-Aboriginal patients and clients who are experiencing housing-related health issues.

Methods

Search strategy

A study protocol was developed, registered with the University of York Centre for Reviews and Dissemination (PROSPERO) and updated on 28/04/2020, registration number CRD42020159816 (https://www.crd.york.ac.uk/prospero/). A search of the international peer-reviewed and grey literature was conducted using the following inclusion criteria: studies that include questions about housing being asked of the population group, children or young people aged 0-18 years old and their families, accessing any health or social service. There were no restrictions on study type, design or method, and studies were not limited by country. Studies were limited to the English language and to published literature in the 10-year period 1 January 2009 to 25 November 2019. Studies about health assessment conducted with clients of housing services were excluded. Review papers were excluded.

The electronic databases MEDLINE, PsycINFO, EMBASE, Ovid Emcare, Scopus and CINAHL from 2009 to 2019 inclusive were systematically searched. The search strategy and key search terms are included in Supplementary Table 1, in line with the PRISMA guidelines. A secondary search of citations from bibliographies of related peer-reviewed articles, and an additional search of the grey

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literature using the search terms “social needs screening tool”; “housing screening tool for children” in Google Scholar were undertaken.

The study titles and abstracts were reviewed independently in Covidence by five authors (NW, AA, SW, MA and KH). Disagreements about inclusion of studies were discussed and decisions made by consensus with four authors (AA, SW, MA, KH). Full texts of papers were reviewed independently by two authors (AA and SW). Divergences were discussed with two authors (AA and SW) and agreement reached.

Data extraction and analysis

Data was extracted from included papers using a standardized data extraction tool by two reviewers (AA and SW). The data extracted included specific details about the study types, aims and methods, populations, screening tools, housing questions, descriptions of actions, such as referral to appropriate services, to address housing issues and other items pertaining to the review question and specific objectives. If housing questions were not available in the papers, but the name of a screening tool was provided, the authors accessed the specific tool through a web search to extract the housing screening questions. Any disagreements that arose between the reviewers were resolved through discussion.

Quality assessment

Two reviewers (AA and SW) assessed the quality of each study using two separate tools, dependent on study type. The Mixed Methods Appraisal Tool (MMAT) tool (57) was used to appraise quantitative RCTs, quantitative non-randomized trials, quantitative descriptive and mixed methods studies. Quality improvement (QI) studies were appraised using the Quality Improvement Minimum Quality Criteria Set (QI-MQCS) version 1.0 (58). The reviewers did not give studies an overall score for methodological quality, as it is discouraged to calculate an overall quality score using the MMAT tool (57). Consensus on quality was reached through discussion.

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Patient and public involvement

This study was conceptualised, in part, with community members who have experienced housing issues, and people who are in regular patient contact with families experiencing housing issues impacting on health and wellbeing.

Results

Papers identified

The search strategy identified 8,323 titles (excluding duplicates). Of those, 352 full text papers were screened and 315 were excluded. Upon a second screen of the included 37 articles, a further 14 were excluded with reasons provided in **Error! Reference source not found.**. A total of 23 papers were included representing 21 studies.

Included papers

The 23 included papers are listed in Table 1. All papers were published in peer-reviewed journals. Studies were primarily conducted in clinical settings (n=21); paediatric primary care clinics, children's hospitals and emergency departments, urban community health centres, general paediatric centres, and antenatal clinics. Two studies were conducted in social service settings including a disability service and a government child welfare unit. All studies were based in high-income countries, with 19 studies conducted in the USA, one study in Australia and one study conducted in Spain.

Studies were heterogenous in terms of method, sample size, age of participants, social screening tools and housing questions included in screening tools. Study types varied; cross-sectional (46, 59-64), randomized controlled trials (42, 43, 65-67), pre and post studies (68, 69), quality improvement (53, 70-72), mixed methods (73, 74), tool adaptation and validation (75), retrospective file audit (76), and secondary data analysis (77). Sample sizes ranged from 45 to 6282 participants. Participants were children, adolescents and/or their families or caregivers. Age ranges of children varied from study to study, and some studies only included children with specific characteristics, such as children with cerebral palsy, and children with open child welfare cases. See Table 1 for characteristics of

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included papers. Social prescribing activities were found in ten of the 21 included studies, see Table 2 for social prescribing for housing issues.

Quality assessment and risk of bias

Assessing studies using the Mixed Methods Appraisal Tool, the three randomised control trials scored highly. One randomized control trial (RCT) met all the criteria (43, 66) and two studies scored highly, meeting the majority of the criteria. It was unclear whether Garg's (42) cluster RCT randomisation was adequately performed and whether participants adhered to the assigned intervention, and Gottlieb's (65) study did not provide adequate information on blinding. There were two robust quantitative non-randomized studies, scoring highly against most criteria. Patel's (2018) study did not provide complete outcome data, however, and it was unclear in Bottino's (2019) study whether participants were representative of the target population nor where there was complete outcome data. Ten studies were categorised as quantitative descriptive, and quality scores were mixed. A common issue related to the representative sample of the target population, with the sample size not being representative in four of the eight studies, and another four studies providing insufficient information to assess whether the sample size was representative. The risk of non response bias was low in three of the quantitative descriptive studies. The two mixed methods studies were of good quality however both did not adequately address inconsistencies between quantitative and qualitative results and did not adhere to the tradition of each method (Farrell, Koita). There were four quality improvement studies, assessed using the Quality Improvement Minimum Quality Criteria Set. Denny and Colvin's studies were of the highest quality, both scoring well in 10 of the 15 criteria. All QI studies had issues with adherence, organizational readiness and spread. See Tables 3a and 3b for quality assessment of included studies.

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Table 1 Included study characteristics

Author (year)	Country	Study type	Participants (age)	Sample size	Name of tool and source of housing questions	Validity, feasibility and/or acceptability of tool	Housing questions (if available)
Badia et al. (2014) (75)	Spain	Quantitative - Tool translation, adaptation, validation	Parents of children and adolescents with cerebral palsy (8-18 years)	221	European Child Environment Questionnaire (ECEQ), based on the International Classification of Functioning, Disability and Health	Construct validity, translation, adaptation and validation of tool conducted specifically for the Spanish context	Enlarged rooms at home (need/availability?); Adapted toilet at home (need/availability?); Modified kitchen at home (need/availability?); Walking aids (need/availability?); Hoists at home (need/availability?)
Beck et al. 2012) (76)	USA	Quantitative - Retrospective review	Newborns and families (age not specified)	639	Developed based on literature review, in consultation with medical staff, social services and medical–legal staff and adapted from the USA's Survey on Income and Program Participation	Food security and domestic violence questions validated; housing questions not validated	"[Are you having] Problems with housing conditions (overcrowding, evictions, lead, utilities, mold, rodents)? Have you spoken with your landlord?"
Bottino et al. (2019) (59)	USA	Quantitative – cross-sectional	Children (3-10 years)	340	Adapted from the American Housing Survey	Food questions validated; no information on housing question validity	"What best describes your current living situation?" [Rent apartment; Rent room(s) within another person's home; Own home; Homeless or living in a shelter; Doubled up with another family; Live with parents/other family; Other]. Selecting 'Rent apartment' or 'Own home' triggered a follow-up item: "In the last 12 months, has the electric or gas company shut off the electricity or gas in your home or threatened to shut off the utilities in your home?" and "Are you concerned about a possible eviction (being "kicked out" from your home?"
Colvin et al. (2016) (53)	USA	Quantitative - postintervention only design	Children (median age 3 years)	45	Income, Housing, Education, Legal status, Language/Immigration, Personal Safety (IHELLP), adapted from Kenyon et	Validity, sensitivity and specificity	"Do you have any concerns about poor housing conditions like mice, mold, cockroaches? Do you have any concerns about being evicted or not being able to pay the rent? Do you have any concerns about not being able to pay your mortgage?"

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					al's IHELLP social history tool (78)		
Denny et al. (2019) (70)	USA	Quality Improvement	Children (0-1 years and 1-5 years)	667	Injury Prevention (IP) + The Safe Environment for Every Kid (SEEK) tool, adapted from Gittelman et al.'s IP tool, and Dubowitz's SEEK tool (79, 80)	Injury Prevention (IP) tool tested for reliability	"Do you have trouble paying utilities or maintaining a safe place to live?"; "Do you use safety gates to protect your child at the top and bottom of all stairways in your house?"; "Are all potentially harmful household cleaners and pesticides either in locked storage OR out of the reach of children?"; "Is the hot water heater in your house adjusted to less than 120 degrees?"
Farrell et al. (2017) (73)	USA	Quantitative - mixed methods population survey	Families with open child welfare cases	6828	Quick Risks and Assets for Family Triage (QRAFT), adapted from Risks and Assets for Family Triage (RAFT) assessment instrument	Some indication of content, construct, and predictive validity, however inconclusive information	Current Housing, Housing Condition, and Housing History, with 5-item risk scale for each, measuring level of housing instability and homelessness
Garg et al. (2009) (60)	USA	Quantitative – cross-sectional descriptive	Parents of children 2 months old to 10 years	100	Well Child Care, Evaluation, Community Resources, Advocacy, Referral, Education (WE CARE), developed collaboratively among paediatric clinic staff and initially guided by Bright Futures paediatric intake form (29)	Face validity	"Do you think you are at risk of becoming homeless?"
Garg et al. (2015) (42)	USA	Quantitative – cluster RCT	Families of infants <6 months old	336	Well Child Care, Evaluation, Community Resources, Advocacy, Referral, Education (WE CARE), adapted from a larger psychosocial screening instrument; three housing instability questions derived from the Children's HealthWatch survey (81)	Validity	Home heating and Housing instability (housing instability questions: having a steady place to sleep; household crowding i.e. >2 people per bedroom, or having moved >2 times in the past year)

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Gottlieb et al. (2014) (65)	USA	Quantitative - RCT	Caregivers of paediatric emergency department patients	538	Some items adapted from the Medical Advocacy Screening questionnaire; other items adapted from other existing validated surveys	Neighbourhood safety question validated	Concerns about the physical condition of your housing; Concerns about the cost or stability of your housing; Threats to your child's safety at school or in the neighbourhood
Gottlieb et al. (2016) (43)	USA	Quantitative - RCT	Children (mean age 5 years)	1809	Used survey instrument from 2014 RCT	Neighbourhood safety question validated	Not having a place to live; unhealthy living environment; other concerns with housing
Gottlieb et al. (2018) (66)	USA	Quantitative - RCT	Children (mean age 5 years) and families in acute care setting	718	Used survey instrument from 2014 RCT	Neighbourhood safety question validated	Not having a place to live; unhealthy living environment; other concerns with housing
Hassan et al. (2013) (61)	USA	Quantitative – cross-sectional	Young adult patients (15-25 years)	401	The Online Advocate, adapted from the American Housing Survey	Used previously validated survey questions	Major: Homeless; utilities shut off; structural problems such as leaking roof, non-functional plumbing, or rodent infestation. Minor: Need fuel/heating assistance; threat of utilities shut off in past 12 months; if renting or receiving subsidized housing, concern about transfer or eviction
Hassan et al. (2015) (46)	USA	Quantitative – prospective intervention	Young adult patients (15-25 years)	401	The Online Advocate, adapted from the American Housing Survey	Used previously validated survey questions	Homeless; having utilities shut off; or having existing structural problems such as leaking roof, non-functional plumbing, or rodent infestation
Hensley et al. (2017) (71)	USA	Quality Improvement	Paediatric patients and families	114	HealthBegins Upstream Risks Screening Tool	Feasibility	"In the last month, have you slept outside, in a shelter, or in a place not meant for sleeping? Y/N; In the last month, have you had concerns about the condition or quality of your housing? Y/N; In the last 12 months, how many times have you or your family moved from one home to another?"
Higginbotham et al. (2019) (72)	USA	Quality Improvement	Children (1 week to 5 years old)	53	Aligned with US Department of Housing and Urban Development housing insecurity definition and previous	Feasibility	Crowding "within the past 12 months there were two or more people per bedroom?"; doubling up "within the past 12 months we were temporarily staying with another family or had another family staying with us?"; and

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					cross-sectional study by Cutts and colleagues (82)		frequent moving "within the past 12 months we moved more than once?"
Koita et al. (2018) (74)	USA	Quantitative - Validation of tool	Caregivers of children < 12 years	28	BARC* Pediatric Adversity and Trauma Questionnaire, modified from Oakland Children's Hospital Family Information and Navigation Desk needs assessment survey	Face validity	"Has your child/family ever been homeless? Has your child ever had problems with housing (for example not having a stable place to live, faced eviction or foreclosure, or lived with multiple families)?"
Kumar et al. (2017) (67)	USA	Quantitative – RCT follow up	Pregnant adolescents < 17 years	106	Modified from the Treatment Services Review assessment tool	No information provided	Residential mobility (number of moves made in the year before prenatal survey, duration of stay, number of people living at their current home, and cohabitation with parents, husband or boyfriend, and other relatives); Need for housing resources at 3 months and 6 months postpartum
Molina et al. (2019) (62)	USA	Quantitative – cross-sectional	Children and families	276	Adapted from health-related social needs questions from the National Academy of Medicine	No information provided	"In the last month, have you slept outside, in a shelter or in a place not meant for sleeping? In the last month, have you had concerns about the condition or quality of your housing? In the last 12 months, how many times have you or your family moved from one home to another? Are you worried that in the next 2 months, you may not have stable housing? In the past 12 months, have you had any utility (electric, gas, water or oil) shut off for not paying your bills? Do you have any concerns about safety in your neighbourhood? Are you afraid you might be hurt in your apartment building or house?"
Patel et al. (2018) (68)	USA	Quantitative – pre and post intervention retrospective chart review	Patients at well-child visit (0-20 years)	322	Income, Housing, Education, Legal status, Language/Immigration, Personal Safety (IHELLP) (78)	No information provided	"Housing, utilities: What kind of housing do you live in (apartment, house)? Do you receive any type of housing subsidy (Section 8, public housing, shelter)? Do you have heat and hot water? Do you have trouble paying rent or utilities at the end of the month?" "Personal

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							safety: Do you feel safe in your neighbourhood?"
Rhodes et al. (2012) (63)	Australia	Quantitative – cross-sectional	Children (4-8 years)	162	Family Psychosocial Resources (FSR) tool, adapted from the Family Resources Scale (83)	No information provided	Basic living needs
Selvaraj et al. (2019) (64)	USA	Quantitative – cross-sectional	Children and adolescents (2 weeks to 17 years)	2569	Addressing Social Key Questions for Health Questionnaire (ASK tool), developed by staff of four academic institutions and literature review of validated paediatric questionnaires on adverse childhood experiences and unmet social needs	Validity and feasibility	Housing and bill insecurity; Witnessed violence in the home or neighbourhood
Uwemedimo (2018) (77)	USA	Secondary data analysis	Caregivers of children < 18 years	148	Family Wellness Screen (FAMNEEDS), developed from previously published SDH screening tools (84)	Validity	"Do you worry that in the next 2 month, you/your family may not have a safe or stable place to live? Y/N. Where is your child living now, Options provided: private house/ apartment, room (in apartment/house), shelter, hotel/motel, no regular place, car, other]. Do you worry that the place you're living now is making you sick? Y/N. If YES choose all that apply: Cigarette smoke, mold or dampness, rodents/bugs, peeling paint, broken appliances, open cracks/holes/wires, not enough heat, water leaks, none" Other questions: "Do you need help from a lawyer with housing, immigration, custody or child support problems? Y/N."
Zielinski et al. (2017) (69)	USA	Quantitative - pre and post	Children	602	WE CARE tool (29)	Validity, feasibility and acceptability	Housing insecurity

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Screening tools and housing questions

Our systematic review found 18 social screening tools that include housing questions. The tools used in clinical and social service settings varied widely. The purpose of all tools was to screen for health- or wellbeing-related social needs among children, adolescents and/or families. The tools were used by healthcare or social service workers to facilitate asking patients or clients questions about social needs, including housing. There were two distinct approaches to screening; one approach was a general social determinants of health screening where housing was one of multiple domains among other social needs, and a second approach was to focus on housing as one domain, and only ask detailed housing-related questions. In the first approach, social needs assessed in tools included housing, food security, income, employment, education, childcare needs, transportation needs, health care access, legal concerns, medical insurance, violence, social connection and isolation.

The housing questions asked of participants varied across studies. Housing questions extracted from the included studies were divided into nine categories, see **Error! Reference source not found.**.

Housing instability, insecurity or homelessness were the most common housing-related screening questions across the tools used, with questions about homelessness, sleeping in a shelter, sleeping outside or in a place not meant for sleeping, sleeping in unsafe housing, frequent moving and temporary or unsafe housing (42, 43, 59, 61, 62, 65, 67, 69, 71-73). Problems with utilities, such as utilities being shut off, threats of utilities being shut off or concern about not being able to pay for utilities, was asked in six (n=6) screening tools (42, 46, 59-62, 64, 76).

Crowding was a common item in the social screening tools, with the following terms being used: “overcrowding” (76), “doubled up or overcrowded in an unsustainable way” (73) and “need enlarged rooms at home” (75). Crowding was considered a part of housing instability in one tool, i.e. “Housing instability: Doubled up with another family” (59). Some tools asked families about general housing problems and either did not specify the nature of the housing problem, or asked about multiple housing-related issues within the one general question (43, 46, 53, 60, 61, 73, 76). Structural and

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physical housing problems (61) included questions specific to the quality or condition of the home (62, 65, 71), and mould or damp (53, 76, 77). See Table 1 for further detail on the housing questions asked in screening tools.

Validity, feasibility and acceptability

Nine (n=9) screening tools had been validated, including the Family Wellness Screening tool or FAMNEEDS (77), The Online Advocate (46) as well as other tools used across studies (42, 60, 65, 74, 75). The studies included in this review did not provide detailed information on validity. It is not clear whether any of the screening tools are robust in producing their intended result, and to what extent the housing questions in particular are suitable for their aims.

The ASK tool (Addressing Social Key Questions for Health questionnaire) had been tested for validity and feasibility (64). The Well Child Care, Evaluation, Community Resources, Advocacy, Referral, Education or WE CARE tool had been validated and tested for feasibility and acceptability (29, 69). The Income, Housing, Education, Legal status, Language/Immigration, Personal Safety or IHELLP tool was tested for validity, sensitivity and specificity (53, 78). Some tools were not validated however underwent other forms of testing such as feasibility, reliability or sensitivity. The screening tool used in Higginbotham's study (2019) was tested for feasibility, as was Hensley's (2017) HealthBegins Upstream Screening tool. The Injury Prevention tool was tested for reliability (70). Insufficient information was provided on the validity, feasibility or acceptability of five tools (59, 62, 63, 67, 68). The information provided for the Quick Risks and Assets Family Triage or QRAFT tool showed some indication of content, construct and predictive validity, however it is unclear whether it was validated (73). The tool used in Beck's (2012) study did not validate the housing questions, however other questions in the tool were validated.

The studies did not explore testing the cultural acceptability or appropriateness of the screening tools. Badia's (2014) study analysed the cultural acceptability of the European Child Environment Questionnaire in order to adapt it to the Spanish context. Uwemedimo's (2018) study took into

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consideration unique challenges faced by immigrant families when screening for social needs and recommended screening tools to be culturally sensitive through availability in multiple languages and for bilingual or multilingual patient navigators to support patients from immigrant backgrounds.

Social prescribing

Ten studies involved social prescribing, i.e. an action was taken upon identifying a social need as part of screening. Actions included providing patients with a community resource guide or community resource information (42, 43, 53, 64, 71, 72), referral to a social worker (53, 61, 69), social service agency (77), or referral to a housing and child welfare demonstration project (73). In several studies, an in-person navigator helped patients with contacting resources (43, 66, 71, 77). Actions specific to housing-related problems, such as a social work referral specific to homelessness, were evident in seven of the ten studies. Screening for social needs resulted in an increase in referrals to appropriate services or resources, and some studies found that screening and subsequent referral led to an improvement in the housing-related issue (42), or an improvement in child health (43, 66). See Table 2 for a summary of information relating to actions as a result of screening for housing issues, or social prescribing.

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Table 2 Social prescribing for housing issues

Author (year)	Country	Screening tool	Action as a result of screening	Referral rates and outcome (if available)
Colvin et al. (2016) (53)	USA	IHELLP	Social work referral or resources provided to families who screened positively to unmet social needs	77.8% of families screening positively for unmet social needs were provided social work resources or referrals; 13.3% already had social needs addressed; <10% had an unmet social need but were unable to be connected with a resource or referral
Farrell et al. (2017) (73)	USA	QRAFT: Quick Risks and Assets for Family Triage	Assess eligibility to a housing and child welfare demonstration project, a supportive housing intervention, and for broader supportive housing referrals. Eligibility criteria: client scoring a 3 (significant risk) or 4 (severe risk) on any of the three housing items	5.4% of families scored 3 or 4 on housing items and upon further assessment, 5.3% were referred to the housing and child welfare demonstration project
Garg et al. (2015) (42)	USA	WE CARE: Well Child Care, Evaluation, Community Resources, Advocacy, Referral, Education	Referral made by clinician using the WE CARE Family Resource Book containing tear-out information sheets listing 2 to 4 free community resources available for each need. The information sheets contain the program name, a brief description, contact information, program hours, and eligibility criteria. Follow up one month after visit; staff telephoned mothers to assess contact of resources and update notes in the child's medical record	Mothers receiving WE CARE screening and referral had lower odds of being in a homeless shelter (aOR = 0.2; 95% CI, 0.1–0.9) and more mothers had enrolled in a new community resource at the 12-month visit (39% vs 24%; aOR = 2.1; 95% CI, 1.2–3.7) compared to families who did not receive WE CARE screening and referral
Glottlieb et al. (2018); Gottlieb et al. (2016) (43, 66)	USA	-	Caregivers who indicated at least one social need received either written information on relevant community resources (active control group) or received support from an in-person navigator immediately after the child's visit and offered follow-up meetings (navigation intervention group)	Both groups showed improved child health after receiving either written information or an in-person navigator; child global health scores (lower scores indicate better health) improved a mean (SE) of -0.36 (0.05) in the navigator group and a mean (SE) of -0.12 (0.05) in the active control group
Hassan et al. (2013); Hassan et al. (2015) (46, 61)	USA	The Online Advocate	Resource specialist reviewed referrals; if questionnaire responses indicated acute concerns regarding homelessness, results were immediately shared with the provider and social worker to facilitate urgent intervention.	75% of participants screened had at least one referral need; 27% required referral relating to housing problem and 14% received referral relating to housing problem. 85% of participants with housing problems reached for follow-up at 1 or 2 months, and 30% of those with housing problems selected that the problem was "completely" or "mostly" resolved
Hensley et al. (2017) (71)	USA	HealthBegins Upstream Risks Screening Tool	Patients with at-risk results were provided a community resources guide with an educational handout, in order to identify local agencies and programs that addressed social	-

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			needs. The guide listed supporting programs or agencies, and included resource eligibility requirements, contact information. Patients were helped in contacting listed community resources.	
Higginbotham et al. (2019) (72)	USA	-	Families screening positively for food insecurity and/or housing insecurity were provided a community resource guide to facilitate referral	Of 13 families screening positive for food insecurity and/or housing insecurity, 85% were given a resource guide
Selvaraj et al. (2019) (64)	USA	ASK Tool: Addressing Social Key Questions for Health Questionnaire	Referral made to community resources in relation to unmet social needs, including referrals made in relation to housing/bill insecurity	284 (11%) total referrals made in relation to unmet social needs; 111 (4.3%) referrals made in relation to housing/bill insecurity
Uwemedimo (2018) (77)	USA	FAMNEEDS Family Wellness Screen	Referral to local community resources by trained navigators and follow-up to ensure linkage to resource. Trained navigators either searched online social service databases or referred to social service case managers at designated partner community organisations	Approximately one-third (30.9%) successfully utilized program-provided resources at 12-week follow-up
Zielinski et al. (2017) (69)	USA	WE-CARE screening tool	Provision of additional resources on homelessness and/or social work referral relating to homelessness	14 (93%) provision of additional resources on homelessness; 11 (73%) social work referral relating to homelessness

Table 3a Quality assessment using the MMAT tool

Author (year)	Category of study design	Are there clear research questions?	Do collected data allow to address research question?	Is randomization adequately performed?	Are groups comparable at baseline?	Are there complete outcome data?	Are outcome assessors blinded to intervention provided?	Did participants adhere to the assigned intervention?
Garg (2015)	Quantitative RCT	Y	Y	Unclear	Y	Y	Y	Unclear
Gottlieb (2016 & 2018)	Quantitative RCT	Y	Y	Y	Y	Y	Y	Y
Gottlieb (2014)	Quantitative RCT	Y	Y	Y	Y	Y	Unclear	Y
Author (year)	**Category of study design**	**Are there clear**	**Do collected data allow to address**	**Are participants representative of target population?**	**Are measurements appropriate regarding both the outcome**	**Are there complete outcome data?**	**Are the confounders accounted for in**	**During study period, is the intervention administered (or**

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		research questions?	**research question?**		**and intervention (or exposure)?**		**the design and analysis?**	**exposure occurred) as intended?**
Bottino (2019)	Quantitative non-randomized	Y	Y	Unclear	Y	Unclear	Y	Y
Patel (2018)	Quantitative non-randomized	Y	Y	Y	Y	N	Y	Y
Author (year)	**Category of study design**	**Are there clear research questions?**	**Do collected data allow to address research question?**	**Is the sampling strategy relevant to address the research question?**	**Is the sample representative of the target population?**	**Are the measurements appropriate?**	**Is the risk of non response bias low?**	**Is the statistical analysis appropriate to answer the research question?**
Badia (2014) (75)	Quantitative descriptive	Y	Y	Y	N	Y	Y	Y
Beck (2012)	Quantitative descriptive	Y	Y	Unclear	Unclear	Y	Unclear	Y
Garg (2009)	Quantitative descriptive	Y	Y	N	N	Y	N	Y
Hassan (2013 & 2015)	Quantitative descriptive	Y	Y	Y	Unclear	Y	Unclear	Y
Kumar (2017)	Quantitative descriptive	Y	Y	Y	Y	Y	Y	Y
Molina (2019)	Quantitative descriptive	Y	Y	Unclear	Unclear	Y	Unclear	Unclear
Rhodes (2012)	Quantitative descriptive	Y	Y	Y	N	Y	N	Y
Selvaraj (2019)	Quantitative descriptive	Y	Y	Y	Y	Y	Unclear	Y
Uwemdimo (2018)	Quantitative descriptive	Y	Y	Y	N	Y	N	Y
Zielisnki (2017)	Quantitative descriptive	Y	Y	Unclear	Unclear	Y	Y	Y
Author (year)	**Category of study design**	**Are there clear**	**Do collected data allow to address**	**Is there adequate rationale for using**	**Are there different components of the**	**Are the outputs of integration of**	**Are divergences and**	**Do the different components of the**

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		research questions?	research question?	a mixed methods design to address the research question?	study effectively integrated to answer the research question?	qualitative and quantitative components adequately interpreted?	inconsistencies between quantitative and qualitative results adequately addressed?	study adhere to the quality criteria of each tradition of the methods involved?
Farrell (2017)	Mixed methods	Y	Y	Y	Y	Y	N	N
Koita (2018)	Mixed methods	Y	Y	Y	Y	Y	N	N

Table 3b Quality assessment using the Quality Improvement Minimum Quality Criteria Set (QI-MQCS) version 1.0

Author (year)	Organizational motivation	Intervention rationale	Intervention description	Organizational Characteristics	Implementation	Study design	Comparator	Data source	Timing	Adherence	Health outcomes	Organizational readiness	Sustainability	Spread	Limitations
Colin (2016)	Y	Y	Y	Y	Y	Y	Y	Y	N	N	Unclear	Unclear	Y	N	Y
Hensley (2017)	Unclear	Y	Y	Y	Y	N	N	Y	N	N	Y	N	N	N	Y
Higginbotham (2019)	N	Y	Y	N	Y	Y	Unclear	Y	Y	N	Unclear	N	N	N	Y
Denny (2019)	N	Y	Y	Y	Y	Y	N	Y	Y	N	Y	Unclear	Y	N	Y

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Discussion

This systematic review provides a synthesis of housing-related screening questions asked of families in healthcare and social service settings, and the subsequent actions taken to support families with housing problems as part of screening. This synthesis demonstrates that the housing questions asked in screening tools are highly variable and there is no standard way to ask families about housing in the healthcare and social service setting. Not all housing screening tools provide a social prescribing avenue, and there are ethical considerations in relation to screening and social prescribing. Based on our findings, we provide some recommendations for screening and social prescribing for families experiencing housing issues, however it is important to note that screening and social prescribing for social determinants of health is an emerging area of healthcare and there is limited information on the validity of the screening tools included in this review. To our knowledge, this is the first time a systematic review has been conducted to catalogue the social screening tools used by healthcare and social service staff to ask patients or clients about their housing and address housing problems through social prescribing.

Housing screening questions asked in healthcare and social service settings

There was variation in the wording of screening questions and type of housing issues asked about when screening for housing problems. None of the included studies commented on standard ways to ask about housing. These findings indicate that there are no standards for asking families about housing problems in healthcare and social services. This is an unsurprising finding, as social prescribing to address social determinants of health in the healthcare and social service context is an emerging practice (51, 85). While it is widely recognised that social determinants such as housing, education, income and food security have an impact on health and wellbeing, the best way to address social needs in healthcare is unclear (47, 48, 51, 85). Eleven studies (n=11) in this review provided the exact wording of housing questions of the screening tools, while some studies only provided prompts rather than the exact questions, for example 'Home heating' (42), 'Residential

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mobility' (67), or 'Concerns about the physical condition of your housing' (65). It was not clear how the questions were asked, for example, whether healthcare and social service staff asked questions word-for-word, or used the screening tool questions as prompts or as a guide. The screening tools captured different housing aspects, from physical housing conditions to the social conditions of housing instability, housing affordability and neighbourhood safety. These different aspects align with the four pillars of housing and health equity; cost, conditions, consistency and context (86), however each screening tool asked a different combination of questions.

It was common for the screening tool questions included in this review to be tailored to groups of people deemed more likely to be experiencing housing issues, such as people experiencing poverty. Tailoring screening to specific communities by identifying the most common problems experienced by that community is considered a critical element of screening for social determinants of health (54). It is possible that the housing screening questions in the identified screening tools are highly variable because they are tailored to local contexts. While there is no standardized way to screen families for housing issues, variability in housing screening questions is potentially positive, as it allows healthcare and social service providers to ensure they capture and attempt to address the most commonly occurring issues experienced by their patients. Unfortunately, this may mean that less common housing aspects are not captured.

Validated housing screening tools

Most screening tools had been tested for construct validity, face validity, feasibility, reliability, sensitivity, specificity and/or acceptability. Limited information was provided on the testing of the specific housing questions embedded in screening tools. Some studies provided information on the validity of housing questions, however the majority did not. It is unclear whether the housing questions used in the screening tools of this review are rigorously tested and robust in producing the intended result of screening. Housing questions were derived or adapted from different sources and were developed using various methods. Housing questions were often adapted from existing surveys

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such as the American Household Survey and were not necessarily validated, which provides some concern for the robustness of the tools. Future research should rigorously test the quality of housing questions embedded in social screening tools and ensure validity in the specific contexts and among the populations the tools are being used.

Recommendations for practice and research

While all screening tools aimed to identify one or more housing issues, only 10 of the 21 studies included in this review offered a subsequent action as a result of identifying a housing need. This finding is important, as it shows that not all screening for housing issues leads to a potential action by the healthcare or social service provider. There was no information provided in the studies as to why some tools including an action and others did not.

Most studies focused on testing social screening in the healthcare or social service setting and did not aim to examine social prescribing after screening. Several studies highlighted the need for appropriate resources to be provided to families (60, 70, 71, 77), a finding consistent with the literature which highlights the need for clinicians to be aware of community resources available for the effective and appropriate treatment of social determinants of health (87). This raises an ethical issue of screening and referral, versus screening only; identifying an issue without offering an action to support a family experiencing housing issues may be considered unethical. However, if appropriate resources are not available, or providers are unclear whether the action they offer is effective, it may be preferable not to offer an action. Identifying a housing issue without providing support may thus be acceptable in cases where a suitable pathway to address the issue is unavailable. While research has found that providing an action upon screening for social determinants of health may lead to addressing social issues, health and social service providers should ensure that the actions and referral pathways are appropriate for their patients and clients.

Several screening tools led to the automatic referral of patients to community services when a patient indicated an unmet need. Garg (88) suggests avoiding this approach. It is critical that

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healthcare providers gauge whether patients wish to be assisted (88). As screening for social determinants of health and social prescribing is an emerging area of practice, and those screened are often from 'disadvantaged' groups, services are encouraged to ensure patients are provided ample opportunity to share decision-making in the screening process, including decision-making on any subsequent material assistance. Furthermore, it is recommended to tailor questions to specific communities by identifying the most common problems experienced by that community (54). Understanding the needs and unique context of the population group being screened is vital in order to increase the potential for a housing screening and referral tool to effectively support patients or clients with their housing needs.

Screening for social needs such as housing is considered sensitive. It is worth noting that several studies discussed the need for clinicians to be adequately trained in asking sensitive questions (53, 61, 76). Selvaraj (64) suggested clinicians establish relationships with families before screening for social needs. Some screening tools provided introductory framing to social screening to provide context behind a clinician asking sensitive questions, to ensure the patient does not feel targeted or singled out, and to ensure the patient understands that the health worker is asking so they can help address any issues. For example, the HealthBegins Upstream Risks Screening Tool and Guide begins with the opening sentence: "*Everyone deserves the opportunity to have a safe, healthy place to live, work, eat, sleep, learn and play. Problems or stress in these areas can affect health. We ask our patients about these issues because we may be able to help*." The SEEK tool frames screening in the following way: "*Being a parent is not always easy. We want to help families have a safe environment for kids. So, we're asking everyone these questions. They are about problems that affect many families. If there's a problem, we'll try to help.*" Introducing social screening with reasoning, specifically linking screening to action, may support patients in answering sensitive questions and alleviate stigma associated with experiencing social hardship.

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Four studies screened high risk populations, such as children with a disability, teenage parents, families with open child welfare cases, and immigrant families. This indicates that it may be important to screen those who may be more likely to experience housing problems. However, Garg (88) suggests screening all patients rather than targeting families based on demographic characteristics, as screening only those at increased risk may reinforce preconceived ideas about particular groups and lead to stigmatisation. While most studies captured race and/or ethnicity of study participants, only two studies specifically examined the cultural acceptability and appropriateness of the screening tool used with families. Further research is needed to ascertain a targeted versus a general approach to screening.

According to the studies in this review, actions to address housing issues were referral to a social worker or social service, provision of a community resource guide or community resource information, and support by an in-person navigator. It is important to note that not all screening tools and referral pathways were tested for effectiveness, however the referral pathways and actions in these studies could be useful in supporting patients or clients with housing issues.

While not an aim of this review, it was found that housing issues are prevalent among those who are screened for housing in healthcare and social service settings (see Supplementary Table 2). Housing issues are prevalent and also vary widely, from physical quality of housing to issues of housing stability. However, there is no standard way to quantify housing problems. Information collected about housing problems are highly variable and there is no consistent method of measuring the severity of housing issues. This calls for a need for greater consistency in how to quantify housing problems.

Housing is an important domain as part of screening for social needs among children and families in healthcare and social services. There is a need for professionals to screen or assess for housing problems in these settings to improve identification of housing problems and to make efforts to address them. Screening for housing issues in healthcare or social services provides an opportunity

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to identify housing-related problems, and often can lead to the referral of children, adolescents and families to relevant services to address their needs. Garg (42) found "that a simple primary care screening and referral system for unmet basic needs increased families' receipt of community based resources". Screening and referral also allows for the ongoing capture of the housing history of a patient or client over time, and provides healthcare workers with a formal process to assess social needs, and "a concrete measure and rationale to use when referring clients" (73). Furthermore, it draws clinicians' attention to the housing programs available (73). Screening for social needs has been found to increase community resource referrals. Before universal screening for social needs at one paediatric clinic, 14 of 700 (2.0%) patients at well child visits were referred to community resources, and upon universal screening for social needs being introduced this increased 6 times (101 of 759, 13.3%) (64).

Our systematic review shows that currently there is high variability in the housing questions asked of families in healthcare and social service settings. It also shows that currently not all screening of housing issues in healthcare and social services leads to action. Based on these findings, there is potentially a need for guidelines for health and social service providers on how to screen families and address their housing-related needs. While some social screening tools in this review were tested for validity, there is currently a lack of strong evidence on the most appropriate and effective way to screen and address housing-related problems from the health and social service perspective. Furthermore, 21 of 23 articles in this systematic review were from studies in the USA and thus the screening tools would not necessarily be generalisable to other high-income countries because of the particular healthcare, social service and housing systems operating in the USA.

Conclusion

This systematic review provides recommendations and important considerations for clinicians and social service providers when screening families for housing issues to improve child health. It catalogues the housing questions asked of families in the context of screening for social

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determinants of health in healthcare and social service settings in high-income countries, and the social prescribing options clinicians and social service providers may offer families to address housing issues. Our systematic review found 23 peer-reviewed articles that described housing as a part of social screening and/or referral used with families in healthcare and social service settings. Nine of the screening tools identified were validated. There are no standard housing-related questions that healthcare and social service workers ask families, and the housing questions in social screening tools vary widely. An action and/or a referral pathway was part of 10 of 21 identified social screening tools, and 7 of the 10 screening tools specifically provided a social prescribing option for housing problems. Referral to a social worker or social service agency, provision of a community resource guide to families, or referral to a specific housing program (housing and child welfare demonstration project) were common and accepted actions to support families with housing problems. There is a need to develop guidelines for health and social service providers to effectively and appropriately screen families and address their housing-related issues.

Figure legend

Figure 1 Search flowchart

Figure 2 Housing aspects asked in screening tools

Abbreviations

ASK	Addressing Social Key Questions for Health
FAMNEEDS	Family Wellness Screening
IHELLP	Income, Housing, Education, Legal status, Language/Immigration, Personal Safety
MMAT	Mixed Methods Appraisal Tool
QI-MQCS	Quality Improvement Minimum Quality Criteria Set
QRAFT	Quick Risks and Assets Family Triage
RCT	Randomized Control Trial
SEEK	Safe Environment for Every Kid

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USA	United States of America
WE CARE	Well Child Care, Evaluation, Community Resources, Advocacy, Referral, Education

Ethics Declarations

Ethics approval not required.

Consent for publication

Not required.

Competing interests

The authors declare no competing interests

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Data Availability Statement

All data relevant to the study are included in the article or uploaded as supplementary information.

Author contributions

SW, MA, KH, JC, MM, LR, MJ, AM, JS, CS, RL, CE and KC contributed to the conceptualisation of the study. MA led the development of the study protocol, supported by SW and KH. NW, AA, SW, MA and KH reviewed study titles and abstracts. AA, SW, MA and KH discussed disagreements about inclusion of studies. AA and SW reviewed full texts of papers, extracted data from included papers and assessed the quality of each study. LT, SR, MM, LR, MJ, AM, KED, KZ, SSL and KC contributed to the interpretation of findings. AA was the major contributor in the review of studies, data extraction, quality assessment, interpretation of findings as well as writing up the manuscript. All authors read the final manuscript, provided edits and approved the final manuscript.

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Figure 1 Search flowchart

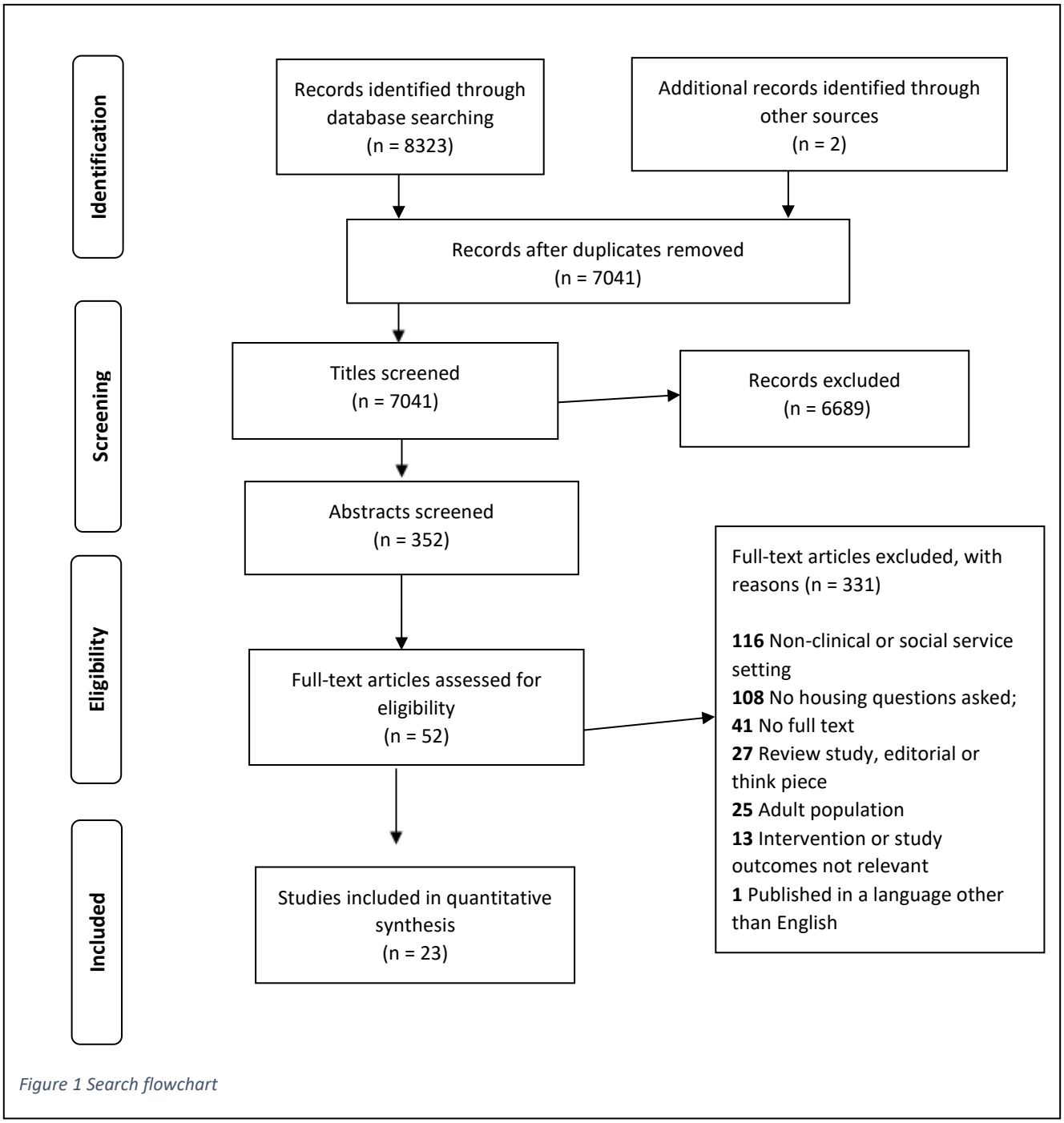


Figure 1 Search flowchart

Figure 2 Housing aspects asked in screening tools

1) General housing problems (housing problems not specified, or multiple housing problems captured);
2) Physical or structural housing problems;
3) Crowding;
4) Pests;
5) Utilities;
6) Housing instability, insecurity or homelessness;
7) Need for modifications in the home;
8) Unsafe neighbourhoods and;
9) Other (housing subsidy, basic living needs, other concerns with housing).

Figure 1 Housing aspects asked in screening tools

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Supplementary Table 1 Search strategy and search terms

Table 1 Search strategy and search terms

1. (House or Housing or home or Homeless or living conditions or social determinants or social situation or psychosocial or socioeconomic factors or basic needs).mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms]* *2. social conditions/ or social environment/ or socioeconomic factors/ or economic status/ or poverty/ or child welfare/ or social welfare/ or residence characteristics* *3. 1 or 2* *4. (screen* or detect or assess*or tool or survey* or question* or questionnaire or standardised or standardized or surveillance or ask*).mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms]* *5. exp Surveys/ and Questionnaires/* *6. 4 or 5* *7. (health service or community health or community service or clinical setting or clinical care or social service or primary care or medical home).mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms]* *8. Health Services/* *9. 7 or 8* *10. 3 and 6 and 9* *11. limit 10 to (English language and humans and yr="2009-2019" and "all child (0 to 19 years)")*
[PUBMED: Terms used for social services abstracts] ("Residence Characteristics"[Mesh] OR (residence[text word] OR housing[text word] OR house*[text word] OR home*[text word]) AND (condition[text word] OR state[text word]) **[CINAHL: Terms used for social services abstracts]** ((MH "Residence Characteristics+") AND (condition OR state)) OR "residence condition" OR "residence state" OR "housing condition" OR "housing state" OR "house* condition" OR "House* state" OR "home condition" OR "home state") **[ProQuest: Terms used for social services abstracts]** ((SU.EXACT("Housing") OR SU.EXACT("Residence")) AND (condition OR state)) OR "residence condition" OR "residence state" OR "housing condition" OR "housing state" OR "house* condition" OR "House* state" OR "home* condition" OR "home state")

Supplementary Table 2 Prevalence of housing problems

Table 2 Prevalence of housing problems

Housing problem category	**Description of housing problem (Housing question)**	**Prevalence n (%)**	**Study**
General housing problems or multiple problems	Housing problems "([Are you having] Problems with housing conditions (overcrowding, evictions, lead, utilities, mold, rodents)? Have you spoken with your landlord?)"	10	Garg et al. (2009)
	Problems with housing conditions e.g. overcrowding, evictions, lead, utilities, mold, rodents among all infants in sample	3.4	Beck (2012)
	Problems with housing conditions e.g. overcrowding, evictions, lead, utilities, mold, rodents among infants with complete documentation	5.6	Beck (2012)
	Housing major problem: Homeless; utilities shut off; structural problems	34	Hassan (2013 & 2015)
	Housing minor problem: Need fuel/heating assistance; threat of utilities shut off in past 12 months; if renting or receiving subsidized housing, concern about transfer or eviction	22	Hassan (2015)
	Living in an unhealthy environment	22.7	Gottlieb (2016)
	Housing problems ("Do you have any concerns about poor housing conditions like mice, mold, cockroaches? Do you have any concerns about being evicted or not being able to pay the rent? Do you have any concerns about not being able to pay your mortgage?")	22	Colvin (2016)
Housing instability/ insecurity and/or homelessness	Slept outside, in a shelter or in a place not meant for sleeping in the last month	7	Molina (2019)
	Moved house in the last 12 months	6	Molina (2019)
	Worried about unstable housing in next 2 months	14	Molina (2019)
	Housing insecurity	23	Molina (2019)
	Housing instability	43	Garg (2015)
	Housing instability: Total	40	Bottino (2019)
	Housing instability: Utilities threatened or shut off	33.2	Bottino (2019)
	Housing instability: Doubled up with another family	1.2	Bottino (2019)

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Supplementary Table 2 Prevalence of housing problems

	Housing instability: Concerned for eviction	10.6	Bottino (2019)
	Housing instability: Homeless or in sheltered housing	1.8	Bottino (2019)
	Concerns about the cost or stability of your housing	43.8	Gottlieb (2014)
	Not having a place to live	29.2	Gottlieb (2016)
	Unstable housing	28.6	Gottlieb (2018)
	Homeless	2	Hassan (2013)
	Housing insecurity, Moving from one home to another	1.8	Hensley (2017)
	Housing insecurity, Slept outside, in a shelter, or in a place not meant for sleeping	1.8	Hensley (2017)
	Housing insecurity	18.8	Higginbotham (2019)
	Current housing: Transitional, temporary or unsafe housing; utility shutoff; exiting residential, hospital or institution; eviction notice; doubled up or overcrowded in an unsustainable way, significant risk, level 3	4	Farrell (2017)
	Current housing: Unsheltered, residence not ordinarily used as sleeping accommodation; i.e., car, park, train station); temporary shelter; property condemned, severe risk, level 4	1.4	Farrell (2017)
	Housing history: Chronic housing instability, for example, multiple shelter placements; 3 or more moves in the past year, significant risk, level 3	2.9	Farrell (2017)
	Housing history: Two or more shelter stays; multiple episodes of living unsheltered in locations not ordinarily used as sleeping accommodations i.e., car, park, train station, severe risk, level 4	0.7	Farrell (2017)
	Housing instability, No. of moves in past year, at prenatal visit: 0	40.6	Kumar (2017)
	Housing instability, No. of moves in past year, at prenatal visit: 1	26.4	Kumar (2017)
	Housing instability, No. of moves in past year, at prenatal visit: >3	15.1	Kumar (2017)
	Homeless	2	Zielinski (2017)
	Structural problems	29	Hassan (2013)

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Supplementary Table 2 Prevalence of housing problems

Physical or structural housing problems	Housing condition: Potential for threat or harm; safety planning essential; unsustainable housing for any reason, significant risk, level 3; and Housing condition: Unsafe; immediate danger/risk; CPS call; needs urgent safety planning; or chronic, severe problems, severe risk, level 4	2.1	Farrell (2017)
	Concerns about the condition or quality of housing	11.5	Hensley (2017)
	Concerns about the physical condition of your housing	35.9	Gottlieb (2014)
	Concerns about the condition or quality of housing	13	Molina (2019)
Crowding	Need enlarged rooms at home	56	Badia (2014)
Utilities and home heating	Utilities shut off	4	Hassan (2013)
	Housing and bill insecurity	15.9	Selvaraj (2019)
	Unmet basic needs: Home heating	9	Garg (2015)
	“In the past 12 months have you had any utility shut off for not paying your bills?”	4	Molina (2019)
Need for modifications in home	Need adapted toilet at home	61	Badia (2014)
	Need modified kitchen at home	21	Badia (2014)
	Need walking aids	39	Badia (2014)
	Need hoists at home	30	Badia (2014)
Unsafe neighbourhoods	Threats to your child’s safety at school or in the neighborhood	51.8	Gottlieb (2014)
	Exposure to violence ("Do you have concerns about safety in your neighborhood?")	16.1	Hensley (2017)
	Do you have concerns about safety in your neighbourhood?	15	Molina (2019)
	Are you afraid you might be hurt in your apartment building or house?	5	Molina (2019)
	Personal safety (“Is there any domestic violence in the home? Do you feel safe in your neighborhood?”)	9	Patel (2018)
Other	Housing subsidy	4	Patel (2018)

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Supplementary Table 2 Prevalence of housing problems

Basic living needs	6	Rhodes (2012)
Other concerns with housing	16.5	Gottlieb (2016)
Needed housing resources at: 3 months postpartum; 6 months postpartum	35.4; 42.3	Kumar (2017)

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Reporting checklist for systematic review (with or without a meta-analysis).

Based on the PRISMA guidelines.

Instructions to authors

Complete this checklist by entering the page numbers from your manuscript where readers will find each of the items listed below.

Your article may not currently address all the items on the checklist. Please modify your text to include the missing information. If you are certain that an item does not apply, please write "n/a" and provide a short explanation.

Upload your completed checklist as an extra file when you submit to a journal.

In your methods section, say that you used the PRISMAreporting guidelines, and cite them as:

Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, Shamseer L, Tetzlaff JM, Akl EA, Brennan SE, Chou R, Glanville J, Grimshaw JM, Hróbjartsson A, Lalu MM, Li T, Loder EW, Mayo-Wilson E, McDonald S, McGuinness LA, Stewart LA, Thomas J, Tricco AC, Welch VA, Whiting P, Moher D. The PRISMA 2020 statement: An updated guideline for reporting systematic reviews

		Reporting Item	Page Number
Title			
Title	#1	Identify the report as a systematic review	1
Abstract			

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Abstract	#2	Report an abstract addressing each item in the PRISMA 2020 for Abstracts checklist	1
Introduction			
Background/rationale	#3	Describe the rationale for the review in the context of existing knowledge	1
Objectives	#4	Provide an explicit statement of the objective(s) or question(s) the review addresses	3 and 4
Methods			
Eligibility criteria	#5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses	4 and 5
Information sources	#6	Specify all databases, registers, websites, organisations, reference lists, and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted	4 and 5
Search strategy	#7	Present the full search strategies for all databases, registers, and websites, including any filters and limits used	4 and 5 and Supplementary Table 1
Selection process	#8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each	4 and 5

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		record and each report retrieved, whether they worked independently, and, if applicable, details of automation tools used in the process	
Data collection process	#9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and, if applicable, details of automation tools used in the process	4 and 5
Data items	#10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (for example, for all measures, time points, analyses), and, if not, the methods used to decide which results to collect	4 and 5, and Tables 1 and 2 Pages 24-30
Study risk of bias assessment	#11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and, if applicable, details of automation tools used in the process	Pages 6-7 and Tables 3a and 3b pages 30-32

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Effect measures	#12	Specify for each outcome the effect measure(s) (such as risk ratio, mean difference) used in the synthesis or presentation of results	NA - effect measures not relevant to this study
Synthesis methods	#13a	Describe the processes used to decide which studies were eligible for each synthesis (such as tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item #5))	NA - did not complete a synthesis of quantitative data and did not have an intervention
Synthesis methods	#13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics or data conversions	NA - did not complete a synthesis of quantitative data and did not have an intervention
Synthesis methods	#13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses	NA - did not complete a synthesis of quantitative data and did

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			not have an intervention
Synthesis methods	#13d	Describe any methods used to synthesise results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used	NA - did not complete a synthesis of quantitative data and did not have an intervention
Synthesis methods	#13e	Describe any methods used to explore possible causes of heterogeneity among study results (such as subgroup analysis, meta-regression)	NA - did not complete a synthesis of quantitative data and did not have an intervention
Synthesis methods	#13f	Describe any sensitivity analyses conducted to assess robustness of the synthesised results	NA - did not complete a synthesis of quantitative data and did not have an intervention

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Reporting bias assessment	#14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases)	NA - results were qualitative and did not have missing information
Certainty assessment	#15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome	NA - confidence intervals not relevant
Data items	#10b	List and define all other variables for which data were sought (such as participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information	4 and 5, and Tables 1 and 2 Pages 24-30
Results			
Study selection	#16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram (http://www.prisma-statement.org/PRISMAStatement/FlowDiagram)	6

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Study selection	#16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded	6
Study characteristics	#17	Cite each included study and present its characteristics	6 and Table 1 page 24
Risk of bias in studies	#18	Present assessments of risk of bias for each included study	Pages 6-7 and Tables 3a and 3b pages 30-32
Results of individual studies	#19	For all outcomes, present for each study (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (such as confidence/credible interval), ideally using structured tables or plots	NA - not quantitative measures
Results of syntheses	#20a	For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies	Pages 6-7 and Tables 3a and 3b pages 30-32
Results of syntheses	#20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (such as confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect	NA - no synthesis conducted

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Results of syntheses	#20c	Present results of all investigations of possible causes of heterogeneity among study results	10
Results of syntheses	#20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesised results	NA - no synthesis conducted and no sensitivity analysis conducted due to nature of study
Risk of reporting biases in syntheses	#31	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed	NA - not applicable to this study
Certainty of evidence	#22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed	NA - study did not involve quantitative measures
Discussion			
Results in context	#23a	Provide a general interpretation of the results in the context of other evidence	Pages 10-16
Limitations of included studies	#23b	Discuss any limitations of the evidence included in the review	Pages 10-16

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Limitations of the review methods	#23c	Discuss any limitations of the review processes used	Pages 10-16
Implications	#23d	Discuss implications of the results for practice, policy, and future research	Pages 10-16
Other information			
Registration and protocol	#24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered	4
Registration and protocol	#24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared	4
Registration and protocol	#24c	Describe and explain any amendments to information provided at registration or in the protocol	NA - no amendments made
Support	#25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review	17
Competing interests	#26	Declare any competing interests of review authors	17
Availability of data, code, and other materials	#27	Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included	17

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studies; data used for all analyses; analytic code;

any other materials used in the review

Notes:

- 7: 4 and 5 and Supplementary Table 1
- 10a: 4 and 5, and Tables 1 and 2 Pages 24-30
- 11: Pages 6-7 and Tables 3a and 3b pages 30-32
- 12: NA - effect measures not relevant to this study
- 13a: NA - did not complete a synthesis of quantitative data and did not have an intervention
- 13b: NA - did not complete a synthesis of quantitative data and did not have an intervention
- 13c: NA - did not complete a synthesis of quantitative data and did not have an intervention
- 13d: NA - did not complete a synthesis of quantitative data and did not have an intervention
- 13e: NA - did not complete a synthesis of quantitative data and did not have an intervention
- 13f: NA - did not complete a synthesis of quantitative data and did not have an intervention
- 14: NA - results were qualitative and did not have missing information
- 15: NA - confidence intervals not relevant
- 17: 6 and Table 1 page 24
- 18: Pages 6-7 and Tables 3a and 3b pages 30-32
- 19: NA - not quantitative measures
- 20a: Pages 6-7 and Tables 3a and 3b pages 30-32

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- 10b: 4 and 5, and Tables 1 and 2 Pages 24-30
- 20b: NA - no synthesis conducted
- 20d: NA - no synthesis conducted and no sensitivity analysis conducted due to nature of study
- 31: NA - not applicable to this study
- 22: NA - study did not involve quantitative measures
- 23a: Pages 10-16
- 23b: Pages 10-16
- 23c: Pages 10-16
- 23d: Pages 10-16
- 24c: NA - no amendments made The PRISMA checklist is distributed under the terms of the Creative Commons Attribution License CC-BY. This checklist was completed on 08. June 2021 using https://www.goodreports.org/, a tool made by the EQUATOR Network in collaboration with Penelope.ai

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Screening and social prescribing in healthcare and social services to address housing issues among children and families: a systematic review

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Title

Screening and social prescribing in healthcare and social services to address housing issues among children and families: a systematic review

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Abstract

Objectives: Housing is a social determinant of health that impacts the health and wellbeing of children and families. Screening and referral to address social determinants of health in clinical and social service settings has been proposed to support families with housing problems. This study aims to identify housing screening questions asked of families in healthcare and social services, determine validated screening tools and extract information about recommendations for action after screening for housing issues. Methods: The electronic databases MEDLINE, PsycINFO, EMBASE, Ovid Emcare, Scopus and CINAHL were searched from 2009 to 2021. Inclusion criteria were peer-reviewed literature that included questions about housing being asked of children or young people aged 0-18 years old and their families accessing any healthcare or social service. We extracted data on the housing questions asked, source of housing questions, validity, and descriptions of actions to address housing issues. Results: Forty-eight peer-reviewed papers met the inclusion criteria. The housing questions in social screening tools vary widely. There are no standard housing-related questions that clinical and social service providers ask families. Fourteen screening tools were validated. An action was embedded as part of social screening activities in 26 of 41 studies. Actions for identified housing problems included provision of a community or clinic-based resource guide, and social prescribing included referral to a social worker, care coordinator, community health worker, social service agency, or referral to a housing and child welfare demonstration project. Conclusion: This review provides a catalogue of housing questions that can be asked of families in the clinical and/or social service setting, and potential subsequent actions.

Strengths and Limitations

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- The first systematic review to catalogue housing questions asked of families in clinical and social service settings, including information on validity of screening tools.
- Independent review of study selection, quality assessment and data extraction.
- Search terms may have been too narrow and may not have captured relevant papers that included information on housing questions in broader social screening tools.
- The quality of studies varied across the studies.
- Heterogeneity in study design, sample size, participant age, housing questions asked across studies, limiting ability to compare studies.

Introduction

Social determinants of health are the environments and conditions in which people are born, grow, work, live and age (1). The role of housing as a social determinant of health is well-established (2, 3). Healthy housing is described by WHO as "shelter that supports a state of complete physical, mental and social well-being" and "provides a feeling of home, including a sense of belonging, security and privacy" (3). Adequate housing is more than shelter and provision of quality physical dwelling conditions (4-6). Housing tenure and housing affordability are important factors in supporting psychological and developmental wellbeing (6-11). Similar to other Western countries, in Australia, where this review was conducted, almost one million people reside in housing considered to be in poor condition and there is an overrepresentation of young people, people in low-income households, and Aboriginal and Torres Strait Islander people living in poor housing conditions (12).

There is growing evidence that housing aspects such as housing quality and environments materially affect health, wellbeing and developmental trajectories well into adulthood (13-16). Housing issues in childhood have a particularly detrimental effect on health and wellbeing. Poor housing conditions in childhood have been predicted to lead to poorer health and increased mortality later in life, after controlling for the confounding effect of socioeconomic conditions (13, 14). Material and social factors that impact on children's wellbeing include poor physical housing quality (mould and damp,

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utilities, toxicants, disrepair, lack of heating, injury hazards), crowding, lack of outdoor space for children to play, accessibility within the home, housing affordability, tenure type, frequent residential moves, homelessness, lack of cultural appropriateness and complex neighbourhood quality (6, 16-23). Psychological and emotional distress, as well as behavioural problems, have been independently associated with children living in poor housing conditions and experiencing housing instability (20, 24-29). Housing instability, frequent moving and homelessness also negatively affect children's health (30) and children's educational outcomes (31). A range of physical health issues such as asthma, acute respiratory symptoms and low birthweight have been independently associated with damp and mouldy housing (22, 32, 33), and living in crowded conditions has been found to lead to infectious diseases such as tuberculosis, meningococcal disease, acute rheumatic fever and otitis media and gastrointestinal infections in New Zealand and Australia (34-41). Multiple forms of housing problems can coexist and have a compounded, cumulative impact on physical and mental health, particularly after prolonged exposure during childhood (11, 42).

Improving the quality of housing and the immediate environment around the home "can save lives, prevent disease, and increase quality of life" (3). There is evidence to support screening and social prescribing for social determinants in healthcare, including housing issues, to address basic resource needs, inequity and improve child health (29, 43-50). Screening involves incorporating standardized tools, such as surveys or questionnaires, completed with families in paediatric or primary care practices (51, 52). Social prescribing, sometimes known as community referral, is a mechanism for clinicians to link or refer patients to non-medical sources of support, such as to the community sector, to address social determinants of health and improve community wellbeing (49, 53-56). Social prescribing has been recognised as a pathway to "address physiological, physical, psychological, psychosocial or socioeconomic issues, as well as enhancing community wellbeing and social inclusion" (55). There is evidence to show that caregivers in the paediatric healthcare setting are interested in assistance if screening positively to housing instability (57). Screening for housing issues has been demonstrated to increase the occurrence of social prescribing. Screening increases

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referral of families to appropriate professionals or providing families with resources (58, 59), and can lead to improved child health (44). While many social screening tools addressing housing issues do exist (60, 61), to date there are no widely accepted guidelines for health and social service professionals to systematically identify specific housing problems for children and address them (29, 59, 62). The aims of this systematic review were to catalogue housing screening questions asked of families in healthcare and social service settings, as described in the international literature; determine if any validated screening tools exist; and extract information about any recommendations for action after screening described in the literature, including social prescribing such as referral through health, housing and social care pathways.

Findings from this review will inform the development of an integrated detection, referral and action pathway to be used in healthcare and social service settings with Aboriginal and Torres Strait Islander children and families who may be experiencing poor health due to housing problems in Australia. The detection, referral and action pathway will also be used with non-Aboriginal patients and clients who are experiencing housing-related health issues.

Methods

Search strategy

A study protocol was developed, registered with the University of York Centre for Reviews and Dissemination (PROSPERO) and updated on 28/04/2020, registration number CRD42020159816 (https://www.crd.york.ac.uk/prospero/). A search of the international peer-reviewed and grey literature was conducted using the following inclusion criteria: studies that include questions about housing being asked of the population group, children or young people aged 0-18 years old and their families, accessing any health or social service. There were no restrictions on study type, design or method, and studies were not limited by country. Studies were limited to the English language and to published literature in the 10-year period 1 January 2009 to 14 December 2021. Studies about

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health assessment conducted with clients of housing services were excluded. Review papers were excluded.

The electronic databases MEDLINE, PsycINFO, EMBASE, Ovid Emcare, Scopus and CINAHL from 2009 to 2021 inclusive were systematically searched. The search strategy and key search terms are included in Supplementary Table 1, in line with the PRISMA guidelines. A secondary search of citations from bibliographies of related peer-reviewed articles, and an additional search of the grey literature using the search terms “social needs screening tool”; “housing screening tool for children” in Google Scholar were undertaken.

All study titles and abstracts were reviewed independently in Covidence by two reviewers, this task was shared across six reviewers (NW, AA, SW, MA, KH and CKS). Disagreements about inclusion of studies were discussed and decisions made by consensus with four authors (AA, SW, MA and KH). Full texts of papers were reviewed independently by three reviewers (AA, SW and KH). Divergences were discussed with three reviewers (AA, SW and KH) and agreement reached.

Data extraction and analysis

Data was extracted from included papers using a standardized data extraction tool by three reviewers (AA,SW and CKS). The data extracted included specific details about the study types, aims and methods, populations, screening tools, housing questions, descriptions of actions, such as referral to appropriate services, to address housing issues and other items pertaining to the review question and specific objectives. If housing questions were not available in the papers, but the name of a screening tool was provided, the authors accessed the specific tool through a web search to extract the housing screening questions. Any disagreements that arose between the reviewers were resolved through discussion.

Quality assessment

Three reviewers (AA, KH and SW) assessed the quality of each study using two separate tools, dependent on study type. The Mixed Methods Appraisal Tool (MMAT) tool (63) was used to appraise

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quantitative RCTs, quantitative non-randomized trials, quantitative descriptive and mixed methods studies. Quality improvement (QI) studies were appraised using the Quality Improvement Minimum Quality Criteria Set (QI-MQCS) version 1.0 (64). The reviewers did not give studies an overall score for methodological quality, as it is discouraged to calculate an overall quality score using the MMAT tool (63). Consensus on quality was reached through discussion.

Patient and public involvement

This study was conceptualised, in part, with community members who have experienced housing issues, and people who are in regular patient contact with families experiencing housing issues impacting on health and wellbeing.

Results

Papers identified

The search strategy identified 10,922 titles (excluding duplicates) and an additional 10 papers were found through a grey literature search and through citations. 9077 titles and abstracts were screened, and 8609 of these were excluded, as these titles and abstracts did not provide any information that led screeners to believe that the full text papers would have questions about housing being asked of people aged 0-18 years old and their families accessing healthcare or social services. 468 full text papers were identified as potentially meeting inclusion criteria and were assessed for eligibility, and 420 were excluded. Reasons for exclusion are provided in **Figure 2**. The two most common reasons for exclusion were that housing questions were being asked outside of a clinical or social service setting (n= 127) and there were no questions about housing being asked (n=145). A total of 48 papers were included representing 41 studies.

Included papers

The 48 included papers are listed in Table 1. All papers were published in peer-reviewed journals. Studies were primarily conducted in clinical settings (n=38); paediatric primary care clinics, children's hospitals and emergency departments, urban community health centres, general paediatric centres,

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ambulatory healthcare practices and antenatal clinics. Three studies were conducted in social service settings including a disability service, a government child welfare unit and a government-funded housing and health support service. All studies were based in high-income countries, with 38 studies conducted in the USA, one study in Australia, one in New Zealand, and one study conducted in Spain.

Studies were heterogenous in terms of method, design, sample size, age of participants, social screening tools and housing questions included in screening tools. Study types varied; cross-sectional (30, 47, 57, 65-79), randomized controlled trials (43, 44, 80-83), pre and post studies (84-87), prospective (88, 89), quality improvement (59, 90-95), mixed methods (96, 97), tool adaptation and validation (98), retrospective (99-101), and secondary data analysis (102). Sample sizes ranged from 45 to 56,253 participants. Participants were newborns, children, adolescents and/or their families or caregivers. Age ranges of children varied from study to study, and some studies only included children with specific characteristics, such as children with cerebral palsy, children with a diagnosis of type 1 Diabetes, children with hearing loss, and children with open child welfare cases. See Table 1 for characteristics of included papers. An action, such a provision of a resource or a social prescribing activity was found in 26 of the 41 included studies, see Table 2 for social prescribing for housing issues.

Quality assessment and risk of bias

Assessing studies using the Mixed Methods Appraisal Tool, the three randomised control trials scored highly. One randomized control trial (RCT) met all the criteria (44, 81) and two studies scored highly, meeting the majority of the criteria. It was unclear whether Garg's (43) cluster RCT randomisation was adequately performed and whether participants adhered to the assigned intervention, and Gottlieb's (80) study did not provide adequate information on blinding. There were three robust quantitative non-randomized studies, scoring highly against most criteria (65, 83, 84). Patel's study did not provide complete outcome data (84), and it was unclear in Bottino's study

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whether participants were representative of the target population nor where there was complete outcome data (65). Twenty-eight studies were categorised as quantitative descriptive, and quality scores were mixed. A common issue related to the representative sample of the target population, with the sample size not being representative in seven of the twenty-eight studies, and another five studies providing insufficient information to assess whether the sample size was representative. The risk of non response bias was low in fifteen of the quantitative descriptive studies (73, 75, 76, 79, 82, 85-88, 98, 103-106). The two mixed methods studies were of good quality however both did not adequately address inconsistencies between quantitative and qualitative results and did not adhere to the tradition of each method (96, 97). There were seven quality improvement studies, assessed using the Quality Improvement Minimum Quality Criteria Set. Five of the seven studies scored highly, scoring well in 10 of the 15 criteria (59, 90, 93-95). Issues with QI studies included adherence, reporting data on health-related outcomes, organizational readiness and spread. See Tables 3a and 3b for quality assessment of included studies.

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Table 1 Included study characteristics

Author (year)	Country	Study type	Participants (age)	Sample size	Name of tool and source of housing questions	Validity, feasibility and/or acceptability of tool	Housing questions (if available)
Arbour et al. (2021) (93)	USA	Quality improvement	Newborns and their families	692	Not provided	Used validated, standardized screening questions	Housing instability; unhealthy housing
Badia et al. (2014) (98)	Spain	Quantitative - Tool translation, adaptation, validation	Parents of children and adolescents with cerebral palsy (8-18 years)	221	European Child Environment Questionnaire (ECEQ), based on the International Classification of Functioning, Disability and Health	Construct validity, translation, adaptation and validation of tool conducted specifically for the Spanish context	Enlarged rooms at home (need/availability?); Adapted toilet at home (need/availability?); Modified kitchen at home (need/availability?); Walking aids (need/availability?); Hoists at home (need/availability?)
Beck et al. 2012 (99)	USA	Quantitative - Retrospective review	Newborns and families (age not specified)	639	Developed based on literature review, in consultation with medical staff, social services and medical–legal staff and adapted from the USA's Survey on Income and Program Participation	Food security and domestic violence questions validated; housing questions not validated	"[Are you having] Problems with housing conditions (overcrowding, evictions, lead, utilities, mold, rodents)? Have you spoken with your landlord?"
Bottino et al. (2019) (65)	USA	Quantitative – cross-sectional	Children (3-10 years)	340	Adapted from the American Housing Survey	Food questions validated; no information on housing question validity	"What best describes your current living situation?" [Rent apartment; Rent room(s) within another person's home; Own home; Homeless or living in a shelter; Doubled up with another family; Live with parents/other family; Other]. Selecting 'Rent apartment' or 'Own home' triggered a follow-up item: "In the last 12 months, has the electric or gas company shut off the electricity or gas in your home or threatened to shut off the utilities in your home?" and "Are you concerned about a possible eviction (being "kicked out" from your home?"
Colvin et al. (2016) (59)	USA	Quantitative - postintervention only design	Children (median age 3 years)	45	Income, Housing, Education, Legal status, Language/Immigration,	Validity, sensitivity and specificity	"Do you have any concerns about poor housing conditions like mice, mold, cockroaches? Do you have any concerns about being evicted or not

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					Personal Safety (IHELLP), adapted from Kenyon et al's IHELLP social history tool (107)		being able to pay the rent? Do you have any concerns about not being able to pay your mortgage?"
Costich et al. (2019) (86)	USA	Quantitative - Retrospective pre-post analysis	Caregivers of children (4-12 years) with special health care needs	80	US Department of Agriculture 18-item Household Food Security Survey and Hager et al.'s (108) Food Insecurity screening tool	Sensitivity, specificity, and convergent validity	"Is it hard for you to access any of the following: food, housing, medical care?"
De Marchis et al. (2019) (109)	USA	Quantitative - cross-sectional	Adult caregivers of paediatric patients	969	See De Marchis 2021	See De Marchis 2021	See De Marchis 2021
De Marchis et al. (2020) (57)	USA	Quantitative - cross-sectional	Adult caregivers of paediatric patients	1021	See De Marchis 2021	See De Marchis 2021	See De Marchis 2021
De Marchis et al. (2021) (71)	USA	Quantitative - cross-sectional	Adult caregivers of paediatric patients	835	Accountable Health Communities (AHC) social risk screening tool and Children's HealthWatch (CHW) Housing Stability Vital Sign questions	First two AHC housing questions adapted from the validated Protocol for Responding to and Assessing Patients' Assets, Risks, and Experiences tool (110); Children's HealthWatch questions validated (111)	AHC: "Think about the place you live. Do you have problems with any of the following? (Check all that apply). Answers: (a) Pests such as bugs, ants, or mice (b) Mold (c) Lead paint or pipes (d) Lack of heat (e) Oven or stove not working (f) Smoke detectors missing or not working (g) Water leaks (h) None of the above"; "What is your housing situation today? (a) I have a steady place to live (b) I have a place to live today, but I am worried about losing it in the future (c) I do not have a steady place to live (I am temporarily staying with others, in a hotel, in a shelter, living outside on the street, on a beach, in a car, abandoned building, bus or train station, or in a park". CHW: "In the past 12 months, was there a time when you were not able to pay the mortgage or rent on time?"; "In the past 12 months, how many times have you moved where you were living?"; "At any time in the

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							past 12 months, were you homeless or living in shelter (including now)?"
Denny et al. (2019) (90)	USA	Quality Improvement	Children (0-1 years and 1-5 years)	667	Injury Prevention (IP) + The Safe Environment for Every Kid (SEEK) tool, adapted from Gittelman et al.'s IP tool, and Dubowitz's SEEK tool (112, 113)	Injury Prevention (IP) tool tested for reliability	"Do you have trouble paying utilities or maintaining a safe place to live?"; "Do you use safety gates to protect your child at the top and bottom of all stairways in your house?"; "Are all potentially harmful household cleaners and pesticides either in locked storage OR out of the reach of children?"; "Is the hot water heater in your house adjusted to less than 120 degrees?"
Farrell et al. (2017) (96)	USA	Quantitative - mixed methods population survey	Families with open child welfare cases	6828	Quick Risks and Assets for Family Triage (QRAFT), adapted from Risks and Assets for Family Triage (RAFT) assessment instrument	Some indication of content, construct, and predictive validity, however inconclusive information	Current Housing, Housing Condition, and Housing History, with 5-item risk scale for each, measuring level of housing instability and homelessness
Fiori et al. (2019) (94)	USA	Quality improvement	Paediatric patients (37-144 months)	7266	Adapted from the Health Leads USA screening toolkit (114)	Validity	Exact questions used were not provided
Fiori et al. (2020) (88)	USA	Quantitative – prospective pragmatic	Paediatric patients (2-13 years)	4948	Adapted from the Health Leads USA screening toolkit (114)	Validity	Exact questions used were not provided
Fiori et al. (2020) (115)	USA	Quantitative	Children (0-17 years)	15,503	Adapted from the Health Leads USA screening toolkit (114)	Validity	Exact questions used were not provided
Garg et al. (2009) (66)	USA	Quantitative – cross-sectional descriptive	Parents of children 2 months old to 10 years	100	Well Child Care, Evaluation, Community Resources, Advocacy, Referral, Education (WE CARE), developed collaboratively among paediatric clinic staff and initially guided by Bright Futures paediatric intake form (29)	Face validity	"Do you think you are at risk of becoming homeless?"

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Garg et al. (2015) (43)	USA	Quantitative – cluster RCT	Families of infants <6 months old	336	Well Child Care, Evaluation, Community Resources, Advocacy, Referral, Education (WE CARE), adapted from a larger psychosocial screening instrument; three housing instability questions derived from the Children's HealthWatch survey (111)	Validity	Home heating and Housing instability (housing instability questions: having a steady place to sleep; household crowding i.e. >2 people per bedroom, or having moved >2 times in the past year)
Gottlieb et al. (2014) (80)	USA	Quantitative - RCT	Caregivers of paediatric emergency department patients	538	Some items adapted from the Medical Advocacy Screening questionnaire; other items adapted from other existing validated surveys	Neighbourhood safety question validated	Concerns about the physical condition of your housing; Concerns about the cost or stability of your housing; Threats to your child's safety at school or in the neighbourhood
Gottlieb et al. (2016) (44)	USA	Quantitative - RCT	Children (mean age 5 years)	1809	Used survey instrument from 2014 RCT	Neighbourhood safety question validated	Not having a place to live; unhealthy living environment; other concerns with housing
Gottlieb et al. (2018) (81)	USA	Quantitative - RCT	Children (mean age 5 years) and families in acute care setting	718	Used survey instrument from 2014 RCT	Neighbourhood safety question validated	Not having a place to live; unhealthy living environment; other concerns with housing
Hardy et al. (2021) (87)	USA	Quantitative - retrospective pre-post	Children and youth (0-21 years)	56,253	Social need domains informed by what are known to affect patients/families and selected based on the consensus of institutional stakeholders	No information provided	"Do you think you are at risk of becoming homeless?"; "Are you currently homeless?"
Hassan et al. (2013) (67)	USA	Quantitative – cross-sectional	Young adult patients (15-25 years)	401	The Online Advocate, adapted from the American Housing Survey	Used previously validated survey questions	Major: Homeless; utilities shut off; structural problems such as leaking roof, non-functional plumbing, or rodent infestation. Minor: Need fuel/heating assistance; threat of utilities shut off in past 12 months; if renting or receiving

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							subsidized housing, concern about transfer or eviction
Hassan et al. (2015) (47)	USA	Quantitative – prospective intervention	Young adult patients (15-25 years)	401	The Online Advocate, adapted from the American Housing Survey	Used previously validated survey questions	Homeless; having utilities shut off; or having existing structural problems such as leaking roof, non-functional plumbing, or rodent infestation
Heller et al. (2020) (73)	USA	Quantitative – cross-sectional	Children under 18 years, and adults	24,633	Tool derived from modified version of Health Leads screening toolkit (114) and homegrown tool	Validity (Health Leads only)	"Are you worried that the place you are living now is making you sick? (has mold, bugs/rodents, water leaks, not enough heat)?"; "Are you worried that in the next 2 months, you may not have a safe or stable place to live? (eviction, being kicked out, homelessness)"
Hensley et al. (2017) (91)	USA	Quality Improvement	Paediatric patients and families	114	HealthBegins Upstream Risks Screening Tool	Feasibility	"In the last month, have you slept outside, in a shelter, or in a place not meant for sleeping? Y/N; In the last month, have you had concerns about the condition or quality of your housing? Y/N; In the last 12 months, how many times have you or your family moved from one home to another?"
Hershey et al. (2021)	USA	Quantitative – prospective	Caregivers of children and youth (4-18 years) with a diagnosis of T1D for >1 year and poor glycemic control or high care utilisation	53	Health Leads USA questionnaire	Validity	"Are you worried that, in the next 2 months, you may not have stable housing?"
Higginbotham et al. (2019) (92)	USA	Quality Improvement	Children (1 week to 5 years old)	53	Aligned with US Department of Housing and Urban Development housing insecurity definition and previous cross-sectional study by Cutts and colleagues (116)	Feasibility	Crowding "within the past 12 months there were two or more people per bedroom?"; doubling up "within the past 12 months we were temporarily staying with another family or had another family staying with us?"; and frequent moving "within the past 12 months we moved more than once?"

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Koita et al. (2018) (97)	USA	Quantitative - Validation of tool	Caregivers of children < 12 years	28	BARC* Pediatric Adversity and Trauma Questionnaire, modified from Oakland Children's Hospital Family Information and Navigation Desk needs assessment survey	Face validity	"Has your child/family ever been homeless? Has your child ever had problems with housing (for example not having a stable place to live, faced eviction or foreclosure, or lived with multiple families)?"
Kumar et al. (2017) (82)	USA	Quantitative – RCT follow up	Pregnant adolescents < 17 years	106	Modified from the Treatment Services Review assessment tool	No information provided	Residential mobility (number of moves made in the year before prenatal survey, duration of stay, number of people living at their current home, and cohabitation with parents, husband or boyfriend, and other relatives); Need for housing resources at 3 months and 6 months postpartum
Matiz et al. (2022)	USA	Quantitative – retrospective chart review	Children with hearing loss (0.6 – 7.9 years)	30	No information provided	No information provided	"Is it hard for you to access any of the following: Food, Housing, Medical Care, N/A.?"
Messmer et al. (2020)	USA	Quantitative – RCT	Parents of children with social needs	414	WE CARE	Validity	See WE CARE questions
Molina et al. (2019) (68)	USA	Quantitative – cross-sectional	Children and families	276	Adapted from health-related social needs questions from the National Academy of Medicine	No information provided	"In the last month, have you slept outside, in a shelter or in a place not meant for sleeping? In the last month, have you had concerns about the condition or quality of your housing? In the last 12 months, how many times have you or your family moved from one home to another? Are you worried that in the next 2 months, you may not have stable housing? In the past 12 months, have you had any utility (electric, gas, water or oil) shut off for not paying your bills? Do you have any concerns about safety in your neighbourhood? Are you afraid you might be hurt in your apartment building or house?"
Oldfield et al. (2021) (74)	USA	Quantitative – cross-sectional	Parents of children of all	154	WE CARE tool and the AHC instrument	Validity (WE CARE); AHC housing questions	WE CARE: "Do you think you are at risk of becoming homeless?"; "Do you have trouble

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			ages, and adolescents (>13 years)			adapted from the validated Protocol for Responding to and Assessing Patients' Assets, Risks, and Experiences tool (110)	paying your heating bill and/or electricity bill?"; AHC: "What is your living situation today? (I have a steady place to live, I have a place to live today, but I am worried about losing it in the future, I do not have a steady place to live (I am temporarily staying with others, in a hotel, in a shelter, living outside on the street, on a beach, in a car, abandoned building, bus or train station, or in a park)"; "Think about the place you live. Do you have problems with any of the following? (pests, mold, lead paint or pipes, lack of heat, oven or stove not working, smoke detectors missing or not working, water leaks, none)"
Patel et al. (2018) (84)	USA	Quantitative – pre and post intervention retrospective chart review	Patients at well-child visit (0-20 years)	322	Income, Housing, Education, Legal status, Language/Immigration, Personal Safety (IHELLP) (107)	No information provided	"Housing, utilities: What kind of housing do you live in (apartment, house)? Do you receive any type of housing subsidy (Section 8, public housing, shelter)? Do you have heat and hot water? Do you have trouble paying rent or utilities at the end of the month?" "Personal safety: Do you feel safe in your neighbourhood?"
Pierse et al. (2020) (106)	New Zealand	Quantitative	Low-income families with children (0-14), hospitalized for health conditions attributable to home environment, and pregnant women with newborns	895	Housing Concerns Survey; developed in partnership with researchers to identify areas of housing need	No information provided	"Is your home usually colder than you would like?"; "During the winter months, was your house so cold that you shivered inside?"; "Does your home smell moldy or musty?"; "Is there mold on the walls in bedrooms or living areas of your home?"; "Are there damp walls in the bedrooms or living areas of your home?"

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Polk et al. (2020) (75)	USA	Quantitative – cross-sectional evaluation	Low-income families	10,916 households	Health Leads universal social needs screener (114)	Validity	"Would you like help with any of the following: Finding housing search resources and emergency shelters?"
Power-Hays et al. (2020) (95)	USA	Quality improvement	Families of children with sickle cell disease	132	WE CARE	Validity	"Do you currently live in a shelter or have no steady place to sleep at night?"; "Do you think you are at risk of becoming homeless? If yes, is this an emergency?"; "Would you like help connecting to resources? Housing/shelter"
Ray et al. (2020) (104)	USA	Quantitative - observational	Caregivers accompanying a child (< 5 years) who received a low-acuity triage score	146	WE CARE and PRAPARE	Validity	See WE CARE and PRAPARE questions
Razani et al. (2020) (77)	USA	Quantitative - observational	Adult caregivers of paediatric patients	890	14-item social and mental health needs screening questionnaire (44)	Neighbourhood safety question validated	"Not having a place to live for example, concerns about eviction, foreclosure, staying with friends/family, and current homelessness"; "Unhealthy living environments, for example, problems such as mold, insects, rats or mice, excess trash"; "Other concerns with your housing"
Rhodes et al. (2012) (69)	Australia	Quantitative – cross-sectional	Children (4-8 years)	162	Family Psychosocial Resources (FSR) tool, adapted from the Family Resources Scale (117)	No information provided	Basic living needs
Sandel et al. (2018) (30)	USA	Quantitative – cross-sectional	Caregivers of children (0 to 48 months of age) in renter households	22,324 families	Not provided	Not tested	"During the last 12 months, was there a time when you were not able to pay the mortgage or rent on time?"; "In the past 12 months, how many places has the child lived?"; "What type of housing does the child live in?"; "Since the child was born, has she or he ever been homeless or lived in a shelter?"
Sandoval et al. (2020) (101)	USA	Quantitative – retrospective chart review	Mother-child dyads	268	MAMA's neighbourhood program prenatal screening questionnaire; postnatal	No information provided	Exact questions used were not provided

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					housing insecurity measured using Homelessness Screening Clinical Reminder Tool		
Selvaraj et al. (2019) (70)	USA	Quantitative – cross-sectional	Children and adolescents (2 weeks to 17 years)	2569	Addressing Social Key Questions for Health Questionnaire (ASK tool), developed by staff of four academic institutions and literature review of validated paediatric questionnaires on adverse childhood experiences and unmet social needs	Validity and feasibility	Housing and bill insecurity; Witnessed violence in the home or neighbourhood
Semple-Hess et al. (2019) (105)	USA	Quantitative – cross-sectional	English- and Spanish-speaking caregivers	768	Questions developed through existing screening tools reported in literature and from paediatric and social worker' knowledge of common needs of patients	Survey tested and refined	"Which of the following services have you/your child ever used?" and "In the next 12 months, which of the following services will you/your child need?" Options included: "6) temporary housing and shelters; 7) safe housing services"
Sokol et al. (2021) (118)	USA	Quantitative – evaluation	Outpatient paediatric patients (0-18y)	30,485	Developed from PRAPARE (110)	Housing question validity; no testing for referral question	"In the next 2 months, are you worried that you may not have stable housing?" For referral and resource support: "Do you want to be contacted by the assistance program for help on any of your responses?"
Spencer et al. (2020) (76)	USA	Quantitative – cross-sectional	Caregivers of school-age children	943	WE CARE	Face and content validity	"Do you think you are at risk of becoming homeless?"; "Do you have trouble paying your heating bill for the winter?"
Uwemedimo (2018) (102)	USA	Secondary data analysis	Caregivers of children < 18 years	148	Family Wellness Screen (FAMNEEDS), developed from previously published SDH screening tools (119)	Validity	"Do you worry that in the next 2 month, you/your family may not have a safe or stable place to live? Y/N. Where is your child living now, Options provided: private house/ apartment, room (in apartment/house), shelter, hotel/motel, no regular place, car, other]. Do you worry that the place you're living now is

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							making you sick? Y/N. If YES choose all that apply: Cigarette smoke, mold or dampness, rodents/bugs, peeling paint, broken appliances, open cracks/holes/wires, not enough heat, water leaks, none" Other questions: "Do you need help from a lawyer with housing, immigration, custody or child support problems? Y/N."
Vaz et al. (2020) (79)	USA	Quantitative - observational	Child–caregiver dyads	249	Social risk questionnaire based on the National Survey of Child Health, Children's HealthWatch, social risk tools created by Gottlieb (80) and Harris (120)	Children's HealthWatch questions validated (111); Gottlieb neighbourhood safety questions validated (80)	Exact questions used were not provided
Zielinski et al. (2017) (85)	USA	Quantitative - pre and post	Children	602	WE CARE tool (29)	Validity, feasibility and acceptability	Housing insecurity

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Screening tools and housing questions

Our systematic review found 28 social screening tools that include housing questions. The tools used in clinical and social service settings varied widely. The purpose of all tools was to screen for health- or wellbeing-related social needs among children, adolescents and/or families. The tools were used by healthcare or social service workers to facilitate asking patients or clients questions about social needs, including housing. There were two distinct approaches to screening; one approach was a general social determinants of health screening where housing was one of multiple domains among other social needs, and a second approach was to focus on housing as one domain, and only ask detailed housing-related questions. In the first approach, social needs assessed in tools included housing, food security, income, employment, education, childcare needs, transportation needs, health care access, legal concerns, medical insurance, violence, social connection and isolation.

The housing questions asked of participants varied across studies. Housing questions extracted from the included studies were divided into nine categories, see **Figure 1.**

Housing instability, insecurity or homelessness were the most common housing-related screening questions across the studies, with questions about homelessness, risk of homelessness, sleeping in a shelter, sleeping outside or in a place not meant for sleeping, sleeping in unsafe housing, frequent moving and temporary or unsafe housing (30, 43, 44, 57, 65, 67, 68, 71-80, 82, 83, 85-89, 91-96, 100, 101, 105, 118). Utilities as an unmet need, especially in relation to home heating, were included as a common housing issue captured in the social screening tools of studies in this review. Problems with utilities, such as utilities being shut off, threats of utilities being shut off or concern about not being able to pay for utilities, and in relation to heating the home (43, 73), was asked as part of social screening in eighteen (n=18) studies (43, 47, 65-68, 70, 71, 74-76, 78, 79, 87, 89, 90, 99, 106). The questions about utilities were often asked alongside broader housing question in screening tools, for example "[Are you having] Problems with housing conditions (overcrowding, evictions, lead, utilities,

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mold, rodents)?" (99) and "Do you have trouble paying utilities or maintaining a safe place to live?" (90).

Crowding was a common item in the social screening tools, with the following terms being used: "overcrowding" (99), "doubled up or overcrowded in an unsustainable way" (96) and "need enlarged rooms at home" (98). Crowding was considered a part of housing instability in one tool, i.e. "Housing instability: Doubled up with another family" (65). Some tools asked families about general housing problems and either did not specify the nature of the housing problem, or asked about multiple housing-related issues within the one general question (44, 47, 59, 66, 67, 96, 99). Structural and physical housing problems (67) included questions specific to the quality or condition of the home (68, 73, 80, 88, 91, 94, 106), mould or damp (59, 99, 102, 106), and unhealthy living environments (77). See Table 1 for further detail on the housing questions asked in screening tools.

Validity, feasibility and acceptability

Fourteen (n=14) screening tools had been validated, including the Family Wellness Screening tool or FAMNEEDS (102), Well Child Care, Evaluation, Community Resources, Advocacy, Referral, Education (WE CARE) tool(66), The Online Advocate (47), The Protocol for Responding to and Assessing Patients' Assets, Risks, and Experiences (PRAPARE) tool (110), The Health Leads Screening Toolkit (114), as well as other tools used across studies (43, 66, 77, 80, 93, 97, 98, 110). The studies included in this review did not provide detailed information on validity. It is not clear whether any of the screening tools are robust in producing their intended result, and to what extent the housing questions in particular are suitable for their aims.

The ASK tool (Addressing Social Key Questions for Health questionnaire) had been tested for validity and feasibility (70). The WE CARE tool used across multiple studies included in this review had been validated and tested for feasibility and acceptability (29, 85). The Income, Housing, Education, Legal status, Language/Immigration, Personal Safety or IHELLP tool was tested for validity, sensitivity and specificity (59, 107). Some tools were not validated however underwent other forms of testing such

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as feasibility, reliability or sensitivity. The screening tool used in Higginbotham's study was tested for feasibility (92), as was Hensley's HealthBegins Upstream Screening tool (91). The Injury Prevention tool was tested for reliability (90). Insufficient information was provided on the validity, feasibility or acceptability of the tools used in fourteen studies (30, 65, 68, 69, 73, 82, 84, 87, 90, 92, 99-101, 105). The information provided for the Quick Risks and Assets Family Triage or QRAFT tool showed some indication of content, construct and predictive validity, however it is unclear whether it was validated (96). The tool used in Beck's 2012 study did not validate the housing questions, however other questions in the tool were validated (99). The first two housing questions of the Accountable Health Communities instrument, used in two studies (71, 74), had been adapted from the validated PRAPARE tool (110).

The studies did not explore testing the cultural acceptability or appropriateness of the screening tools. Badia's (2014) study analysed the cultural acceptability of the European Child Environment Questionnaire in order to adapt it to the Spanish context. Uwemedimo's (2018) study took into consideration unique challenges faced by immigrant families when screening for social needs and recommended screening tools to be culturally sensitive through availability in multiple languages and for bilingual or multilingual patient navigators to support patients from immigrant backgrounds. Some studies involved screening in languages other than English, for example, in Spanish (68, 70, 74, 77, 79, 80, 97, 98). Power-Hays' study offered screening in English, Spanish, and Haitian Creole, and, provided patients with low literacy to have the screener read to them (95).

Social prescribing

Twenty-six studies reported either provision of information, or social prescribing, i.e. a linking or referral occurred upon identifying a social need as part of screening. Several studies provided patients with a community resource guide, community resource information or clinic-based resources (43, 44, 57, 59, 70-73, 75, 78, 86-88, 91-94) after screening for housing and other social needs. Social prescribing was also common, with patients screening positively being referred to a

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social worker, care coordinator, community health worker or advocate (59, 67, 75, 83, 85, 87, 89, 94, 95, 100, 101, 118), social service agency or community organisation (86, 95, 102),or referral to a housing and child welfare demonstration project (96). In Pierse's study families were referred from healthcare settings to the Well Homes program and asked about their perception of their housing conditions, leading to all families receiving information about keeping warm, dry and safe, and some being offered an intervention such as a mould kit, or a housing relocation referral (106). De Marchis' study (57) offered information about local services and resources to all patients, regardless of whether the patient participated in their study on social risk screening. In several studies, a patient navigator helped patients with contacting resources or community-based health and social services (44, 81, 83, 91, 102). Messmer's study investigated the impact of on-site versus remote patient navigators to address parent's unmet social needs, finding that referrals for housing were similar when an on-site patient navigator was present versus a remote patient navigator (27.2% vs 30.7) (83). Actions specific to housing-related problems, such as a social work referral specific to homelessness, were evident in ten of the 26 studies. In Hershey's study of children with high-risk type 1 Diabetes, families screened for social needs were offered assistance in the form of goal-setting with the support of a community health worker (89). Housing-specific goals included access affordable housing or affordable home renovations, help parent enrol in first-time homeowners' program, and assist in negotiating rental arrears (89). Screening for social needs resulted in an increase in referrals to appropriate services or resources, and some studies found that screening and subsequent referral led to an improvement in the housing-related issue (43, 86), or an improvement in child health (44, 81). Interestingly, social screening and social prescribing studies appear to be increasing, with the majority of studies found to be published from 2019 onwards. See Table 2 for a summary of information relating to actions as a result of screening for housing issues, or social prescribing.

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Table 2 Social prescribing for housing issues

Author (year)	Screening tool name (if available)	Action as a result of screening	Referral rates and outcome (if available)
Arbour et al. (2021) (93)	-	Provision of resource information for concrete supports	86% of families screening positively for general health-related social needs were provided resources information for concrete supports
Colvin et al. (2016) (59)	IHELLP	Social work referral or resources provided to families who screened positively to unmet social needs	77.8% of families screening positively for unmet social needs were provided social work resources or referrals; 13.3% already had social needs addressed; <10% had an unmet social need but were unable to be connected with a resource or referral
Costich et al. (2019) (86)	-	Social service referral, goal setting and resource navigation upon screening and participating in the Special Kids Achieving Their Everything (SKATE) Community Health Worker program	The number of caregivers reporting that it was hard to access housing was reduced from 23% to 9.5%
Farrell et al. (2017) (96)	QRAFT: Quick Risks and Assets for Family Triage	Assess eligibility to a housing and child welfare demonstration project, a supportive housing intervention, and for broader supportive housing referrals. Eligibility criteria: client scoring a 3 (significant risk) or 4 (severe risk) on any of the three housing items	5.4% of families scored 3 or 4 on housing items and upon further assessment, 5.3% were referred to the housing and child welfare demonstration project
Fiori et al. (2019); Fiori et al. (2020) (88, 94)	Adapted from Health Leads Screening Toolkit	For non-urgent issues: handoff with Community Health Worker who provides resources and schedule a follow-up. For urgent issues: referral to onsite social worker	Not reported
Garg et al. (2015) (43)	WE CARE: Well Child Care, Evaluation, Community Resources, Advocacy, Referral, Education	Referral made by clinician using the WE CARE Family Resource Book containing tear-out information sheets listing 2 to 4 free community resources available for each need. The information sheets contain the program name, a brief description, contact information, program hours, and eligibility criteria. Follow up one month after visit; staff telephoned mothers to assess contact of resources and update notes in the child's medical record	Mothers receiving WE CARE screening and referral had lower odds of being in a homeless shelter (aOR = 0.2; 95% CI, 0.1–0.9) and more mothers had enrolled in a new community resource at the 12-month visit (39% vs 24%; aOR = 2.1; 95% CI, 1.2–3.7) compared to families who did not receive WE CARE screening and referral
Gottlieb et al. (2018); Gottlieb et al. (2016) (44, 81)	-	Caregivers who indicated at least one social need received either written information on relevant community resources (active control group) or received support from an in-person navigator immediately after the child's visit and offered follow-up meetings (navigation intervention group)	Both groups showed improved child health after receiving either written information or an in-person navigator; child global health scores (lower scores indicate better health) improved a mean (SE) of -0.36 (0.05) in the navigator group and a mean (SE) of -0.12 (0.05) in the active control group

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Hardy et al. (2021) (87)	-	A resource sheet on community resource linkages was provided if general social need identified. Any urgent social need received a social work consult.	Not reported
Hassan et al. (2013); Hassan et al. (2015) (47, 67)	The Online Advocate	Resource specialist reviewed referrals; if questionnaire responses indicated acute concerns regarding homelessness, results were immediately shared with the provider and social worker to facilitate urgent intervention.	75% of participants screened had at least one referral need; 27% required referral relating to housing problem and 14% received referral relating to housing problem. 85% of participants with housing problems reached for follow-up at 1 or 2 months, and 30% of those with housing problems selected that the problem was "completely" or "mostly" resolved
Heller et al. (2020) (73)	-	Providers offered to connect patients to clinic-based resources.	Not reported
Hensley et al. (2017) (91)	HealthBegins Upstream Risks Screening Tool	Patients with at-risk results were provided a community resources guide with an educational handout, in order to identify local agencies and programs that addressed social needs. The guide listed supporting programs or agencies, and included resource eligibility requirements, contact information. Patients were helped in contacting listed community resources.	Not reported
Hershey et al. (2020) (89)	Health Leads USA	Families asked if they would like assistance with any reported needs and if the need was urgent, and connected to community health worker. Goals were set with families with support from a community health worker. Housing related goals included: Access affordable housing or affordable home renovations; Help parent enrol in first-time homeowners' program; Assist in negotiating rental arrears	41% of families requested assistance with housing; 16% of families selected goals relating to living situation
Higginbotham et al. (2019) (92)	-	Families screening positively for food insecurity and/or housing insecurity were provided a community resource guide to facilitate referral.	Of 13 families screening positive for food insecurity and/or housing insecurity, 85% were given a resource guide
Matiz et al. (2022)	-	Referral to a community health worker. Community health workers supported caregivers to navigate the complexities of social services, education, and healthcare after initial assessment.	93% of patients required social service referrals
Messmer et al. (2020)	WE CARE	In-person or offsite patient navigator referral to community-based health and social services	27.2% of families screening positively to housing as unmet need referred by on-site patient navigator, 30.7% of families screening positively to housing as unmet need referred by remote patient navigator

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Oldfield et al. (2021) (74)	WE CARE and AHC	Positive screens added to patient's medical record, notifying patients' primary paediatrician of positive screens, allowing paediatrician to make targeted referrals as needed	Not reported
Pierse et al. (2020) (106)	Housing Concerns Survey	Provision of information for families on how best to keep their home warm, dry, and safe, provision of a housing-related intervention e.g. mould kit, heater, and/or referral for a housing relocation, or health or social referral	A total of 5,537 interventions were delivered; Bedding, heaters, and draft stopping delivered over 90% of the time
Polk et al. (2020) (75)	Health Leads	Depending on category of need, patient provided rapid resource referral, i.e. information only or enrolled in Health Leads program where patient is contacted by a Health Leads advocate	6.3% of successful resource connection for housing
Power-Hays et al. (2020) (95)	WE CARE	Provision of relevant resource sheet and referral to local community organizations for the specific needs endorsed. Providers and patients decided whether families also required referral to social worker; social workers called families with positive screens 2-3 weeks after their visit	80% of patients were referred to a relevant community organization; 45% of patients available via follow-up phone call reached out to the community organization; 69% of patients who reached out stated that the community organization was helpful
Ray et al. (2020)	WE CARE and PRAPARE	Universal provision of community resource packet at two-week follow-up	Resource packet used by 37% of those who had reported a social need
Sandoval et al. (2021) (101)	MAMA's neighbourhood program prenatal screening questionnaire	Care coordinator providers referral or facilitates contact with agencies or community organizations	Not reported
Selvaraj et al. (2019) (70)	ASK Tool: Addressing Social Key Questions for Health Questionnaire	Referral made to community resources in relation to unmet social needs, including referrals made in relation to housing/bill insecurity	284 (11%) total referrals made in relation to unmet social needs; 111 (4.3%) referrals made in relation to housing/bill insecurity
Semple-Hess et al. (2019) (105)	Questions developed through existing screening tools	Caregivers with any emergent unmet need provided a social work consultation, as per institutional practice	Not reported
Sokol et al. (2021) (118)	Developed from PRAPARE	Referral to a social worker to provide support with identified need	14% of parents and youths requested a referral for identified needs
Uwemedimo (2018) (102)	FAMNEEDS Family Wellness Screen	Referral to local community resources by trained navigators and follow-up to ensure linkage to resource. Trained navigators either searched online social service databases or referred to social service case managers at designated partner	Approximately one-third (30.9%) successfully utilized program-provided resources at 12-week follow-up

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		community organisations	
Zielinski et al. (2017) (85)	WE CARE screening tool	Provision of additional resources on homelessness and/or social work referral relating to homelessness	14 (93%) provision of additional resources on homelessness; 11 (73%) social work referral relating to homelessness

Table 3a Quality assessment using the MMAT tool

Author (year)	Category of study design	Are there clear research questions?	Do collected data allow to address research question?	Is randomization adequately performed?	Are groups comparable at baseline?	Are there complete outcome data?	Are outcome assessors blinded to intervention provided?	Did participants adhere to the assigned intervention?
Garg (2015)	Quantitative RCT	Y	Y	Unclear	Y	Y	Y	Unclear
Gottlieb (2016 & 2018)	Quantitative RCT	Y	Y	Y	Y	Y	Y	Y
Gottlieb (2014)	Quantitative RCT	Y	Y	Y	Y	Y	Unclear	Y
Author (year)	**Category of study design**	**Are there clear research questions?**	**Do collected data allow to address research question?**	**Are participants representative of target population?**	**Are measurements appropriate regarding both the outcome and intervention (or exposure)?**	**Are there complete outcome data?**	**Are the confounders accounted for in the design and analysis?**	**During study period, is the intervention administered (or exposure occurred) as intended?**
Bottino (2019)	Quantitative non-randomized	Y	Y	Unclear	Y	Unclear	Y	Y
Messmer (2020)	Quantitative non-randomized	Y	Y	Y	Y	Y	N	Y
Patel (2018)	Quantitative non-randomized	Y	Y	Y	Y	N	Y	Y

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Author (year)	Category of study design	Are there clear research questions?	Do collected data allow to address research question?	Is the sampling strategy relevant to address the research question?	Is the sample representative of the target population?	Are the measurements appropriate?	Is the risk of non response bias low?	Is the statistical analysis appropriate to answer the research question?
Badia (2014)	Quantitative descriptive	Y	Y	Y	N	Y	Y	Y
Beck (2012)	Quantitative descriptive	Y	Y	Unclear	Unclear	Y	Unclear	Y
Costich (2019)	Quantitative descriptive	Y	Y	Y	Y	Y	Y	Y
De Marchis (2019, 2020, 2021)	Quantitative descriptive	Y	Y	Y	N	Y	N	Y
Garg (2009)	Quantitative descriptive	Y	Y	N	N	Y	N	Y
Hardy (2020)	Quantitative descriptive	Y	Y	Y	Y	Y	Y	Y
Hassan (2013 & 2015)	Quantitative descriptive	Y	Y	Y	Unclear	Y	Unclear	Y
Heller (2020)	Quantitative descriptive	Y	Y	Y	Y	Y	Y	Y
Hershey (2021)	Quantitative descriptive	Y	Y	Y	Unclear	Y	Unclear	Y
Fiori (2020) (88)	Quantitative descriptive	Y	Y	Y	Y	Y	Y	Y
Fiori (2020) (103)	Quantitative descriptive	Y	Y	Y	Y	Y	Y	Y
Kumar (2017)	Quantitative descriptive	Y	Y	Y	Y	Y	Y	Y
Matiz (2022)	Quantitative descriptive	Y	Y	Y	N	Y	N	
Molina (2019)	Quantitative descriptive	Y	Y	Unclear	Unclear	Y	Unclear	Unclear

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Author (year)	Category of study design	Are there clear research questions?	Do collected data allow to address research question?	Is there adequate rationale for using a mixed methods design to address	Are there different components of the study effectively integrated to answer	Are the outputs of integration of qualitative and quantitative	Are divergences and inconsistencies between	Do the different components of the study adhere to the quality criteria of each
Oldfield (2020)	Quantitative descriptive	Y	Y	Y	Y	Y	Y	Y
Pierse (2020)	Quantitative descriptive	Y	Y	Y	Y	Y	Unclear	Y
Polk (2020)	Quantitative descriptive	Y	Y	Y	Y	Y	Y	Y
Ray (2020)	Quantitative descriptive	Y	Y	Y	Y	Y	Y	Y
Razani (2020)	Quantitative descriptive	Y	Y	Y	Unclear	Y	Unclear	Y
Rhodes (2012)	Quantitative descriptive	Y	Y	Y	N	Y	N	Y
Sandoval (2020)	Quantitative descriptive	Y	Y	Y	Y	Y	Y	Y
Selvaraj (2019)	Quantitative descriptive	Y	Y	Y	Y	Y	Unclear	Y
Semple-Hess (2019)	Quantitative descriptive	Y	Y	Y	Y	Y	Y	Y
Sokol (2021)	Quantitative descriptive	Y	Y	Y	Y	Y	Y	Y
Spencer (2020)	Quantitative descriptive	Y	Y	Y	Y	Y	N	Y
Uwemedimo (2018)	Quantitative descriptive	Y	Y	Y	N	Y	N	Y
Vaz (2020)	Quantitative descriptive	Y	Y	Y	Y	Y	Y	Y
Zielinski (2017)	Quantitative descriptive	Y	Y	Unclear	Unclear	Y	Y	Y

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				the research question?	the research question?	components adequately interpreted?	quantitative and qualitative results adequately addressed?	tradition of the methods involved?
Farrell (2017)	Mixed methods	Y	Y	Y	Y	Y	N	N
Koita (2018)	Mixed methods	Y	Y	Y	Y	Y	N	N

Table 3b Quality assessment using the Quality Improvement Minimum Quality Criteria Set (QI-MQCS) version 1.0

Author (year)	Organizational motivation	Intervention rationale	Intervention description	Organizational Characteristics	Implementation	Study design	Comparator	Data source	Timing	Adherence	Health outcomes	Organizational readiness	Sustainability	Spread	Limitations
Arbour (2021)	Y	Y	Y	Y	Y	Y	Y	Y	Y	N	Y	Y	N	Y	Y
Colin (2016)	Y	Y	Y	Y	Y	Y	Y	Y	N	N	Unclear	Unclear	Y	N	Y
Fiori (2019)	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
Hensley (2017)	Unclear	Y	Y	Y	Y	N	N	Y	N	N	Y	N	N	N	Y
Higginbotham (2019)	N	Y	Y	N	Y	Y	Unclear	Y	Y	N	Unclear	N	N	N	Y
Denny (2019)	N	Y	Y	Y	Y	Y	N	Y	Y	N	Y	Unclear	Y	N	Y

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Power-Hays (2020)	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	N	Y	Y	N	Y

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Discussion

This systematic review provides a synthesis of housing-related screening questions asked of families in healthcare and social service settings, and the subsequent actions taken to support families with housing problems as part of screening. This synthesis demonstrates that the housing questions asked in screening tools are highly variable and there is no standard way to ask families about housing in the healthcare and social service setting. Not all housing screening tools are designed with provision of a resource or social prescribing embedded, for example, some housing screening tools are simply a set of questions to ask families about housing and practitioners using these tools are not prompted to provide information, referral or links to the community sector after screening. This raises ethical considerations in relation to screening and social prescribing. For example, is it ethical for health professionals and social service workers to collect information about a patient's housing without providing support if the housing condition is affecting the patient's health? Another question raised is what is the benefit of screening for housing, especially if there is no subsequent action taken to address housing issues? Based on our findings, we provide some recommendations for screening and social prescribing for families experiencing housing issues, however it is important to note that screening and social prescribing for social determinants of health is an emerging area of healthcare and there is limited information on the validity of the screening tools included in this review. Furthermore, the evidence base for social prescribing is currently lacking due to difficulties in the generation of robust studies evaluating social prescribing in healthcare (121). To our knowledge, this is the first time a systematic review has been conducted to catalogue the social screening tools used by healthcare and social service staff to ask patients or clients about their housing and address housing problems through an action such as provision of information or social prescribing.

Housing screening questions asked in healthcare and social service settings

There was variation in the wording of screening questions and type of housing issues asked about when screening for housing problems. None of the included studies commented on standard ways

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to ask about housing. These findings indicate that there are no standards for asking families about housing problems in healthcare and social services. This is an unsurprising finding, as screening and social prescribing to address social determinants of health in the healthcare and social service context is an emerging practice (53, 122). As more screening tools emerge and are validated, clinicians and social service staff may be more likely to choose from a range of pre-developed validated screening tools rather than design their own. This is evident in the use of the WE CARE tool, with 7 studies included in this review utilising this tool. A complexity with screening for housing is that definitions and measurements of housing issues are highly variable, inconsistent and/or are rarely defined explicitly across the literature, leading to problems with their adequate measurement and capture. As an example, housing insecurity is measured, defined and characterised in many different ways. Varying definitions of housing insecurity capture dimensions of housing stability, affordability, quality, safety, and/or homelessness, and the inconsistency and incompleteness of these definitions mean that people actually experiencing housing insecurity may not be captured within existing housing insecurity measures (123). This is likely to be true of other housing measures in clinical and social service settings and calls for greater consistency, completeness and standardisation of housing issues (123).

While it is widely recognised that social determinants such as housing, education, income and food security have an impact on health and wellbeing, the best way to ask about social needs in healthcare remains unclear (48, 49, 53, 122). There was no rationale provided for the choice of housing questions, and for subsequent actions. Twenty-five papers (n=25) in this review provided the exact wording of housing questions used in their studies, while some studies only provided prompts rather than the exact questions, for example ‘Home heating’ (43), ‘Residential mobility’ (82), ‘Concerns about the physical condition of your housing’ (80), or ‘unhealthy housing’ (93). In the latter examples it was not clear how the questions were asked, for example, whether healthcare and social service staff asked questions word-for-word, or used the screening tool questions as prompts or as a guide. Some studies reported the name of the screening tool they used, but did not list the

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questions. The screening tools captured different housing aspects, from physical housing conditions to the social conditions of housing instability, housing affordability and neighbourhood safety. These different aspects align with the four pillars of housing and health equity; cost, conditions, consistency and context (124), however each screening tool asked a different combination of questions, and sometimes asked several questions within one broad housing question.

Validated housing screening tools

Most screening tools used in the studies had been tested for construct validity, face validity, feasibility, reliability, sensitivity, specificity and/or acceptability. Limited information was provided on the testing of the specific housing questions embedded in screening tools. Some studies provided information on the validity of housing questions, however the majority did not. It is unclear whether the housing questions used in the screening tools of this review are rigorously tested and robust in producing the intended result of screening. Housing questions were often adapted from existing surveys such as the American Household Survey and were not necessarily validated, which provides some concern for the robustness of the tools. Future research should rigorously test the quality of housing questions embedded in social screening tools and ensure cultural appropriateness and validity in the specific contexts and among the populations the tools are being used.

Recommendations for practice and research

While all screening tools aimed to identify one or more housing issues, 26 of the 41 studies (63%) included in this review offered a subsequent action as a result of identifying a social need. This finding is important, as it shows that not all screening tools for social determinants of health are designed to facilitate provision of information or subsequent social prescribing by the healthcare or social service provider. There was no information provided in the studies as to why some tools included an action and others did not.

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Most studies focused on testing social screening in the healthcare or social service setting and did not aim to examine social prescribing after screening. Several studies highlighted the need for appropriate resources to be provided to families (66, 74, 90, 91, 102), a finding consistent with the literature which highlights the need for clinicians to be aware of community resources available for the effective and appropriate treatment of social determinants of health (125). This raises an ethical issue of screening and referral, versus screening only; identifying an issue without offering an action to support a family experiencing housing issues may be considered unethical. However, if appropriate resources are not available, or providers are unclear whether the action they offer is effective, it may be preferable not to offer an action. Identifying a housing issue without providing support may be acceptable in cases where a suitable pathway to address the issue is unavailable. In Hardy's study, the tool was developed based on the social worker's capacity to adequately address the identified social need (87). While research has found that providing an action upon screening for social determinants of health may lead to addressing social issues, health and social service providers should ensure that the actions and referral pathways are appropriate for their patients and clients.

Regarding the type of information provided to patients who screened positively to general social needs or housing, studies did not provide details on the information provided. Oldfield found that parents prefer to receive information on social needs through e-mail, text message or a paper print-out, rather than an in-person consult with a community health worker, however this was a small sample size and may not be generalisable (74).

Several screening practices led to the automatic referral of patients to community services when a patient indicated an unmet need. Garg (126) suggests avoiding this approach. It is critical that healthcare providers gauge whether patients wish to be assisted (126). As screening for social determinants of health and social prescribing is an emerging area of practice, and those screened are often from 'disadvantaged' groups, services are encouraged to ensure patients are provided ample opportunity to share decision-making in the screening process, including decision-making on

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any subsequent material assistance. Furthermore, it is recommended to tailor questions to specific communities by identifying the most common problems experienced by that community (60). Understanding the needs and unique context of the population group being screened is vital to increase the potential for a housing screening and referral tool to effectively support patients or clients with their housing needs.

Screening for social needs such as housing is considered sensitive. It is worth noting that several studies discussed the need for clinicians to be adequately trained in asking sensitive questions (59, 67, 99). Selvaraj (70) suggested clinicians establish relationships with families before screening for social needs. Some screening tools provided introductory framing to social screening to provide context behind a clinician asking sensitive questions, to ensure the patient does not feel targeted or singled out, and to ensure the patient understands that the health worker is asking so they can help address any issues. For example, the HealthBegins Upstream Risks Screening Tool and Guide begins with the opening sentence: "*Everyone deserves the opportunity to have a safe, healthy place to live, work, eat, sleep, learn and play. Problems or stress in these areas can affect health. We ask our patients about these issues because we may be able to help.*" The SEEK tool frames screening in the following way: "*Being a parent is not always easy. We want to help families have a safe environment for kids. So, we're asking everyone these questions. They are about problems that affect many families. If there's a problem, we'll try to help.*" Introducing social screening with reasoning, specifically linking screening to action, may support patients in answering sensitive questions and alleviate stigma associated with experiencing social hardship.

The majority of studies screened high risk populations, such as families on low incomes, children with a disability, teenage parents, families with open child welfare cases, and immigrant families. This indicates that it may be important to screen those who may be more likely to experience housing problems. Tailoring screening to specific communities by identifying the most common problems experienced by that community is considered a critical element of screening for social

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determinants of health (60). It is possible that the housing screening questions in the identified screening tools are highly variable because they are tailored to local contexts. Semple-Hess' study, for example, attempted to address the most commonly occurring issues experienced by their patients (105). Unfortunately, this approach may mean that less common housing aspects are not captured. While most studies captured race and/or ethnicity of study participants, only two studies specifically examined the cultural acceptability and appropriateness of the screening tool used with families. This is interesting given the high proportion of Hispanic/Latino participants (65, 73, 74, 77, 80, 86, 93, 97, 100) and African American participants in multiple studies (30, 65-67, 70, 76, 83, 87), and the overrepresentation of Maori and Pacific people in one study (106). However, Garg (126) suggests screening all patients rather than targeting families based on demographic characteristics, as screening only those at increased risk may reinforce preconceived ideas about particular groups and lead to stigmatisation. Further research is needed to ascertain a targeted versus a general approach to screening.

Studies did not report whether people with lived experience of housing problems were consulted in the development of questions. Rather, it was common for research teams, healthcare and social workers and institutional stakeholders to develop questions. Engaging directly with people who have lived experience of housing problems may lead to more appropriate and acceptable questions to specific communities.

According to the studies in this review, social prescribing actions to address housing issues were referral to a social worker or social service, care coordinator, community health worker or advocate and support by an in-person navigator. These types of social prescribing activities have been described as signposting/information referral (56). It is important to note that not all screening tools and referral pathways were tested for effectiveness, however the referral pathways and actions in these studies could be useful in supporting patients or clients with housing issues. While not an aim of this review, it was found that housing issues are prevalent among those who are screened for

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housing in healthcare and social service settings (see Supplementary Table 2). Housing issues are prevalent and also vary widely, from physical quality of housing to issues of housing stability. However, there is no standard way to quantify housing problems. Information collected about housing problems are highly variable and there is no consistent method of measuring the severity of housing issues. This calls for a need for greater consistency and completeness in how to quantify and define housing problems (123).

Housing is an important domain as part of screening for social needs among children and families in healthcare and social services. There is a need for professionals to screen or assess for housing problems in these settings to improve identification of housing problems and to make efforts to address them. Screening for housing issues in healthcare or social services provides an opportunity to identify housing-related problems, and often can lead to the referral of children, adolescents and families to relevant services to address their needs. Garg (43) found "that a simple primary care screening and referral system for unmet basic needs increased families' receipt of community based resources". Screening and referral also allows for the ongoing capture of the housing history of a patient or client over time, and provides healthcare workers with a formal process to assess social needs, and "a concrete measure and rationale to use when referring clients" (96). Furthermore, it draws clinicians' attention to the housing programs available (96). Screening for social needs has been found to increase community resource referrals. Before universal screening for social needs at one paediatric clinic, 14 of 700 (2.0%) patients at well child visits were referred to community resources, and upon universal screening for social needs being introduced this increased 6 times (101 of 759, 13.3%) (70). Given the current lack of evidence on the impact of screening and social prescribing, it would be useful to specifically generate evidence on the reach, scope and acceptability of social prescribing to understand its benefit, as suggested by Husk (121). A quantitative study of the acceptability of social risk screening in patients and caregivers, including asking about housing instability, found that 79% of participants reported that social risk screening was appropriate (72). A qualitative study linked to this aforementioned quantitative study found that

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social risks screening was considered important, and participants understood the link between social risks and health (127).

Our systematic review shows that currently there is high variability in the housing questions asked of families in healthcare and social service settings. It also shows that currently not all screening of housing issues in healthcare and social services leads to action. Based on these findings, there is potentially a need for guidelines for health and social service providers on how to screen families and address their housing-related needs. While some social screening tools in this review were tested for validity, there is currently a lack of strong evidence on the most appropriate and effective way to screen and address housing-related problems from the health and social service perspective.

The key recommendations based on our systematic review are as follows:

- We encourage screening tools for housing needs as part of a clinical or social service pathway to be followed by referral pathways
- It is important for clinical and social service staff to be trained in asking appropriate social needs questions in a sensitive manner, especially as housing issues are prevalent and also vary widely
- Housing needs should form part of holistic social needs assessment in healthcare and social services
- We encourage further studies from diverse settings to finalise the domains that are critical to document as part of social needs screening

There were limitations to the review, notably the search terms selected and inclusion criteria being too narrow. It is possible that a search that expanded beyond housing questions being asked of children or young people and their carers may have yielded additional housing questions that could be incorporated into use for families. It is possible that wider search terms may have captured more papers on broader social screening tools that included housing questions. Strengths of the review include rigorous critical appraisal of each study, allowing a true exploration of the strength of evidence and robust understanding by evaluating studies for internal and external validity.

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Furthermore, 45 of 48 articles in this systematic review were from studies in the USA and thus the screening tools would not necessarily be generalisable to other high-income countries because of the particular healthcare, social service and housing systems operating in the USA.

Conclusion

This systematic review provides recommendations and important considerations for clinicians and social service providers when screening families for housing issues to improve child health. It catalogues the housing questions asked of families in the context of screening for social determinants of health in healthcare and social service settings in high-income countries, and the social prescribing options clinicians and social service providers may offer families to address housing issues. Our systematic review found 48 peer-reviewed articles that described housing as a part of social screening and/or referral used with families in healthcare and social service settings. Fourteen of the twenty-eight screening tools identified were validated. There are no standard housing-related questions that healthcare and social service workers ask families, and the housing questions in social screening tools vary widely. An action and/or a referral was part of 26 of 41 identified studies, and 10 of the 26 studies specifically provided an action or social prescribing option for housing problems. Referral to a social worker or social service agency, provision of a community resource guide to families, or referral to a specific housing program (housing and child welfare demonstration project) were common and accepted actions to support families with housing problems. There is a need to develop guidelines for health and social service providers to effectively and appropriately screen families and address their housing-related issues.

Figure legend

Figure 1 Search flowchart

Figure 2 Housing aspects asked in screening tools

Abbreviations

AHC The Accountable Health Communities instrument

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ASK	Addressing Social Key Questions for Health
CHW	Children's HealthWatch
FAMNEEDS	Family Wellness Screening
IHELLP	Income, Housing, Education, Legal status, Language/Immigration, Personal Safety
MMAT	Mixed Methods Appraisal Tool
PRAPARE	Protocol for Responding to and Assessing Patients' Assets, Risks, and Experiences
QI-MQCS	Quality Improvement Minimum Quality Criteria Set
QRAFT	Quick Risks and Assets Family Triage
RCT	Randomized Control Trial
SEEK	Safe Environment for Every Kid
USA	United States of America
WE CARE	Well Child Care, Evaluation, Community Resources, Advocacy, Referral, Education

Ethics Declarations

Ethics approval not required.

Consent for publication

Not required.

Competing interests

The authors declare no competing interests

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Data Availability Statement

All data relevant to the study are included in the article or uploaded as supplementary information.

Author contributions

SW, MA, KH, JC, MM, LR, MJ, AM, JS, CS, RL, CE and KC contributed to the conceptualisation of the study. MA led the development of the study protocol, supported by SW and KH. NW, AA, SW, MA and KH reviewed study titles and abstracts. AA, SW, MA and KH discussed disagreements about

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inclusion of studies. AA and SW reviewed full texts of papers, extracted data from included papers and assessed the quality of each study. LT, SR, MM, LR, MJ, AM, KED, KZ, SSL and KC contributed to the interpretation of findings. AA was the major contributor in the review of studies, data extraction, quality assessment, interpretation of findings as well as writing up the manuscript. All authors read the final manuscript, provided edits and approved the final manuscript.

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Figure 1 Search flowchart

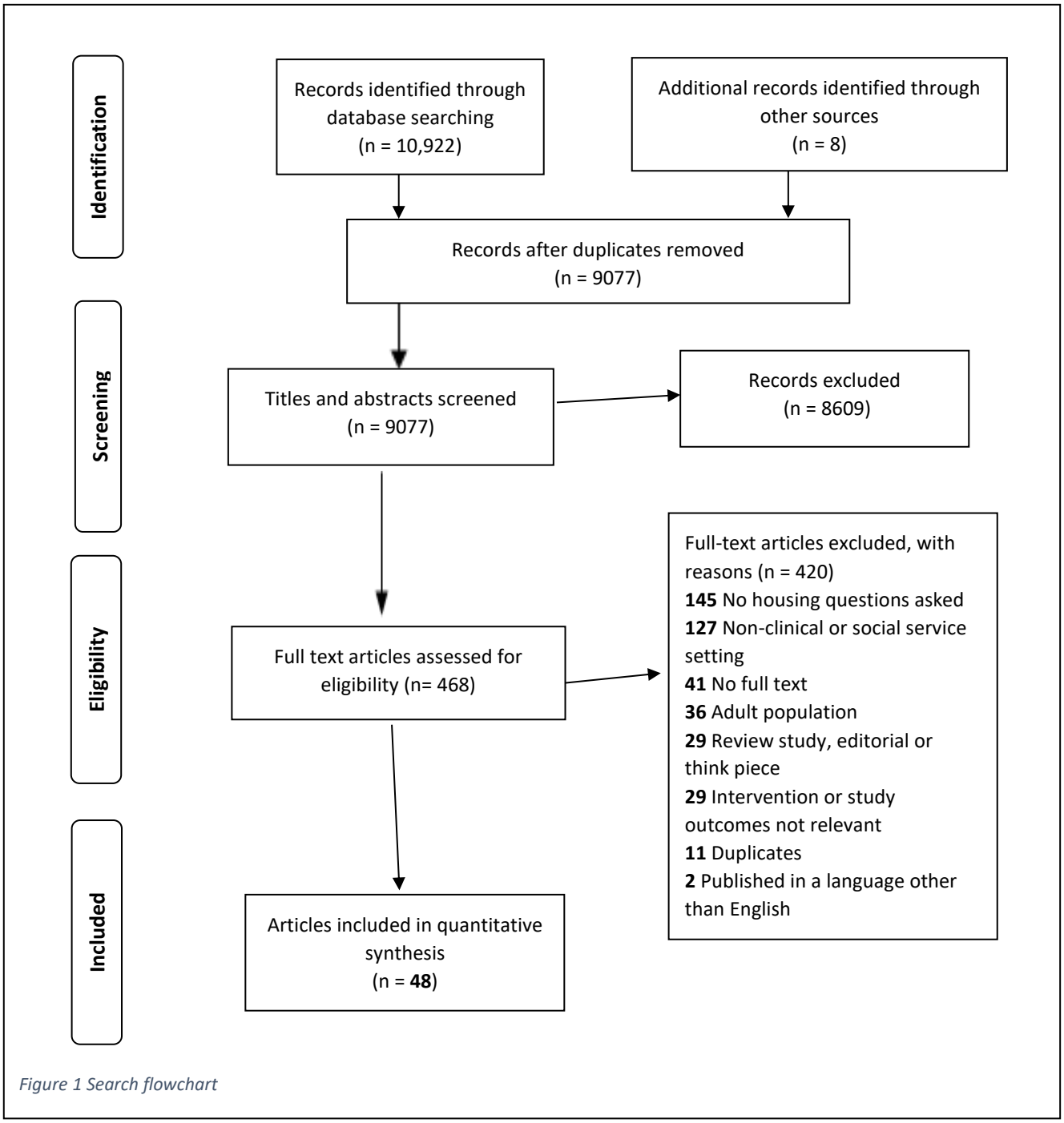


Figure 1 Search flowchart

Figure 2 Housing aspects asked in screening tools

1) General housing problems (housing problems not specified, or multiple housing problems captured);
2) Physical or structural housing problems;
3) Crowding;
4) Pests;
5) Utilities;
6) Housing instability, insecurity or homelessness;
7) Need for modifications in the home;
8) Unsafe neighbourhoods and;
9) Other (housing subsidy, basic living needs, other concerns with housing).

Figure 1 Housing aspects asked in screening tools

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Supplementary Table 1 Search strategy and search terms

Table 1 Search strategy and search terms (Medline, Emcare, EMBASE)

1. *housing.mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms]*
2. *homeless.mp.*
3. *living conditions.mp.*
4. *Socioeconomic Factors/*
5. *social situation.mp.*
6. *Social Environment/*
7. *psychosocial.mp.*
8. *basic needs.mp.*
9. *Child welfare/*
10. *Social welfare/*
11. *Social determinants.mp.*
12. *social conditions/ or economic status/ or poverty/*
13. *1 or 2 or 3 or 4 or 5 or 6 or 7 or 8 or 9 or 10 or 11 or 12*
14. *screen*.mp.*
15. *detect.mp.*
16. *assess*.mp.*
17. *tool.mp.*
18. *survey*.mp. or exp "Surveys and Questionnaires"/*
19. *question*.mp.*
20. *(standardised or standardized).mp.*
21. *surveillance.mp.*
22. *14 or 15 or 16 or 17 or 18 or 19 or 20 or 21*
23. *Health Services/*
24. *community health.mp.*
25. *((community or social) adj4 service*).mp.*

Supplementary Table 1 Search strategy and search terms

26. *clinical setting.mp.*
27. *Primary Health Care/*
28. *clinical care.mp.*
29. *medical home.mp.*
30. *23 or 24 or 25 or 26 or 27 or 28 or 29*31. 13 and 22 and 30
31. limit 31 to *(English language and humans and yr="2009-2021" and ("all infant (birth to 23 months)" or "all child (0 to 18 years)" or "newborn infant (birth to 1 month)" or "preschool child (2 to 5 years)" or "child (6 to 12 years)" or "adolescent (13 to 18 years)")*

[SCOPUS: Terms used] (TITLE-ABS-KEY (housing OR homeless OR "living conditions" OR "social determinants" OR (social W/3 (situation OR environment OR condition*)) OR psychosocial OR "socioeconomic factor*" OR "basic needs" OR "economic status" OR poverty OR (welfare W/3 (child OR social))) AND TITLE-ABS-KEY (screen* OR detect* OR assess* OR tool OR survey OR question* OR standardised OR standardized OR surveillance) AND TITLE-ABS-KEY ((care W/3 (health OR community OR social)) OR "medical home") AND KEY (child* OR infant OR toddler OR pediatr* OR paediatr* OR baby OR babies)) AND PUBYEAR > 2019 AND PUBYEAR < 2022 AND (LIMIT-TO (DOCTYPE , "ar")) AND (LIMIT-TO (LANGUAGE , "English")) AND (EXCLUDE (LANGUAGE , "French")) AND (EXCLUDE (SUBJAREA , "BIOC") OR EXCLUDE (SUBJAREA , "AGRI")) AND (EXCLUDE (SUBJAREA , "IMMU") OR EXCLUDE (SUBJAREA , "PHAR")) AND (EXCLUDE (PUBSTAGE , "aip")) AND (EXCLUDE (SUBJAREA , "ARTS") OR EXCLUDE (SUBJAREA , "DENT")) AND (EXCLUDE (SRCTYPE , "k")) AND (EXCLUDE (SUBJAREA , "ECON"))

[CINAHL: Terms used] housing OR homeless OR "living conditions" OR "social determinants" OR (social N3 (situation OR environment OR condition*)) OR psychosocial OR "socioeconomic factor*" OR "basic needs" OR "economic status" OR poverty OR (welfare N3 (child OR social)) AND screen* OR detect* OR assess* OR tool OR survey OR question* OR standardised OR standardized OR (surveillance care N3 (health OR community OR social)) OR "medical home" AND child* OR infant OR toddler OR pediatr* OR paediatr* OR baby OR babies. English only. child, preschool: 2-5 years, child: 6-12 years, all infant, infant: 1-23 months, adolescent: 13-18 years, infant, newborn: birth-1

[PsycINFO: Terms used] ((noft(housing) OR MAINSUBJECT.EXACT("Homeless") OR noft("living conditions" OR "psychosocial" OR "basic needs" OR "social determinants" OR "economic status") OR MAINSUBJECT.EXACT("Psychosocial Factors") OR MAINSUBJECT.EXACT("Poverty") OR MAINSUBJECT.EXACT("Socioeconomic status") OR noft(social NEAR/3 (environment OR situation OR condition*)) OR noft(welfare NEAR/3 (child OR social))) AND (noft(screen* OR detect OR assess* OR tool OR question* OR ask* OR surveillance OR standardised OR standardized) OR MAINSUBJECT.EXACT("Questionnaires") OR MAINSUBJECT.EXACT("Surveys")) AND ((MAINSUBJECT.EXACT("Primary Health Care") OR MAINSUBJECT.EXACT("Health Care Services")) OR MAINSUBJECT.EXACT("Community Health") OR MAINSUBJECT.EXACT("social services") OR noft("clinical setting" OR "medical home" OR "clinical care")) AND noft(child* OR infant* OR toddler OR pediatr* OR paediatr* OR adolescen* OR baby OR babies)) AND (la.exact("ENG")

Supplementary Table 2 Prevalence of housing problems

Table 2 Prevalence of housing problems

Housing problem category	**Description of housing problem (Housing question)**	**Prevalence (%)**
General housing problems or multiple problems	Problems with housing conditions e.g. overcrowding, evictions, lead, utilities, mold, rodents among all infants in sample (Beck 2012)	3.4
	Problems with housing conditions e.g. overcrowding, evictions, lead, utilities, mold, rodents among infants with complete documentation (Beck 2012)	5.6
	Housing problems (Colvin 2016; Garg 2009)	22; 10
	Housing quality and/or availability Fiori (2019); Fiori (2020)	40
	Housing major problem: Homeless; utilities shut off; structural problems (Hassan 2013 & 2015)	34
	Housing minor problem: Need fuel/heating assistance; threat of utilities shut off in past 12 months; if renting or receiving subsidized housing, concern about transfer or eviction (Hassan 2015)	22
Housing instability/ insecurity and/or homelessness	Housing insecurity (Arbour 2021; Higginbotham 2019; Matiz 2022; Molina 2019; Oldfield 2021; Hensley 2017; Sokol 2021)	13; 18.8; 10; 23; 6.7-12; 1.8; 10
	Housing instability (Bottino 2019; De Marchis 2020; Farrell 2017; Garg 2015; Gottlieb 2018; Heller 2020; Kumar 2017; Ray 2020; Sandel 2018; Vaz 2020)	40; 52; 2.9; 43; 28.6; 4.8; 15.1-40.6; 89; 8; 5
	Homeless (Bottino 2019; Farrell 2017; Gottlieb 2016; Hardy 2021; Hassan 2013; Molina 2019; Razani 2020; Zielinski 2017)	1.8; 1.4; 29.2; 29.5; 2; 7; 25; 2
	Homeless in child's lifetime (Sandel 2018)	12
	Risk of homelessness (Hardy 2021; Spencer 2020	1.8; 5.3
	Hard to access housing (Costich 2019)	23
	Concerns about the cost or stability of your housing (Gottlieb 2014)	43.8
	Concern for stable housing (Hershey 2021; Molina 2019)	17; 14
	Concerned for eviction (Bottino 2019)	10.6

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Supplementary Table 2 Prevalence of housing problems

	Housing as unmet need (Power-Hays 2020; Messmer 2020)	11; 29
	Behind on rent in past 12 months (Sandel 2018)	27
	Would like support finding housing search resources and emergency shelters (Polk 2020)	11.4 - 24.6
	Needed housing resources at: 3 months postpartum; 6 months postpartum (Kumar 2017)	35.4; 42.3
Physical or structural housing problems	Housing condition: Potential for threat or harm; safety planning essential; unsustainable housing for any reason, significant risk, level 3; and Housing condition: Unsafe; immediate danger/risk; CPS call; needs urgent safety planning; or chronic, severe problems, severe risk, level 4 (Farrell 2017)	2.1
	Concerns about the physical condition of your housing (Gottlieb 2014)	35.9
	Structural problems (Hassan 2013)	29
	Housing quality risk (Heller 2020)	6.5
	Concerns about the condition or quality of housing (Hensley 2017; Molina 2019)	11.5; 13
	Mold on the walls of bedrooms or living areas (Pierse 2020)	88
	Unhealthy living environments (Gottlieb 2016; Razani 2020)	22.7; 17
	Problems like mold, insects, rats or mice (Vaz 2020)	8
Crowding	Need enlarged rooms at home (Badia 2014)	56
	Doubled up with another family (Bottino 2019)	1.2
Utilities and home heating	Unmet basic needs: Home heating (Garg 2015; Sokol 2021)	9; 24.2
	Inadequate heating and draughts (Pierse 2020)	81
	“Is your home usually colder than you would like?” (Pierse 2020)	85
	“During the winter months, was your house so cold that you shivered inside?” (Pierse 2020)	67
	Utilities shut off (Hassan 2013; Molina 2019)	4; 4

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Supplementary Table 2 Prevalence of housing problems

	Utilities as unmet need (Messmer 2020; Ray 2020)	21; 84
	Housing and bill insecurity (Selvaraj 2019)	15.9
	Utilities threatened or shut off (Bottino 2019)	33.2
	Not enough money for utilities (Vaz 2020)	26
Need for modifications in home	Need adapted toilet at home (Badia 2014)	61
	Need modified kitchen at home (Badia 2014)	21
	Need walking aids (Badia 2014)	39
	Need hoists at home (Badia 2014)	30
Unsafe neighbourhoods	Threats to your child's safety at school or in the neighborhood (Gottlieb 2014)	51.8
	Do you have concerns about safety in your neighbourhood? (Hensley 2017; Molina 2019)	16.1; 15
	Are you afraid you might be hurt in your apartment building or house? (Molina 2019)	5
	Personal safety ("Is there any domestic violence in the home? Do you feel safe in your neighborhood?") (Patel 2018)	9
Other	Unhealthy housing (Arbour 2021)	3.5
	Other concerns with housing (Gottlieb 2016; Razani 2020)	16.5; 13
	Housing subsidy (Patel 2018)	4
	Basic living needs (Rhodes 2012)	6

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Reporting checklist for systematic review (with or without a meta-analysis).

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		Reporting Item	Page Number
Title			
Title	#1	Identify the report as a systematic review	1
Abstract			

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Abstract	#2	Report an abstract addressing each item in the PRISMA 2020 for Abstracts checklist	1
Introduction			
Background/rationale	#3	Describe the rationale for the review in the context of existing knowledge	1
Objectives	#4	Provide an explicit statement of the objective(s) or question(s) the review addresses	3 and 4
Methods			
Eligibility criteria	#5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses	4 and 5
Information sources	#6	Specify all databases, registers, websites, organisations, reference lists, and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted	4 and 5
Search strategy	#7	Present the full search strategies for all databases, registers, and websites, including any filters and limits used	4 and 5 and Supplementary Table 1
Selection process	#8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each	4 and 5

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		record and each report retrieved, whether they worked independently, and, if applicable, details of automation tools used in the process	
Data collection process	#9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and, if applicable, details of automation tools used in the process	4 and 5
Data items	#10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (for example, for all measures, time points, analyses), and, if not, the methods used to decide which results to collect	4 and 5, and Tables 1 and 2 Pages 24-30
Study risk of bias assessment	#11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and, if applicable, details of automation tools used in the process	Pages 6-7 and Tables 3a and 3b pages 30-32

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Effect measures	#12	Specify for each outcome the effect measure(s) (such as risk ratio, mean difference) used in the synthesis or presentation of results	NA - effect measures not relevant to this study
Synthesis methods	#13a	Describe the processes used to decide which studies were eligible for each synthesis (such as tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item #5))	NA - did not complete a synthesis of quantitative data and did not have an intervention
Synthesis methods	#13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics or data conversions	NA - did not complete a synthesis of quantitative data and did not have an intervention
Synthesis methods	#13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses	NA - did not complete a synthesis of quantitative data and did

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			not have an intervention
Synthesis methods	#13d	Describe any methods used to synthesise results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used	NA - did not complete a synthesis of quantitative data and did not have an intervention
Synthesis methods	#13e	Describe any methods used to explore possible causes of heterogeneity among study results (such as subgroup analysis, meta-regression)	NA - did not complete a synthesis of quantitative data and did not have an intervention
Synthesis methods	#13f	Describe any sensitivity analyses conducted to assess robustness of the synthesised results	NA - did not complete a synthesis of quantitative data and did not have an intervention

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Reporting bias assessment	#14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases)	NA - results were qualitative and did not have missing information
Certainty assessment	#15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome	NA - confidence intervals not relevant
Data items	#10b	List and define all other variables for which data were sought (such as participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information	4 and 5, and Tables 1 and 2 Pages 24-30
Results			
Study selection	#16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram (http://www.prisma-statement.org/PRISMAStatement/FlowDiagram)	6

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Study selection	#16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded	6
Study characteristics	#17	Cite each included study and present its characteristics	6 and Table 1 page 24
Risk of bias in studies	#18	Present assessments of risk of bias for each included study	Pages 6-7 and Tables 3a and 3b pages 30-32
Results of individual studies	#19	For all outcomes, present for each study (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (such as confidence/credible interval), ideally using structured tables or plots	NA - not quantitative measures
Results of syntheses	#20a	For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies	Pages 6-7 and Tables 3a and 3b pages 30-32
Results of syntheses	#20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (such as confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect	NA - no synthesis conducted

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Results of syntheses	#20c	Present results of all investigations of possible causes of heterogeneity among study results	10
Results of syntheses	#20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesised results	NA - no synthesis conducted and no sensitivity analysis conducted due to nature of study
Risk of reporting biases in syntheses	#31	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed	NA - not applicable to this study
Certainty of evidence	#22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed	NA - study did not involve quantitative measures
Discussion			
Results in context	#23a	Provide a general interpretation of the results in the context of other evidence	Pages 10-16
Limitations of included studies	#23b	Discuss any limitations of the evidence included in the review	Pages 10-16

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Limitations of the review methods	#23c	Discuss any limitations of the review processes used	Pages 10-16
Implications	#23d	Discuss implications of the results for practice, policy, and future research	Pages 10-16
Other information			
Registration and protocol	#24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered	4
Registration and protocol	#24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared	4
Registration and protocol	#24c	Describe and explain any amendments to information provided at registration or in the protocol	NA - no amendments made
Support	#25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review	17
Competing interests	#26	Declare any competing interests of review authors	17
Availability of data, code, and other materials	#27	Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included	17

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studies; data used for all analyses; analytic code;

any other materials used in the review

Notes:

- 7: 4 and 5 and Supplementary Table 1
- 10a: 4 and 5, and Tables 1 and 2 Pages 24-30
- 11: Pages 6-7 and Tables 3a and 3b pages 30-32
- 12: NA - effect measures not relevant to this study
- 13a: NA - did not complete a synthesis of quantitative data and did not have an intervention
- 13b: NA - did not complete a synthesis of quantitative data and did not have an intervention
- 13c: NA - did not complete a synthesis of quantitative data and did not have an intervention
- 13d: NA - did not complete a synthesis of quantitative data and did not have an intervention
- 13e: NA - did not complete a synthesis of quantitative data and did not have an intervention
- 13f: NA - did not complete a synthesis of quantitative data and did not have an intervention
- 14: NA - results were qualitative and did not have missing information
- 15: NA - confidence intervals not relevant
- 17: 6 and Table 1 page 24
- 18: Pages 6-7 and Tables 3a and 3b pages 30-32
- 19: NA - not quantitative measures
- 20a: Pages 6-7 and Tables 3a and 3b pages 30-32

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- 10b: 4 and 5, and Tables 1 and 2 Pages 24-30
- 20b: NA - no synthesis conducted
- 20d: NA - no synthesis conducted and no sensitivity analysis conducted due to nature of study
- 31: NA - not applicable to this study
- 22: NA - study did not involve quantitative measures
- 23a: Pages 10-16
- 23b: Pages 10-16
- 23c: Pages 10-16
- 23d: Pages 10-16
- 24c: NA - no amendments made The PRISMA checklist is distributed under the terms of the Creative Commons Attribution License CC-BY. This checklist was completed on 08. June 2021 using https://www.goodreports.org/, a tool made by the EQUATOR Network in collaboration with Penelope.ai

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BMJ Open

Screening and social prescribing in healthcare and social services to address housing issues among children and families: a systematic review

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Title

Screening and social prescribing in healthcare and social services to address housing issues among children and families: a systematic review

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Abstract

Objectives: Housing is a social determinant of health that impacts the health and wellbeing of children and families. Screening and referral to address social determinants of health in clinical and social service settings has been proposed to support families with housing problems. This study aims to identify housing screening questions asked of families in healthcare and social services, determine validated screening tools and extract information about recommendations for action after screening for housing issues. Methods: The electronic databases MEDLINE, PsycINFO, EMBASE, Ovid Emcare, Scopus and CINAHL were searched from 2009 to 2021. Inclusion criteria were peer-reviewed literature that included questions about housing being asked of children or young people aged 0-18 years old and their families accessing any healthcare or social service. We extracted data on the housing questions asked, source of housing questions, validity, and descriptions of actions to address housing issues. Results: Forty-nine peer-reviewed papers met the inclusion criteria. The housing questions in social screening tools vary widely. There are no standard housing-related questions that clinical and social service providers ask families. Fourteen screening tools were validated. An action was embedded as part of social screening activities in 27 of 42 studies. Actions for identified housing problems included provision of a community or clinic-based resource guide, and social prescribing included referral to a social worker, care coordinator or care navigation service, community health worker, social service agency, referral to a housing and child welfare demonstration project or provided intensive case management and wraparound services. Conclusion: This review provides a catalogue of housing questions that can be asked of families in the clinical and/or social service setting, and potential subsequent actions.

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Strengths and Limitations

- The first systematic review to catalogue housing questions asked of families in clinical and social service settings, including information on validity of screening tools.
- Independent review of study selection, quality assessment and data extraction.
- Search terms may have been too narrow and may not have captured relevant papers that included information on housing questions in broader social screening tools.
- The quality of studies varied across the studies.
- Heterogeneity in study design, sample size, participant age, housing questions asked across studies, limiting ability to compare studies.

Introduction

Social determinants of health are the environments and conditions in which people are born, grow, work, live and age (1). The role of housing as a social determinant of health is well-established (2, 3). Healthy housing is described by WHO as "shelter that supports a state of complete physical, mental and social well-being" and "provides a feeling of home, including a sense of belonging, security and privacy" (3). Adequate housing is more than shelter and provision of quality physical dwelling conditions (4-6). Housing tenure and housing affordability are important factors in supporting psychological and developmental wellbeing (6-11). Similar to other Western countries, in Australia, where this review was conducted, almost one million people reside in housing considered to be in poor condition and there is an overrepresentation of young people, people in low-income households, and Aboriginal and Torres Strait Islander people living in poor housing conditions (12).

There is growing evidence that housing aspects such as housing quality and environments materially affect health, wellbeing and developmental trajectories well into adulthood (13-16). Housing issues in childhood have a particularly detrimental effect on health and wellbeing. Poor housing conditions in childhood have been predicted to lead to poorer health and increased mortality later in life, after controlling for the confounding effect of socioeconomic conditions (13, 14). Material and social

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factors that impact on children's wellbeing include poor physical housing quality (mould and damp, utilities, toxicants, disrepair, lack of heating, injury hazards), crowding, lack of outdoor space for children to play, accessibility within the home, housing affordability, tenure type, frequent residential moves, homelessness, lack of cultural appropriateness and complex neighbourhood quality (6, 16-23). Psychological and emotional distress, as well as behavioural problems, have been independently associated with children living in poor housing conditions and experiencing housing instability (20, 24-29). Housing instability, frequent moving and homelessness also negatively affect children's health (30) and children's educational outcomes (31). A range of physical health issues such as asthma, acute respiratory symptoms and low birthweight have been independently associated with damp and mouldy housing (22, 32, 33), and living in crowded conditions has been found to lead to infectious diseases such as tuberculosis, meningococcal disease, acute rheumatic fever and otitis media and gastrointestinal infections in New Zealand and Australia (34-41). Multiple forms of housing problems can coexist and have a compounded, cumulative impact on physical and mental health, particularly after prolonged exposure during childhood (11, 42).

Improving the quality of housing and the immediate environment around the home "can save lives, prevent disease, and increase quality of life" (3). There is evidence to support screening and social prescribing for social determinants in healthcare, including housing issues, to address basic resource needs, inequity and improve child health (29, 43-50). Screening involves incorporating standardized tools, such as surveys or questionnaires, completed with families in paediatric or primary care practices (51, 52). Social prescribing, sometimes known as community referral, is a mechanism for clinicians to link or refer patients to non-medical sources of support, such as to the community sector, to address social determinants of health and improve community wellbeing (49, 53-56). Social prescribing has been recognised as a pathway to "address physiological, physical, psychological, psychosocial or socioeconomic issues, as well as enhancing community wellbeing and social inclusion" (55). There is evidence to show that caregivers in the paediatric healthcare setting are interested in assistance if screening positively to housing instability (57). Screening for housing

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issues has been demonstrated to increase the occurrence of social prescribing. Screening increases referral of families to appropriate professionals or providing families with resources (58, 59), and can lead to improved child health (44). While many social screening tools addressing housing issues do exist (60, 61), to date there are no widely accepted guidelines for health and social service professionals to systematically identify specific housing problems for children and address them (29, 59, 62). The aims of this systematic review were to catalogue housing screening questions asked of families in healthcare and social service settings, as described in the international literature; determine if any validated screening tools exist; and extract information about any recommendations for action after screening described in the literature, including social prescribing such as referral through health, housing and social care pathways.

Findings from this review will inform the development of an integrated detection, referral and action pathway to be used in healthcare and social service settings with Aboriginal and Torres Strait Islander children and families who may be experiencing poor health due to housing problems in Australia. The detection, referral and action pathway will also be used with non-Aboriginal patients and clients who are experiencing housing-related health issues.

Methods

Search strategy

A study protocol was developed, registered with the University of York Centre for Reviews and Dissemination (PROSPERO) and updated on 28/04/2020, registration number CRD42020159816 (https://www.crd.york.ac.uk/prospero/). A search of the international peer-reviewed and grey literature was conducted using the following inclusion criteria: studies that include questions about housing being asked of the population group, children or young people aged 0-18 years old and their families, accessing any health or social service. There were no restrictions on study type, design or method, and studies were not limited by country. Studies were limited to the English language and to published literature in the 10-year period 1 January 2009 to 14 December 2021. Studies about

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health assessment conducted with clients of housing services were excluded. Review papers were excluded.

The electronic databases MEDLINE, PsycINFO, EMBASE, Ovid Emcare, Scopus and CINAHL from 2009 to 2021 inclusive were systematically searched. The search strategy and key search terms are included in Supplementary Table 1, in line with the PRISMA guidelines. A secondary search of citations from bibliographies of related peer-reviewed articles, and an additional search of the grey literature using the search terms “social needs screening tool”; “housing screening tool for children” in Google Scholar were undertaken.

All study titles and abstracts were reviewed independently in Covidence by two reviewers, this task was shared across six reviewers (NW, AA, SW, MA, KH and CKS). Disagreements about inclusion of studies were discussed and decisions made by consensus with four authors (AA, SW, MA and KH). Full texts of papers were reviewed independently by three reviewers (AA, SW and KH). Divergences were discussed with three reviewers (AA, SW and KH) and agreement reached.

Data extraction and analysis

Data was extracted from included papers using a standardized data extraction tool by three reviewers (AA, SW and CKS). The data extracted included specific details about the study types, aims and methods, populations, screening tools, housing questions, descriptions of actions, such as referral to appropriate services, to address housing issues and other items pertaining to the review question and specific objectives. If housing questions were not available in the papers, but the name of a screening tool was provided, the authors accessed the specific tool through a web search to extract the housing screening questions. Any disagreements that arose between the reviewers were resolved through discussion.

Quality assessment

Three reviewers (AA, KH and SW) assessed the quality of each study using two separate tools, dependent on study type. The Mixed Methods Appraisal Tool (MMAT) tool (63) was used to appraise

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quantitative RCTs, quantitative non-randomized trials, quantitative descriptive and mixed methods studies. Quality improvement (QI) studies were appraised using the Quality Improvement Minimum Quality Criteria Set (QI-MQCS) version 1.0 (64). The reviewers did not give studies an overall score for methodological quality, as it is discouraged to calculate an overall quality score using the MMAT tool (63). Consensus on quality was reached through discussion.

Patient and public involvement

This study was conceptualised, in part, with community members who have experienced housing issues, and people who are in regular patient contact with families experiencing housing issues impacting on health and wellbeing.

Results

Papers identified

The search strategy identified 10,922 titles (excluding duplicates) and an additional 10 papers were found through a grey literature search and through citations. 9077 titles and abstracts were screened, and 8609 of these were excluded, as these titles and abstracts did not provide any information that led screeners to believe that the full text papers would have questions about housing being asked of people aged 0-18 years old and their families accessing healthcare or social services. 468 full text papers were identified as potentially meeting inclusion criteria and were assessed for eligibility, and 419 were excluded. Reasons for exclusion are provided in **Figure 1**. The two most common reasons for exclusion were that housing questions were being asked outside of a clinical or social service setting (n= 127) and there were no questions about housing being asked (n=145). A total of 49 papers were included representing 42 studies.

Included papers

The 49 included papers are listed in Table 1. All papers were published in peer-reviewed journals. Studies were primarily conducted in clinical settings (n=39); paediatric primary care clinics, children's hospitals and emergency departments, urban community health centres, general paediatric centres,

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ambulatory healthcare practices and antenatal clinics. Three studies were conducted in social service settings including a disability service, a government child welfare unit and a government-funded housing and health support service. All studies were based in high-income countries, with 39 studies conducted in the USA, one study in Australia, one in New Zealand, and one study conducted in Spain.

Studies were heterogenous in terms of method, design, sample size, age of participants, social screening tools and housing questions included in screening tools. Study types varied; cross-sectional (30, 47, 57, 65-79), randomized controlled trials (43, 44, 80-84), pre and post studies (85-88), prospective (89, 90), quality improvement (59, 91-96), mixed methods (97, 98), tool adaptation and validation (99), retrospective (100-102), and secondary data analysis (103). Sample sizes ranged from 45 to 56,253 participants. Participants were newborns, children, adolescents and/or their families or caregivers. Age ranges of children varied from study to study, and some studies only included children with specific characteristics, such as children with cerebral palsy, children with a diagnosis of type 1 Diabetes, children with hearing loss, and children with open child welfare cases. See Table 1 for characteristics of included papers. An action, such a provision of a resource or a social prescribing activity was found in 27 of the 42 included studies, see Table 2 for social prescribing for housing issues.

Quality assessment and risk of bias

Assessing studies using the Mixed Methods Appraisal Tool, the four randomised control trials scored highly. One randomized control trial (RCT) met all the criteria (44, 81) and three studies scored highly, meeting the majority of the criteria. It was unclear whether Garg's (43) cluster RCT randomisation was adequately performed and whether participants adhered to the assigned intervention. Gottlieb's (80) study did not provide adequate information on blinding, and Bovell-Ammon's RCT did not provide information on participant adherence to the intervention, randomisation and blinding (84). There were three robust quantitative non-randomized studies,

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scoring highly against most criteria (65, 83, 85). Patel's study did not provide complete outcome data (85), and it was unclear in Bottino's study whether participants were representative of the target population nor where there was complete outcome data (65). Twenty-nine studies were categorised as quantitative descriptive, and quality scores were mixed. A common issue related to the representative sample of the target population, with the sample size not being representative in seven of the twenty-nine studies, and another five studies providing insufficient information to assess whether the sample size was representative. The risk of non response bias was low in sixteen of the quantitative descriptive studies (30, 73, 75, 76, 79, 82, 86-89, 99, 104-107). The two mixed methods studies were of good quality however both did not adequately address inconsistencies between quantitative and qualitative results and did not adhere to the tradition of each method (97, 98). There were seven quality improvement studies, assessed using the Quality Improvement Minimum Quality Criteria Set. Five of the seven studies scored highly, scoring well in 10 of the 15 criteria (59, 91, 94-96). Issues with QI studies included adherence, reporting data on health-related outcomes, organizational readiness and spread. See Tables 3a and 3b for quality assessment of included studies.

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Table 1 Included study characteristics

Author (year)	Country	Study type	Participants (age)	Sample size	Name of tool and source of housing questions	Validity, feasibility and/or acceptability of tool	Housing questions (if available)
Arbour et al. (2021) (94)	USA	Quality improvement	Newborns and their families	692	Not provided	Used validated, standardized screening questions	Housing instability; unhealthy housing
Badia et al. (2014) (99)	Spain	Quantitative - Tool translation, adaptation, validation	Parents of children and adolescents with cerebral palsy (8-18 years)	221	European Child Environment Questionnaire (ECEQ), based on the International Classification of Functioning, Disability and Health	Construct validity, translation, adaptation and validation of tool conducted specifically for the Spanish context	Enlarged rooms at home (need/availability?); Adapted toilet at home (need/availability?); Modified kitchen at home (need/availability?); Walking aids (need/availability?); Hoists at home (need/availability?)
Beck et al. 2012 (100)	USA	Quantitative - Retrospective review	Newborns and families (age not specified)	639	Developed based on literature review, in consultation with medical staff, social services and medical–legal staff and adapted from the USA's Survey on Income and Program Participation	Food security and domestic violence questions validated; housing questions not validated	"[Are you having] Problems with housing conditions (overcrowding, evictions, lead, utilities, mold, rodents)? Have you spoken with your landlord?"
Bottino et al. (2019) (65)	USA	Quantitative – cross-sectional	Children (3-10 years)	340	Adapted from the American Housing Survey	Food questions validated; no information on housing question validity	"What best describes your current living situation?" [Rent apartment; Rent room(s) within another person's home; Own home; Homeless or living in a shelter; Doubled up with another family; Live with parents/other family; Other]. Selecting 'Rent apartment' or 'Own home' triggered a follow-up item: "In the last 12 months, has the electric or gas company shut off the electricity or gas in your home or threatened to shut off the utilities in your home?" and "Are you concerned about a possible eviction (being "kicked out" from your home?"
Bovell-Ammon et al. (2020) (84)	USA	Quantitative - RCT	Medically complex families with children	78	No information provided	No information provided	Experience of one or more adverse housing circumstances: homelessness in previous year, having moved two or more times in previous

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			under 11 years of age				year (multiple moves), behind on rent in the previous year, and paying more than 50% of family income on rent.
Colvin et al. (2016) (59)	USA	Quantitative - postintervention only design	Children (median age 3 years)	45	Income, Housing, Education, Legal status, Language/Immigration, Personal Safety (IHELLP), adapted from Kenyon et al's IHELLP social history tool (108)	Validity, sensitivity and specificity	"Do you have any concerns about poor housing conditions like mice, mold, cockroaches? Do you have any concerns about being evicted or not being able to pay the rent? Do you have any concerns about not being able to pay your mortgage?"
Costich et al. (2019) (87)	USA	Quantitative - Retrospective pre-post analysis	Caregivers of children (4-12 years) with special health care needs	80	US Department of Agriculture 18-item Household Food Security Survey and Hager et al.'s (109) Food Insecurity screening tool	Sensitivity, specificity, and convergent validity	"Is it hard for you to access any of the following: food, housing, medical care?"
De Marchis et al. (2019) (110)	USA	Quantitative - cross-sectional	Adult caregivers of paediatric patients	969	See De Marchis 2021	See De Marchis 2021	See De Marchis 2021
De Marchis et al. (2020) (57)	USA	Quantitative - cross-sectional	Adult caregivers of paediatric patients	1021	See De Marchis 2021	See De Marchis 2021	See De Marchis 2021
De Marchis et al. (2021) (71)	USA	Quantitative - cross-sectional	Adult caregivers of paediatric patients	835	Accountable Health Communities (AHC) social risk screening tool and Children's HealthWatch (CHW) Housing Stability Vital Sign questions	First two AHC housing questions adapted from the validated Protocol for Responding to and Assessing Patients' Assets, Risks, and Experiences tool (111); Children's HealthWatch questions validated (112)	AHC: "Think about the place you live. Do you have problems with any of the following? (Check all that apply). Answers: (a) Pests such as bugs, ants, or mice (b) Mold (c) Lead paint or pipes (d) Lack of heat (e) Oven or stove not working (f) Smoke detectors missing or not working (g) Water leaks (h) None of the above"; "What is your housing situation today? (a) I have a steady place to live (b) I have a place to live today, but I am worried about losing it in the future (c) I do not have a steady place to live (I am temporarily staying with others, in a hotel, in a shelter, living outside on the street, on a beach, in a car,

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							abandoned building, bus or train station, or in a park". CHW: "In the past 12 months, was there a time when you were not able to pay the mortgage or rent on time?"; "In the past 12 months, how many times have you moved where you were living?"; "At any time in the past 12 months, were you homeless or living in shelter (including now)?"
Denny et al. (2019) (91)	USA	Quality Improvement	Children (0-1 years and 1-5 years)	667	Injury Prevention (IP) + The Safe Environment for Every Kid (SEEK) tool, adapted from Gittelman et al.'s IP tool, and Dubowitz's SEEK tool (113, 114)	Injury Prevention (IP) tool tested for reliability	"Do you have trouble paying utilities or maintaining a safe place to live?"; "Do you use safety gates to protect your child at the top and bottom of all stairways in your house?"; "Are all potentially harmful household cleaners and pesticides either in locked storage OR out of the reach of children?"; "Is the hot water heater in your house adjusted to less than 120 degrees?"
Farrell et al. (2017) (97)	USA	Quantitative - mixed methods population survey	Families with open child welfare cases	6828	Quick Risks and Assets for Family Triage (QRAFT), adapted from Risks and Assets for Family Triage (RAFT) assessment instrument	Some indication of content, construct, and predictive validity, however inconclusive information	Current Housing, Housing Condition, and Housing History, with 5-item risk scale for each, measuring level of housing instability and homelessness
Fiori et al. (2019) (95)	USA	Quality improvement	Paediatric patients (37-144 months)	7266	Adapted from the Health Leads USA screening toolkit (115)	Validity	Exact questions used were not provided
Fiori et al. (2020) (89)	USA	Quantitative – prospective pragmatic	Paediatric patients (2-13 years)	4948	Adapted from the Health Leads USA screening toolkit (115)	Validity	Exact questions used were not provided
Fiori et al. (2020) (116)	USA	Quantitative	Children (0-17 years)	15,503	Adapted from the Health Leads USA screening toolkit (115)	Validity	Exact questions used were not provided
Garg et al. (2009) (66)	USA	Quantitative – cross-sectional descriptive	Parents of children 2 months old to 10 years	100	Well Child Care, Evaluation, Community Resources, Advocacy, Referral, Education (WE CARE),	Face validity	"Do you think you are at risk of becoming homeless?"

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					developed collaboratively among paediatric clinic staff and initially guided by Bright Futures paediatric intake form (29)		
Garg et al. (2015) (43)	USA	Quantitative – cluster RCT	Families of infants <6 months old	336	Well Child Care, Evaluation, Community Resources, Advocacy, Referral, Education (WE CARE), adapted from a larger psychosocial screening instrument; three housing instability questions derived from the Children's HealthWatch survey (112)	Validity	Home heating and Housing instability (housing instability questions: having a steady place to sleep; household crowding i.e. >2 people per bedroom, or having moved >2 times in the past year)
Gottlieb et al. (2014) (80)	USA	Quantitative - RCT	Caregivers of paediatric emergency department patients	538	Some items adapted from the Medical Advocacy Screening questionnaire; other items adapted from other existing validated surveys	Neighbourhood safety question validated	Concerns about the physical condition of your housing; Concerns about the cost or stability of your housing; Threats to your child's safety at school or in the neighbourhood
Gottlieb et al. (2016) (44)	USA	Quantitative - RCT	Children (mean age 5 years)	1809	Used survey instrument from 2014 RCT	Neighbourhood safety question validated	Not having a place to live; unhealthy living environment; other concerns with housing
Gottlieb et al. (2018) (81)	USA	Quantitative - RCT	Children (mean age 5 years) and families in acute care setting	718	Used survey instrument from 2014 RCT	Neighbourhood safety question validated	Not having a place to live; unhealthy living environment; other concerns with housing
Hardy et al. (2021) (88)	USA	Quantitative - retrospective pre-post	Children and youth (0-21 years)	56,253	Social need domains informed by what are known to affect patients/families and selected based on the consensus of institutional stakeholders	No information provided	"Do you think you are at risk of becoming homeless?"; "Are you currently homeless?"

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Hassan et al. (2013) (67)	USA	Quantitative – cross-sectional	Young adult patients (15-25 years)	401	The Online Advocate, adapted from the American Housing Survey	Used previously validated survey questions	Major: Homeless; utilities shut off; structural problems such as leaking roof, non-functional plumbing, or rodent infestation. Minor: Need fuel/heating assistance; threat of utilities shut off in past 12 months; if renting or receiving subsidized housing, concern about transfer or eviction
Hassan et al. (2015) (47)	USA	Quantitative – prospective intervention	Young adult patients (15-25 years)	401	The Online Advocate, adapted from the American Housing Survey	Used previously validated survey questions	Homeless; having utilities shut off; or having existing structural problems such as leaking roof, non-functional plumbing, or rodent infestation
Heller et al. (2020) (73)	USA	Quantitative – cross-sectional	Children under 18 years, and adults	24,633	Tool derived from modified version of Health Leads screening toolkit (115) and homegrown tool	Validity (Health Leads only)	"Are you worried that the place you are living now is making you sick? (has mold, bugs/rodents, water leaks, not enough heat)?"; "Are you worried that in the next 2 months, you may not have a safe or stable place to live? (eviction, being kicked out, homelessness)"
Hensley et al. (2017) (92)	USA	Quality Improvement	Paediatric patients and families	114	HealthBegins Upstream Risks Screening Tool	Feasibility	"In the last month, have you slept outside, in a shelter, or in a place not meant for sleeping? Y/N; In the last month, have you had concerns about the condition or quality of your housing? Y/N; In the last 12 months, how many times have you or your family moved from one home to another?"
Hershey et al. (2021)	USA	Quantitative – prospective	Caregivers of children and youth (4-18 years) with a diagnosis of T1D for >1 year and poor glycemic control or high care utilisation	53	Health Leads USA questionnaire	Validity	"Are you worried that, in the next 2 months, you may not have stable housing?"
Higginbotham et al. (2019) (93)	USA	Quality Improvement	Children (1 week to 5 years old)	53	Aligned with US Department of Housing and	Feasibility	Crowding "within the past 12 months there were two or more people per bedroom?";

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					Urban Development housing insecurity definition and previous cross-sectional study by Cutts and colleagues (117)		doubling up "within the past 12 months we were temporarily staying with another family or had another family staying with us?"; and frequent moving "within the past 12 months we moved more than once?"
Koita et al. (2018) (98)	USA	Quantitative - Validation of tool	Caregivers of children < 12 years	28	BARC* Pediatric Adversity and Trauma Questionnaire, modified from Oakland Children's Hospital Family Information and Navigation Desk needs assessment survey	Face validity	"Has your child/family ever been homeless? Has your child ever had problems with housing (for example not having a stable place to live, faced eviction or foreclosure, or lived with multiple families)?"
Kumar et al. (2017) (82)	USA	Quantitative – RCT follow up	Pregnant adolescents < 17 years	106	Modified from the Treatment Services Review assessment tool	No information provided	Residential mobility (number of moves made in the year before prenatal survey, duration of stay, number of people living at their current home, and cohabitation with parents, husband or boyfriend, and other relatives); Need for housing resources at 3 months and 6 months postpartum
Matiz et al. (2022)	USA	Quantitative – retrospective chart review	Children with hearing loss (0.6 – 7.9 years)	30	No information provided	No information provided	"Is it hard for you to access any of the following: Food, Housing, Medical Care, N/A.?"
Messmer et al. (2020)	USA	Quantitative – RCT	Parents of children with social needs	414	WE CARE	Validity	See WE CARE questions
Molina et al. (2019) (68)	USA	Quantitative – cross-sectional	Children and families	276	Adapted from health-related social needs questions from the National Academy of Medicine	No information provided	"In the last month, have you slept outside, in a shelter or in a place not meant for sleeping? In the last month, have you had concerns about the condition or quality of your housing? In the last 12 months, how many times have you or your family moved from one home to another? Are you worried that in the next 2 months, you may not have stable housing? In the past 12 months, have you had any utility (electric, gas, water or oil) shut off for not paying your bills?

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Oldfield et al. (2021) (74)	USA	Quantitative – cross-sectional	Parents of children of all ages, and adolescents (>13 years)	154	WE CARE tool and the AHC instrument	Validity (WE CARE); AHC housing questions adapted from the validated Protocol for Responding to and Assessing Patients' Assets, Risks, and Experiences tool (111)	Do you have any concerns about safety in your neighbourhood? Are you afraid you might be hurt in your apartment building or house?" WE CARE: "Do you think you are at risk of becoming homeless?"; "Do you have trouble paying your heating bill and/or electricity bill?"; AHC: "What is your living situation today? (I have a steady place to live, I have a place to live today, but I am worried about losing it in the future, I do not have a steady place to live (I am temporarily staying with others, in a hotel, in a shelter, living outside on the street, on a beach, in a car, abandoned building, bus or train station, or in a park)"; "Think about the place you live. Do you have problems with any of the following? (pests, mold, lead paint or pipes, lack of heat, oven or stove not working, smoke detectors missing or not working, water leaks, none)"
Patel et al. (2018) (85)	USA	Quantitative – pre and post intervention retrospective chart review	Patients at well-child visit (0-20 years)	322	Income, Housing, Education, Legal status, Language/Immigration, Personal Safety (IHELLP) (108)	No information provided	"Housing, utilities: What kind of housing do you live in (apartment, house)? Do you receive any type of housing subsidy (Section 8, public housing, shelter)? Do you have heat and hot water? Do you have trouble paying rent or utilities at the end of the month?" "Personal safety: Do you feel safe in your neighbourhood?"
Pierse et al. (2020) (107)	New Zealand	Quantitative	Low-income families with children (0-14), hospitalized for health conditions attributable to home	895	Housing Concerns Survey; developed in partnership with researchers to identify areas of housing need	No information provided	"Is your home usually colder than you would like?"; "During the winter months, was your house so cold that you shivered inside?"; "Does your home smell moldy or musty?"; "Is there mold on the walls in bedrooms or living areas of your home?"; "Are there damp walls in the bedrooms or living areas of your home?"

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			environment, and pregnant women with newborns				
Polk et al. (2020) (75)	USA	Quantitative – cross-sectional evaluation	Low-income families	10,916 households	Health Leads universal social needs screener (115)	Validity	"Would you like help with any of the following: Finding housing search resources and emergency shelters?"
Power-Hays et al. (2020) (96)	USA	Quality improvement	Families of children with sickle cell disease	132	WE CARE	Validity	"Do you currently live in a shelter or have no steady place to sleep at night?"; "Do you think you are at risk of becoming homeless? If yes, is this an emergency?"; "Would you like help connecting to resources? Housing/shelter"
Ray et al. (2020) (105)	USA	Quantitative - observational	Caregivers accompanying a child (< 5 years) who received a low-acuity triage score	146	WE CARE and PRAPARE	Validity	See WE CARE and PRAPARE questions
Razani et al. (2020) (77)	USA	Quantitative - observational	Adult caregivers of paediatric patients	890	14-item social and mental health needs screening questionnaire (44)	Neighbourhood safety question validated	"Not having a place to live for example, concerns about eviction, foreclosure, staying with friends/family, and current homelessness"; "Unhealthy living environments, for example, problems such as mold, insects, rats or mice, excess trash"; "Other concerns with your housing"
Rhodes et al. (2012) (69)	Australia	Quantitative – cross-sectional	Children (4-8 years)	162	Family Psychosocial Resources (FSR) tool, adapted from the Family Resources Scale (118)	No information provided	Basic living needs
Sandel et al. (2018) (30)	USA	Quantitative – cross-sectional	Caregivers of children (0 to 48 months of age) in renter households	22,324 families	Not provided	Not tested	"During the last 12 months, was there a time when you were not able to pay the mortgage or rent on time?"; "In the past 12 months, how many places has the child lived?"; "What type of housing does the child live in?"; "Since the child

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							was born, has she or he ever been homeless or lived in a shelter?"
Sandoval et al. (2020) (102)	USA	Quantitative – retrospective chart review	Mother-child dyads	268	MAMA's neighbourhood program prenatal screening questionnaire; postnatal housing insecurity measured using Homelessness Screening Clinical Reminder Tool	No information provided	Exact questions used were not provided
Selvaraj et al. (2019) (70)	USA	Quantitative – cross-sectional	Children and adolescents (2 weeks to 17 years)	2569	Addressing Social Key Questions for Health Questionnaire (ASK tool), developed by staff of four academic institutions and literature review of validated paediatric questionnaires on adverse childhood experiences and unmet social needs	Validity and feasibility	Housing and bill insecurity; Witnessed violence in the home or neighbourhood
Semple-Hess et al. (2019) (106)	USA	Quantitative – cross-sectional	English- and Spanish-speaking caregivers	768	Questions developed through existing screening tools reported in literature and from paediatric and social worker' knowledge of common needs of patients	Survey tested and refined	"Which of the following services have you/your child ever used?" and "In the next 12 months, which of the following services will you/your child need?" Options included: "6) temporary housing and shelters; 7) safe housing services"
Sokol et al. (2021) (119)	USA	Quantitative – evaluation	Outpatient paediatric patients (0-18y)	30,485	Developed from PRAPARE (111)	Housing question validity; no testing for referral question	"In the next 2 months, are you worried that you may not have stable housing?" For referral and resource support: "Do you want to be contacted by the assistance program for help on any of your responses?"
Spencer et al. (2020) (76)	USA	Quantitative – cross-sectional	Caregivers of school-age children	943	WE CARE	Face and content validity	"Do you think you are at risk of becoming homeless?"; "Do you have trouble paying your heating bill for the winter?"

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Uwemedimo (2018) (103)	USA	Secondary data analysis	Caregivers of children < 18 years	148	Family Wellness Screen (FAMNEEDS), developed from previously published SDH screening tools (120)	Validity	"Do you worry that in the next 2 month, you/your family may not have a safe or stable place to live? Y/N. Where is your child living now, Options provided: private house/ apartment, room (in apartment/house), shelter, hotel/motel, no regular place, car, other]. Do you worry that the place you're living now is making you sick? Y/N. If YES choose all that apply: Cigarette smoke, mold or dampness, rodents/bugs, peeling paint, broken appliances, open cracks/holes/wires, not enough heat, water leaks, none" Other questions: "Do you need help from a lawyer with housing, immigration, custody or child support problems? Y/N."
Vaz et al. (2020) (79)	USA	Quantitative - observational	Child–caregiver dyads	249	Social risk questionnaire based on the National Survey of Child Health, Children's HealthWatch, social risk tools created by Gottlieb (80) and Harris (121)	Children's HealthWatch questions validated (112); Gottlieb neighbourhood safety questions validated (80)	Exact questions used were not provided
Zielinski et al. (2017) (86)	USA	Quantitative - pre and post	Children	602	WE CARE tool (29)	Validity, feasibility and acceptability	Housing insecurity

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Screening tools and housing questions

Our systematic review found 29 social screening tools that include housing questions. The tools used in clinical and social service settings varied widely. The purpose of all tools was to screen for health- or wellbeing-related social needs among children, adolescents and/or families. The tools were used by healthcare or social service workers to facilitate asking patients or clients questions about social needs, including housing. There were two distinct approaches to screening; one approach was a general social determinants of health screening where housing was one of multiple domains among other social needs, and a second approach was to focus on housing as one domain, and only ask detailed housing-related questions. In the first approach, social needs assessed in tools included housing, food security, income, employment, education, childcare needs, transportation needs, health care access, legal concerns, medical insurance, violence, social connection and isolation.

The housing questions asked of participants varied across studies. Housing questions extracted from the included studies were divided into nine categories, see **Figure 2.**

Housing instability, insecurity or homelessness were the most common housing-related screening questions across the studies, with questions about homelessness, risk of homelessness, sleeping in a shelter, sleeping outside or in a place not meant for sleeping, sleeping in unsafe housing, frequent moving and temporary or unsafe housing (30, 43, 44, 57, 65, 67, 68, 71-80, 82-84, 86-90, 92-97, 101, 102, 106, 119). Utilities as an unmet need, especially in relation to home heating, were included as a common housing issue captured in the social screening tools of studies in this review. Problems with utilities, such as utilities being shut off, threats of utilities being shut off or concern about not being able to pay for utilities, and in relation to heating the home (43, 73), was asked as part of social screening in eighteen (n=18) studies (43, 47, 65-68, 70, 71, 74-76, 78, 79, 88, 90, 91, 100, 107). The questions about utilities were often asked alongside broader housing question in screening tools, for example “[Are you having] Problems with housing conditions (overcrowding, evictions, lead, utilities,

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mold, rodents)?" (100) and "Do you have trouble paying utilities or maintaining a safe place to live?" (91).

Crowding was a common item in the social screening tools, with the following terms being used: "overcrowding" (100), "doubled up or overcrowded in an unsustainable way" (97) and "need enlarged rooms at home" (99). Crowding was considered a part of housing instability in one tool, i.e. "Housing instability: Doubled up with another family" (65). Some tools asked families about general housing problems and either did not specify the nature of the housing problem, or asked about multiple housing-related issues within the one general question (44, 47, 59, 66, 67, 97, 100). Structural and physical housing problems (67) included questions specific to the quality or condition of the home (68, 73, 80, 89, 92, 95, 107), mould or damp (59, 100, 103, 107), and unhealthy living environments (77). See Table 1 for further detail on the housing questions asked in screening tools.

Validity, feasibility and acceptability

Fourteen (n=14) screening tools had been validated, including the Family Wellness Screening tool or FAMNEEDS (103), Well Child Care, Evaluation, Community Resources, Advocacy, Referral, Education (WE CARE) tool (66), The Online Advocate (47), The Protocol for Responding to and Assessing Patients' Assets, Risks, and Experiences (PRAPARE) tool (111), The Health Leads Screening Toolkit (115), as well as other tools used across studies (43, 66, 77, 80, 94, 98, 99, 111). The studies included in this review did not provide detailed information on validity. It is not clear whether any of the screening tools are robust in producing their intended result, and to what extent the housing questions in particular are suitable for their aims.

The ASK tool (Addressing Social Key Questions for Health questionnaire) had been tested for validity and feasibility (70). The WE CARE tool used across multiple studies included in this review had been validated and tested for feasibility and acceptability (29, 86). The Income, Housing, Education, Legal status, Language/Immigration, Personal Safety or IHELLP tool was tested for validity, sensitivity and specificity (59, 108). Some tools were not validated however underwent other forms of testing such

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as feasibility, reliability or sensitivity. The screening tool used in Higginbotham's study was tested for feasibility (93), as was Hensley's HealthBegins Upstream Screening tool (92). The Injury Prevention tool was tested for reliability (91). Insufficient information was provided on the validity, feasibility or acceptability of the tools used in fourteen studies (30, 65, 68, 69, 73, 82, 85, 88, 91, 93, 100-102, 106). The information provided for the Quick Risks and Assets Family Triage or QRAFT tool showed some indication of content, construct and predictive validity, however it is unclear whether it was validated (97). The tool used in Beck's 2012 study did not validate the housing questions, however other questions in the tool were validated (100). The first two housing questions of the Accountable Health Communities instrument, used in two studies (71, 74), had been adapted from the validated PRAPARE tool (111).

The studies did not explore testing the cultural acceptability or appropriateness of the screening tools. Badia's (2014) study analysed the cultural acceptability of the European Child Environment Questionnaire in order to adapt it to the Spanish context. Uwemedimo's (2018) study took into consideration unique challenges faced by immigrant families when screening for social needs and recommended screening tools to be culturally sensitive through availability in multiple languages and for bilingual or multilingual patient navigators to support patients from immigrant backgrounds. Some studies involved screening in languages other than English, for example, in Spanish (68, 70, 74, 77, 79, 80, 98, 99). Power-Hays' study offered screening in English, Spanish, and Haitian Creole, and, provided patients with low literacy to have the screener read to them (96).

Social prescribing

Twenty-seven studies reported provision of information, and/or social prescribing, i.e. a linking or referral occurred upon identifying a social need as part of screening. Several studies provided patients with a community resource guide, a list of local housing services, community resource information or clinic-based resources (43, 44, 57, 59, 70-73, 75, 78, 84, 87-89, 92-95) after screening for housing and other social needs. Social prescribing was also common, with patients screening

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positively being referred to a social worker, care coordinator, care navigation service, community health worker or advocate (59, 67, 75, 83, 84, 86, 88, 90, 95, 96, 101, 102, 119), social service agency or community organisation (87, 96, 103), referral to a housing and child welfare demonstration project (97), or provided intensive case management in the form of wraparound services relating directly to a family's specific needs (84). In Pierse's study families were referred from healthcare settings to the Well Homes program and asked about their perception of their housing conditions, leading to all families receiving information about keeping warm, dry and safe, and some being offered an intervention such as a mould kit, or a housing relocation referral (107). Bovell-Ammon's RCT on medically complex families experiencing housing instability or homelessness provided the intervention group with intensive case management and wraparound services, such as support with searching for housing, eviction prevention, legal or financial services, and if eligible, provision of a public housing unit (84). De Marchis' study (57) offered information about local services and resources to all patients, regardless of whether the patient participated in their study on social risk screening. In several studies, a patient navigator helped patients with contacting resources or community-based health and social services (44, 81, 83, 92, 103). Messmer's study investigated the impact of on-site versus remote patient navigators to address parent's unmet social needs, finding that referrals for housing were similar when an on-site patient navigator was present versus a remote patient navigator (27.2% vs 30.7) (83). Actions specific to housing-related problems, such as a social work referral specific to homelessness, were evident in eleven of the 27 studies. In Hershey's study of children with high-risk type 1 Diabetes, families screened for social needs were offered assistance in the form of goal-setting with the support of a community health worker (90). Housing-specific goals included access affordable housing or affordable home renovations, help parent enrol in first-time homeowners' program, and assist in negotiating rental arrears (90). Screening for social needs resulted in an increase in referrals to appropriate services or resources, and some studies found that screening and subsequent referral led to an improvement in the housing-related issue (43, 87), or an improvement in child health (44, 81, 84). Interestingly, social screening and social

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prescribing studies appear to be increasing, with the majority of studies found to be published from 2019 onwards. See Table 2 for a summary of information relating to actions as a result of screening for housing issues, or social prescribing.

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Table 2 Social prescribing for housing issues

Author (year)	Screening tool name (if available)	Action as a result of screening	Referral rates and outcome (if available)
Arbour et al. (2021) (94)	-	Provision of resource information for concrete supports	86% of families screening positively for general health-related social needs were provided resources information for concrete supports
Bovell-Ammon et al. (2020) (84)	-	Intervention group provided with intensive case management and wraparound services to meet specific needs, e.g., support with housing search, eviction prevention, legal or financial services, and if eligible, a public housing unit; Control group provided with a list of resources detailing housing services available in the family's community, as well as hospital-based social work and care navigation services	Significant decreases between baseline and six months in homelessness (30.3% intervention vs 37.9% control) and multiple moves (2.9% intervention vs 7.1% control) at 6 months follow up. Being behind on rent decreased significantly in the intervention group (29.4%) but not the control group (44.8%) at 6 months follow up. Significant changes in child health status and parental anxiety among the intervention group compared to control group: at six-month follow-up.
Colvin et al. (2016) (59)	IHELLP	Social work referral or resources provided to families who screened positively to unmet social needs	77.8% of families screening positively for unmet social needs were provided social work resources or referrals; 13.3% already had social needs addressed; <10% had an unmet social need but were unable to be connected with a resource or referral
Costich et al. (2019) (87)	-	Social service referral, goal setting and resource navigation upon screening and participating in the Special Kids Achieving Their Everything (SKATE) Community Health Worker program	The number of caregivers reporting that it was hard to access housing was reduced from 23% to 9.5%
Farrell et al. (2017) (97)	QRAFT: Quick Risks and Assets for Family Triage	Assess eligibility to a housing and child welfare demonstration project, a supportive housing intervention, and for broader supportive housing referrals. Eligibility criteria: client scoring a 3 (significant risk) or 4 (severe risk) on any of the three housing items	5.4% of families scored 3 or 4 on housing items and upon further assessment, 5.3% were referred to the housing and child welfare demonstration project
Fiori et al. (2019); Fiori et al. (2020) (89, 95)	Adapted from Health Leads Screening Toolkit	For non-urgent issues: handoff with Community Health Worker who provides resources and schedule a follow-up. For urgent issues: referral to onsite social worker	Not reported
Garg et al. (2015) (43)	WE CARE: Well Child Care, Evaluation, Community Resources, Advocacy, Referral, Education	Referral made by clinician using the WE CARE Family Resource Book containing tear-out information sheets listing 2 to 4 free community resources available for each need. The information sheets contain the program name, a brief description, contact information, program hours, and eligibility criteria. Follow up	Mothers receiving WE CARE screening and referral had lower odds of being in a homeless shelter (aOR = 0.2; 95% CI, 0.1–0.9) and more mothers had enrolled in a new community resource at the 12-month visit (39% vs 24%; aOR = 2.1; 95% CI, 1.2–3.7) compared to families who did not receive WE CARE screening and referral

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		one month after visit; staff telephoned mothers to assess contact of resources and update notes in the child's medical record	
Gottlieb et al. (2018); Gottlieb et al. (2016) (44, 81)	-	Caregivers who indicated at least one social need received either written information on relevant community resources (active control group) or received support from an in-person navigator immediately after the child's visit and offered follow-up meetings (navigation intervention group)	Both groups showed improved child health after receiving either written information or an in-person navigator; child global health scores (lower scores indicate better health) improved a mean (SE) of -0.36 (0.05) in the navigator group and a mean (SE) of -0.12 (0.05) in the active control group
Hardy et al. (2021) (88)	-	A resource sheet on community resource linkages was provided if general social need identified. Any urgent social need received a social work consult.	Not reported
Hassan et al. (2013); Hassan et al. (2015) (47, 67)	The Online Advocate	Resource specialist reviewed referrals; if questionnaire responses indicated acute concerns regarding homelessness, results were immediately shared with the provider and social worker to facilitate urgent intervention.	75% of participants screened had at least one referral need; 27% required referral relating to housing problem and 14% received referral relating to housing problem. 85% of participants with housing problems reached for follow-up at 1 or 2 months, and 30% of those with housing problems selected that the problem was "completely" or "mostly" resolved
Heller et al. (2020) (73)	-	Providers offered to connect patients to clinic-based resources.	Not reported
Hensley et al. (2017) (92)	HealthBegins Upstream Risks Screening Tool	Patients with at-risk results were provided a community resources guide with an educational handout, in order to identify local agencies and programs that addressed social needs. The guide listed supporting programs or agencies, and included resource eligibility requirements, contact information. Patients were helped in contacting listed community resources.	Not reported
Hershey et al. (2020) (90)	Health Leads USA	Families asked if they would like assistance with any reported needs and if the need was urgent, and connected to community health worker. Goals were set with families with support from a community health worker. Housing related goals included: Access affordable housing or affordable home renovations; Help parent enrol in first-time homeowners' program; Assist in negotiating rental arrears	41% of families requested assistance with housing; 16% of families selected goals relating to living situation
Higginbotham et al. (2019) (93)	-	Families screening positively for food insecurity and/or housing insecurity were provided a community resource guide to facilitate referral.	Of 13 families screening positive for food insecurity and/or housing insecurity, 85% were given a resource guide

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Matiz et al. (2022)	-	Referral to a community health worker. Community health workers supported caregivers to navigate the complexities of social services, education, and healthcare after initial assessment.	93% of patients required social service referrals
Messmer et al. (2020)	WE CARE	In-person or offsite patient navigator referral to community-based health and social services	27.2% of families screening positively to housing as unmet need referred by on-site patient navigator, 30.7% of families screening positively to housing as unmet need referred by remote patient navigator
Oldfield et al. (2021) (74)	WE CARE and AHC	Positive screens added to patient's medical record, notifying patients' primary paediatrician of positive screens, allowing paediatrician to make targeted referrals as needed	Not reported
Pierse et al. (2020) (107)	Housing Concerns Survey	Provision of information for families on how best to keep their home warm, dry, and safe, provision of a housing-related intervention e.g. mould kit, heater, and/or referral for a housing relocation, or health or social referral	A total of 5,537 interventions were delivered; Bedding, heaters, and draft stopping delivered over 90% of the time
Polk et al. (2020) (75)	Health Leads	Depending on category of need, patient provided rapid resource referral, i.e. information only or enrolled in Health Leads program where patient is contacted by a Health Leads advocate	6.3% of successful resource connection for housing
Power-Hays et al. (2020) (96)	WE CARE	Provision of relevant resource sheet and referral to local community organizations for the specific needs endorsed. Providers and patients decided whether families also required referral to social worker; social workers called families with positive screens 2-3 weeks after their visit	80% of patients were referred to a relevant community organization; 45% of patients available via follow-up phone call reached out to the community organization; 69% of patients who reached out stated that the community organization was helpful
Ray et al. (2020)	WE CARE and PRAPARE	Universal provision of community resource packet at two-week follow-up	Resource packet used by 37% of those who had reported a social need
Sandoval et al. (2021) (102)	MAMA's neighbourhood program prenatal screening questionnaire	Care coordinator providers referral or facilitates contact with agencies or community organizations	Not reported
Selvaraj et al. (2019) (70)	ASK Tool: Addressing Social Key Questions for Health Questionnaire	Referral made to community resources in relation to unmet social needs, including referrals made in relation to housing/bill insecurity	284 (11%) total referrals made in relation to unmet social needs; 111 (4.3%) referrals made in relation to housing/bill insecurity
Semple-Hess et al.	Questions developed through existing	Caregivers with any emergent unmet need provided a social work consultation, as per institutional practice	Not reported

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(2019) (106)	screening tools		
Sokol et al. (2021) (119)	Developed from PRAPARE	Referral to a social worker to provide support with identified need	14% of parents and youths requested a referral for identified needs
Uwemedimo (2018) (103)	FAMNEEDS Family Wellness Screen	Referral to local community resources by trained navigators and follow-up to ensure linkage to resource. Trained navigators either searched online social service databases or referred to social service case managers at designated partner community organisations	Approximately one-third (30.9%) successfully utilized program-provided resources at 12-week follow-up
Zielinski et al. (2017) (86)	WE CARE screening tool	Provision of additional resources on homelessness and/or social work referral relating to homelessness	14 (93%) provision of additional resources on homelessness; 11 (73%) social work referral relating to homelessness

Table 3a Quality assessment using the MMAT tool

Author (year)	Category of study design	Are there clear research questions?	Do collected data allow to address research question?	Is randomization adequately performed?	Are groups comparable at baseline?	Are there complete outcome data?	Are outcome assessors blinded to intervention provided?	Did participants adhere to the assigned intervention?
Bovell-Ammon (2020)	Quantitative RCT	Y	Y	Unclear	Y	Y	Unclear	Unclear
Garg (2015)	Quantitative RCT	Y	Y	Unclear	Y	Y	Y	Unclear
Gottlieb (2016 & 2018)	Quantitative RCT	Y	Y	Y	Y	Y	Y	Y
Gottlieb (2014)	Quantitative RCT	Y	Y	Y	Y	Y	Unclear	Y
Author (year)	**Category of study design**	**Are there clear research questions?**	**Do collected data allow to address research question?**	**Are participants representative of target population?**	**Are measurements appropriate regarding both the outcome and intervention (or exposure)?**	**Are there complete outcome data?**	**Are the confounders accounted for in the design and analysis?**	**During study period, is the intervention administered (or exposure occurred) as intended?**
Bottino (2019)	Quantitative non-randomized	Y	Y	Unclear	Y	Unclear	Y	Y

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Messmer (2020)	Quantitative non-randomized	Y	Y	Y	Y	Y	N	Y
Patel (2018)	Quantitative non-randomized	Y	Y	Y	Y	N	Y	Y
Author (year)	**Category of study design**	**Are there clear research questions?**	**Do collected data allow to address research question?**	**Is the sampling strategy relevant to address the research question?**	**Is the sample representative of the target population?**	**Are the measurements appropriate?**	**Is the risk of non response bias low?**	**Is the statistical analysis appropriate to answer the research question?**
Badia (2014)	Quantitative descriptive	Y	Y	Y	N	Y	Y	Y
Beck (2012)	Quantitative descriptive	Y	Y	Unclear	Unclear	Y	Unclear	Y
Costich (2019)	Quantitative descriptive	Y	Y	Y	Y	Y	Y	Y
De Marchis (2019, 2020, 2021)	Quantitative descriptive	Y	Y	Y	N	Y	N	Y
Garg (2009)	Quantitative descriptive	Y	Y	N	N	Y	N	Y
Hardy (2021)	Quantitative descriptive	Y	Y	Y	Y	Y	Y	Y
Hassan (2013 & 2015)	Quantitative descriptive	Y	Y	Y	Unclear	Y	Unclear	Y
Heller (2020)	Quantitative descriptive	Y	Y	Y	Y	Y	Y	Y
Hershey (2021)	Quantitative descriptive	Y	Y	Y	Unclear	Y	Unclear	Y
Fiori (2020) (89)	Quantitative descriptive	Y	Y	Y	Y	Y	Y	Y
Fiori (2020) (104)	Quantitative descriptive	Y	Y	Y	Y	Y	Y	Y

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Kumar (2017)	Quantitative descriptive	Y	Y	Y	Y	Y	Y	Y
Matiz (2022)	Quantitative descriptive	Y	Y	Y	N	Y	N	
Molina (2019)	Quantitative descriptive	Y	Y	Unclear	Unclear	Y	Unclear	Unclear
Oldfield (2020)	Quantitative descriptive	Y	Y	Y	Y	Y	Y	Y
Pierse (2020)	Quantitative descriptive	Y	Y	Y	Y	Y	Unclear	Y
Polk (2020)	Quantitative descriptive	Y	Y	Y	Y	Y	Y	Y
Ray (2020)	Quantitative descriptive	Y	Y	Y	Y	Y	Y	Y
Razani (2020)	Quantitative descriptive	Y	Y	Y	Unclear	Y	Unclear	Y
Rhodes (2012)	Quantitative descriptive	Y	Y	Y	N	Y	N	Y
Sandel (2018)	Quantitative descriptive	Y	Y	Y	Y	Y	Unclear	Y
Sandoval (2020)	Quantitative descriptive	Y	Y	Y	Y	Y	Y	Y
Selvaraj (2019)	Quantitative descriptive	Y	Y	Y	Y	Y	Unclear	Y
Semple-Hess (2019)	Quantitative descriptive	Y	Y	Y	Y	Y	Y	Y
Sokol (2021)	Quantitative descriptive	Y	Y	Y	Y	Y	Y	Y
Spencer (2020)	Quantitative descriptive	Y	Y	Y	Y	Y	N	Y

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Uwemedimo (2018)	Quantitative descriptive	Y	Y	Y	N	Y	N	Y
Vaz (2020)	Quantitative descriptive	Y	Y	Y	Y	Y	Y	Y
Zielinski (2017)	Quantitative descriptive	Y	Y	Unclear	Unclear	Y	Y	Y
Author (year)	**Category of study design**	**Are there clear research questions?**	**Do collected data allow to address research question?**	**Is there adequate rationale for using a mixed methods design to address the research question?**	**Are there different components of the study effectively integrated to answer the research question?**	**Are the outputs of integration of qualitative and quantitative components adequately interpreted?**	**Are divergences and inconsistencies between quantitative and qualitative results adequately addressed?**	**Do the different components of the study adhere to the quality criteria of each tradition of the methods involved?**
Farrell (2017)	Mixed methods	Y	Y	Y	Y	Y	N	N
Koita (2018)	Mixed methods	Y	Y	Y	Y	Y	N	N

Table 3b Quality assessment using the Quality Improvement Minimum Quality Criteria Set (QI-MQCS) version 1.0

Author (year)	Organizational motivation	Intervention rationale	Intervention description	Organizational Characteristics	Implementation	Study design	Comparator	Data source	Timing	Adherence	Health outcomes	Organizational readiness	Sustainability	Spread	Limitations
Arbour (2021)	Y	Y	Y	Y	Y	Y	Y	Y	Y	N	Y	Y	N	Y	Y
Colvin (2016)	Y	Y	Y	Y	Y	Y	Y	Y	N	N	Unclear	Unclear	Y	N	Y
Fiori (2019)	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y

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Hensley (2017)	Unclear	Y	Y	Y	Y	N	N	Y	N	N	Y	N	N	N	Y
Higginbotham (2019)	N	Y	Y	N	Y	Y	Unclear	Y	Y	N	Unclear	N	N	N	Y
Denny (2019)	N	Y	Y	Y	Y	Y	N	Y	Y	N	Y	Unclear	Y	N	Y
Power-Hays (2020)	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	N	Y	Y	N	Y

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Discussion

This systematic review provides a synthesis of housing-related screening questions asked of families in healthcare and social service settings, and the subsequent actions taken to support families with housing problems as part of screening. This synthesis demonstrates that the housing questions asked in screening tools are highly variable and there is no standard way to ask families about housing in the healthcare and social service setting. Not all housing screening tools are designed with provision of a resource or social prescribing embedded, for example, some housing screening tools are simply a set of questions to ask families about housing and practitioners using these tools are not prompted to provide information, referral or links to the community sector after screening. This raises ethical considerations in relation to screening and social prescribing. For example, is it ethical for health professionals and social service workers to collect information about a patient's housing without providing support if the housing condition is affecting the patient's health? Another question raised is what is the benefit of screening for housing, especially if there is no subsequent action taken to address housing issues? Based on our findings, we provide some recommendations for screening and social prescribing for families experiencing housing issues, however it is important to note that screening and social prescribing for social determinants of health is an emerging area of healthcare and there is limited information on the validity of the screening tools included in this review. Furthermore, the evidence base for social prescribing is currently lacking due to difficulties in the generation of robust studies evaluating social prescribing in healthcare (122). To our knowledge, this is the first time a systematic review has been conducted to catalogue the social screening tools used by healthcare and social service staff to ask patients or clients about their housing and address housing problems through an action such as provision of information or social prescribing.

Housing screening questions asked in healthcare and social service settings

There was variation in the wording of screening questions and type of housing issues asked about when screening for housing problems. None of the included studies commented on standard ways

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to ask about housing. These findings indicate that there are no standards for asking families about housing problems in healthcare and social services. This is an unsurprising finding, as screening and social prescribing to address social determinants of health in the healthcare and social service context is an emerging practice (53, 123). As more screening tools emerge and are validated, clinicians and social service staff may be more likely to choose from a range of pre-developed validated screening tools rather than design their own. This is evident in the use of the WE CARE tool, with 7 studies included in this review utilising this tool. A complexity with screening for housing is that definitions and measurements of housing issues are highly variable, inconsistent and/or are rarely defined explicitly across the literature, leading to problems with their adequate measurement and capture. As an example, housing insecurity is measured, defined and characterised in many different ways. Varying definitions of housing insecurity capture dimensions of housing stability, affordability, quality, safety, and/or homelessness, and the inconsistency and incompleteness of these definitions mean that people actually experiencing housing insecurity may not be captured within existing housing insecurity measures (124). This is likely to be true of other housing measures in clinical and social service settings and calls for greater consistency, completeness and standardisation of housing issues (124).

While it is widely recognised that social determinants such as housing, education, income and food security have an impact on health and wellbeing, the best way to ask about social needs in healthcare remains unclear (48, 49, 53, 123). There was no rationale provided for the choice of housing questions, and for subsequent actions. Twenty-five papers (n=25) in this review provided the exact wording of housing questions used in their studies, while some studies only provided prompts rather than the exact questions, for example ‘Home heating’ (43), ‘Residential mobility’ (82), ‘Concerns about the physical condition of your housing’ (80), or ‘unhealthy housing’ (94). In the latter examples it was not clear how the questions were asked, for example, whether healthcare and social service staff asked questions word-for-word, or used the screening tool questions as prompts or as a guide. Some studies reported the name of the screening tool they used, but did not list the

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questions. The screening tools captured different housing aspects, from physical housing conditions to the social conditions of housing instability, housing affordability and neighbourhood safety. These different aspects align with the four pillars of housing and health equity; cost, conditions, consistency and context (125), however each screening tool asked a different combination of questions, and sometimes asked several questions within one broad housing question.

Validated housing screening tools

Most screening tools used in the studies had been tested for construct validity, face validity, feasibility, reliability, sensitivity, specificity and/or acceptability. Limited information was provided on the testing of the specific housing questions embedded in screening tools. Some studies provided information on the validity of housing questions, however the majority did not. It is unclear whether the housing questions used in the screening tools of this review are rigorously tested and robust in producing the intended result of screening. Housing questions were often adapted from existing surveys such as the American Household Survey and were not necessarily validated, which provides some concern for the robustness of the tools. Future research should rigorously test the quality of housing questions embedded in social screening tools and ensure cultural appropriateness and validity in the specific contexts and among the populations the tools are being used.

Recommendations for practice and research

While all screening tools aimed to identify one or more housing issues, 27 of the 42 studies (64%) included in this review offered a subsequent action as a result of identifying a social need. This finding is important, as it shows that not all screening tools for social determinants of health are designed to facilitate provision of information or subsequent social prescribing by the healthcare or social service provider. There was no information provided in the studies as to why some tools included an action and others did not.

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Most studies focused on testing social screening in the healthcare or social service setting and did not aim to examine social prescribing after screening. Several studies highlighted the need for appropriate resources to be provided to families (66, 74, 91, 92, 103), a finding consistent with the literature which highlights the need for clinicians to be aware of community resources available for the effective and appropriate treatment of social determinants of health (126). This raises an ethical issue of screening and referral, versus screening only; identifying an issue without offering an action to support a family experiencing housing issues may be considered unethical. However, if appropriate resources are not available, or providers are unclear whether the action they offer is effective, it may be preferable not to offer an action. Identifying a housing issue without providing support may be acceptable in cases where a suitable pathway to address the issue is unavailable. In Hardy's study, the tool was developed based on the social worker's capacity to adequately address the identified social need (88). While research has found that providing an action upon screening for social determinants of health may lead to addressing social issues, health and social service providers should ensure that the actions and referral pathways are appropriate for their patients and clients.

Regarding the type of information provided to patients who screened positively to general social needs or housing, studies did not provide details on the information provided. Oldfield found that parents prefer to receive information on social needs through e-mail, text message or a paper print-out, rather than an in-person consult with a community health worker, however this was a small sample size and may not be generalisable (74).

Several screening practices led to the automatic referral of patients to community services when a patient indicated an unmet need. Garg (127) suggests avoiding this approach. It is critical that healthcare providers gauge whether patients wish to be assisted (127). As screening for social determinants of health and social prescribing is an emerging area of practice, and those screened are often from 'disadvantaged' groups, services are encouraged to ensure patients are provided ample opportunity to share decision-making in the screening process, including decision-making on

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any subsequent material assistance. Furthermore, it is recommended to tailor questions to specific communities by identifying the most common problems experienced by that community (60). Understanding the needs and unique context of the population group being screened is vital to increase the potential for a housing screening and referral tool to effectively support patients or clients with their housing needs.

Screening for social needs such as housing is considered sensitive. It is worth noting that several studies discussed the need for clinicians to be adequately trained in asking sensitive questions (59, 67, 100). Selvaraj (70) suggested clinicians establish relationships with families before screening for social needs. Some screening tools provided introductory framing to social screening to provide context behind a clinician asking sensitive questions, to ensure the patient does not feel targeted or singled out, and to ensure the patient understands that the health worker is asking so they can help address any issues. For example, the HealthBegins Upstream Risks Screening Tool and Guide begins with the opening sentence: "*Everyone deserves the opportunity to have a safe, healthy place to live, work, eat, sleep, learn and play. Problems or stress in these areas can affect health. We ask our patients about these issues because we may be able to help.*" The SEEK tool frames screening in the following way: "*Being a parent is not always easy. We want to help families have a safe environment for kids. So, we're asking everyone these questions. They are about problems that affect many families. If there's a problem, we'll try to help.*" Introducing social screening with reasoning, specifically linking screening to action, may support patients in answering sensitive questions and alleviate stigma associated with experiencing social hardship.

The majority of studies screened high risk populations, such as families on low incomes, children with a disability, teenage parents, families with open child welfare cases, medically complex families and immigrant families. This indicates that it may be important to screen those who may be more likely to experience housing problems. Tailoring screening to specific communities by identifying the most common problems experienced by that community is considered a critical element of

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screening for social determinants of health (60). It is possible that the housing screening questions in the identified screening tools are highly variable because they are tailored to local contexts. Semple-Hess' study, for example, attempted to address the most commonly occurring issues experienced by their patients (106). Unfortunately, this approach may mean that less common housing aspects are not captured. While most studies captured race and/or ethnicity of study participants, only two studies specifically examined the cultural acceptability and appropriateness of the screening tool used with families. This is interesting given the high proportion of Hispanic/Latino participants (65, 73, 74, 77, 80, 84, 87, 94, 98, 101) and African American participants in multiple studies (30, 65-67, 70, 76, 83, 84, 88), and the overrepresentation of Maori and Pacific people in one study (107). However, Garg (127) suggests screening all patients rather than targeting families based on demographic characteristics, as screening only those at increased risk may reinforce preconceived ideas about particular groups and lead to stigmatisation. Further research is needed to ascertain a targeted versus a general approach to screening.

Studies did not report whether people with lived experience of housing problems were consulted in the development of questions. Rather, it was common for research teams, healthcare and social workers and institutional stakeholders to develop questions. Engaging directly with people who have lived experience of housing problems may lead to more appropriate and acceptable questions to specific communities.

According to the studies in this review, social prescribing actions to address housing issues were referral to a social worker or social service, care coordinator, community health worker or advocate and support by an in-person navigator. These types of social prescribing activities have been described as signposting/information referral (56). It is important to note that not all screening tools and referral pathways were tested for effectiveness, however the referral pathways and actions in these studies could be useful in supporting patients or clients with housing issues. While not an aim of this review, it was found that housing issues are prevalent among those who are screened for

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housing in healthcare and social service settings (see Supplementary Table 2). Housing issues are prevalent and also vary widely, from physical quality of housing to issues of housing stability. However, there is no standard way to quantify housing problems. Information collected about housing problems are highly variable and there is no consistent method of measuring the severity of housing issues. This calls for a need for greater consistency and completeness in how to quantify and define housing problems (124).

Housing is an important domain as part of screening for social needs among children and families in healthcare and social services. There is a need for professionals to screen or assess for housing problems in these settings to improve identification of housing problems and to make efforts to address them. Screening for housing issues in healthcare or social services provides an opportunity to identify housing-related problems, and often can lead to the referral of children, adolescents and families to relevant services to address their needs. Garg (43) found "that a simple primary care screening and referral system for unmet basic needs increased families' receipt of community based resources". Screening and referral also allows for the ongoing capture of the housing history of a patient or client over time, and provides healthcare workers with a formal process to assess social needs, and "a concrete measure and rationale to use when referring clients" (97). Furthermore, it draws clinicians' attention to the housing programs available (97). Screening for social needs has been found to increase community resource referrals. Before universal screening for social needs at one paediatric clinic, 14 of 700 (2.0%) patients at well child visits were referred to community resources, and upon universal screening for social needs being introduced this increased 6 times (101 of 759, 13.3%) (70). Given the current lack of evidence on the impact of screening and social prescribing, it would be useful to specifically generate evidence on the reach, scope and acceptability of social prescribing to understand its benefit, as suggested by Husk (122). A quantitative study of the acceptability of social risk screening in patients and caregivers, including asking about housing instability, found that 79% of participants reported that social risk screening was appropriate (72). A qualitative study linked to this aforementioned quantitative study found that

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social risks screening was considered important, and participants understood the link between social risks and health (128).

Our systematic review shows that currently there is high variability in the housing questions asked of families in healthcare and social service settings. It also shows that currently not all screening of housing issues in healthcare and social services leads to action. Based on these findings, there is potentially a need for guidelines for health and social service providers on how to screen families and address their housing-related needs. While some social screening tools in this review were tested for validity, there is currently a lack of strong evidence on the most appropriate and effective way to screen and address housing-related problems from the health and social service perspective.

The key recommendations based on our systematic review are as follows:

- We encourage screening tools for housing needs as part of a clinical or social service pathway to be followed by referral pathways
- It is important for clinical and social service staff to be trained in asking appropriate social needs questions in a sensitive manner, especially as housing issues are prevalent and also vary widely
- Housing needs should form part of holistic social needs assessment in healthcare and social services
- We encourage further studies from diverse settings to finalise the domains that are critical to document as part of social needs screening

There were limitations to the review, notably the search terms selected and inclusion criteria being too narrow. It is possible that a search that expanded beyond housing questions being asked of children or young people and their carers may have yielded additional housing questions that could be incorporated into use for families. It is possible that wider search terms may have captured more papers on broader social screening tools that included housing questions. Strengths of the review include rigorous critical appraisal of each study, allowing a true exploration of the strength of evidence and robust understanding by evaluating studies for internal and external validity.

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Furthermore, 45 of 48 articles in this systematic review were from studies in the USA and thus the screening tools would not necessarily be generalisable to other high-income countries because of the particular healthcare, social service and housing systems operating in the USA.

Conclusion

This systematic review provides recommendations and important considerations for clinicians and social service providers when screening families for housing issues to improve child health. It catalogues the housing questions asked of families in the context of screening for social determinants of health in healthcare and social service settings in high-income countries, and the social prescribing options clinicians and social service providers may offer families to address housing issues. Our systematic review found 49 peer-reviewed articles that described housing as a part of social screening and/or referral used with families in healthcare and social service settings. Fourteen of the twenty-eight screening tools identified were validated. There are no standard housing-related questions that healthcare and social service workers ask families, and the housing questions in social screening tools vary widely. An action and/or a referral was part of 27 of 42 identified studies, and 11 of the 27 studies specifically provided an action or social prescribing option for housing problems. Referral to a social worker or social service agency, provision of a community resource guide to families, or referral to a specific housing program were common actions to support families with housing problems in healthcare and social service settings. There is a need to develop guidelines for health and social service providers to effectively and appropriately screen families and address their housing-related issues.

Figure legend

Figure 1 Search flowchart

Figure 2 Housing aspects asked in screening tools

Abbreviations

AHC The Accountable Health Communities instrument

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ASK	Addressing Social Key Questions for Health
CHW	Children's HealthWatch
FAMNEEDS	Family Wellness Screening
IHELLP	Income, Housing, Education, Legal status, Language/Immigration, Personal Safety
MMAT	Mixed Methods Appraisal Tool
PRAPARE	Protocol for Responding to and Assessing Patients' Assets, Risks, and Experiences
QI-MQCS	Quality Improvement Minimum Quality Criteria Set
QRAFT	Quick Risks and Assets Family Triage
RCT	Randomized Control Trial
SEEK	Safe Environment for Every Kid
USA	United States of America
WE CARE	Well Child Care, Evaluation, Community Resources, Advocacy, Referral, Education

Ethics Declarations

Ethics approval not required.

Consent for publication

Not required.

Competing interests

The authors declare no competing interests

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Data Availability Statement

All data relevant to the study are included in the article or uploaded as supplementary information.

Author contributions

SW, MA, KH, JC, MM, LR, MJ, AM, JS, CW, RL, CE and KC contributed to the conceptualisation of the study. MA led the development of the study protocol, supported by SW and KH. NW, AA, SW, MA

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and KH reviewed study titles and abstracts. AA, SW, MA and KH discussed disagreements about inclusion of studies. AA and SW reviewed full texts of papers, extracted data from included papers and assessed the quality of each study. LT, SR, MM, LR, MJ, AM, KED, KZ, SSL and KC contributed to the interpretation of findings. AA was the major contributor in the review of studies, data extraction, quality assessment, interpretation of findings as well as writing up the manuscript. All authors read the final manuscript, provided edits and approved the final manuscript.

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Figure 1 Search flowchart

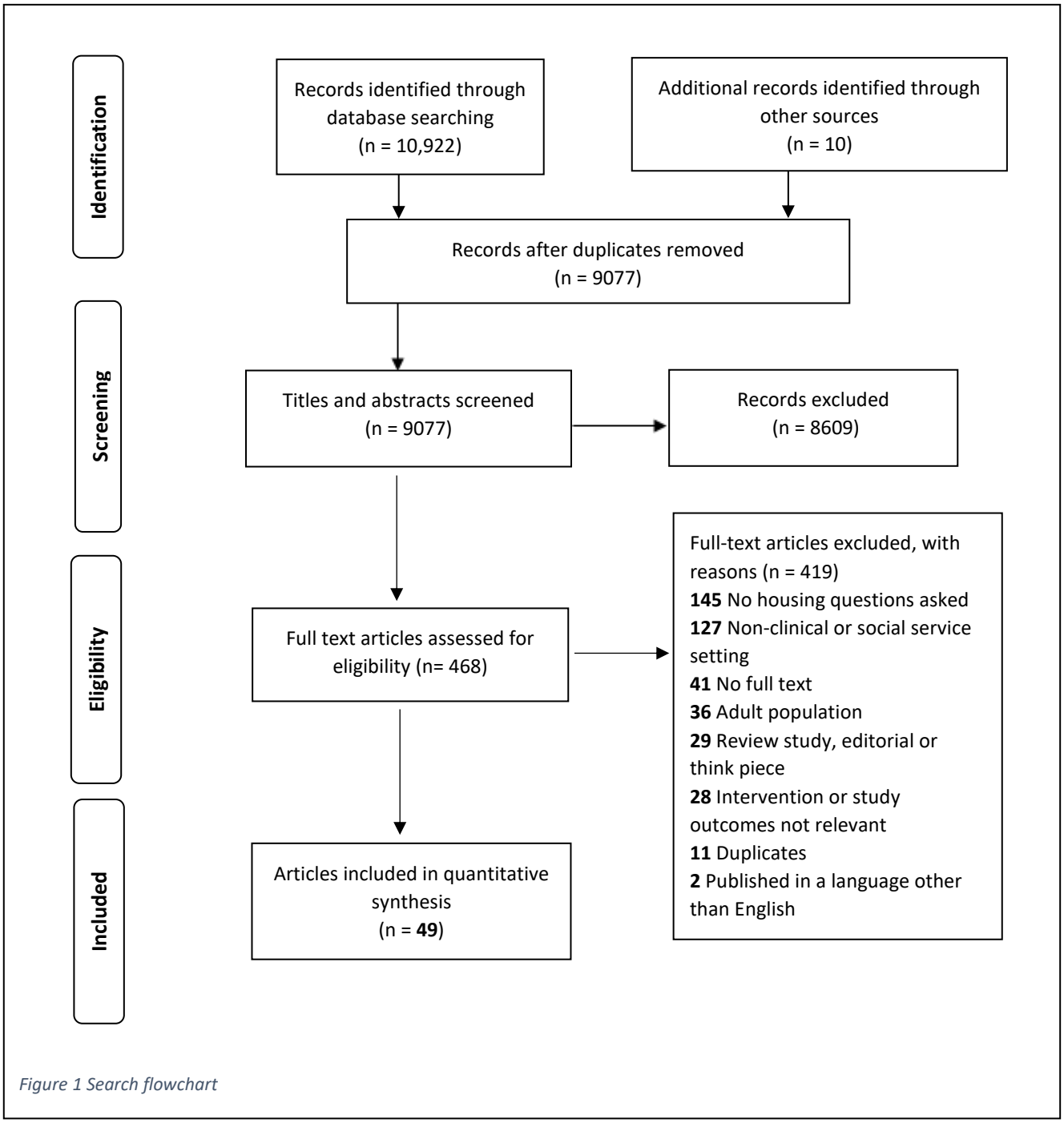


Figure 1 Search flowchart

Figure 2 Housing aspects asked in screening tools

1) General housing problems (housing problems not specified, or multiple housing problems captured);
2) Physical or structural housing problems;
3) Crowding;
4) Pests;
5) Utilities;
6) Housing instability, insecurity or homelessness;
7) Need for modifications in the home;
8) Unsafe neighbourhoods and;
9) Other (housing subsidy, basic living needs, other concerns with housing).

Figure 1 Housing aspects asked in screening tools

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Supplementary Table 1 Search strategy and search terms

Table 1 Search strategy and search terms (Medline, Emcare, EMBASE)

1. *housing.mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms]*
2. *homeless.mp.*
3. *living conditions.mp.*
4. *Socioeconomic Factors/*
5. *social situation.mp.*
6. *Social Environment/*
7. *psychosocial.mp.*
8. *basic needs.mp.*
9. *Child welfare/*
10. *Social welfare/*
11. *Social determinants.mp.*
12. *social conditions/ or economic status/ or poverty/*
13. *1 or 2 or 3 or 4 or 5 or 6 or 7 or 8 or 9 or 10 or 11 or 12*
14. *screen*.mp.*
15. *detect.mp.*
16. *assess*.mp.*
17. *tool.mp.*
18. *survey*.mp. or exp "Surveys and Questionnaires"/*
19. *question*.mp.*
20. *(standardised or standardized).mp.*
21. *surveillance.mp.*
22. *14 or 15 or 16 or 17 or 18 or 19 or 20 or 21*
23. *Health Services/*
24. *community health.mp.*
25. *((community or social) adj4 service*).mp.*

Supplementary Table 1 Search strategy and search terms

26. *clinical setting.mp.*
27. *Primary Health Care/*
28. *clinical care.mp.*
29. *medical home.mp.*
30. *23 or 24 or 25 or 26 or 27 or 28 or 29*31. 13 and 22 and 30
31. limit 31 to *(English language and humans and yr="2009-2021" and ("all infant (birth to 23 months)" or "all child (0 to 18 years)" or "newborn infant (birth to 1 month)" or "preschool child (2 to 5 years)" or "child (6 to 12 years)" or "adolescent (13 to 18 years)")*

[SCOPUS: Terms used] (TITLE-ABS-KEY (housing OR homeless OR "living conditions" OR "social determinants" OR (social W/3 (situation OR environment OR condition*)) OR psychosocial OR "socioeconomic factor*" OR "basic needs" OR "economic status" OR poverty OR (welfare W/3 (child OR social))) AND TITLE-ABS-KEY (screen* OR detect* OR assess* OR tool OR survey OR question* OR standardised OR standardized OR surveillance) AND TITLE-ABS-KEY ((care W/3 (health OR community OR social)) OR "medical home") AND KEY (child* OR infant OR toddler OR pediatr* OR paediatr* OR baby OR babies)) AND PUBYEAR > 2019 AND PUBYEAR < 2022 AND (LIMIT-TO (DOCTYPE , "ar")) AND (LIMIT-TO (LANGUAGE , "English")) AND (EXCLUDE (LANGUAGE , "French")) AND (EXCLUDE (SUBJAREA , "BIOC") OR EXCLUDE (SUBJAREA , "AGRI")) AND (EXCLUDE (SUBJAREA , "IMMU") OR EXCLUDE (SUBJAREA , "PHAR")) AND (EXCLUDE (PUBSTAGE , "aip")) AND (EXCLUDE (SUBJAREA , "ARTS") OR EXCLUDE (SUBJAREA , "DENT")) AND (EXCLUDE (SRCTYPE , "k")) AND (EXCLUDE (SUBJAREA , "ECON"))

[CINAHL: Terms used] housing OR homeless OR "living conditions" OR "social determinants" OR (social N3 (situation OR environment OR condition*)) OR psychosocial OR "socioeconomic factor*" OR "basic needs" OR "economic status" OR poverty OR (welfare N3 (child OR social)) AND screen* OR detect* OR assess* OR tool OR survey OR question* OR standardised OR standardized OR (surveillance care N3 (health OR community OR social)) OR "medical home" AND child* OR infant OR toddler OR pediatr* OR paediatr* OR baby OR babies. English only. child, preschool: 2-5 years, child: 6-12 years, all infant, infant: 1-23 months, adolescent: 13-18 years, infant, newborn: birth-1

[PsycINFO: Terms used] ((noft(housing) OR MAINSUBJECT.EXACT("Homeless") OR noft("living conditions" OR "psychosocial" OR "basic needs" OR "social determinants" OR "economic status") OR MAINSUBJECT.EXACT("Psychosocial Factors") OR MAINSUBJECT.EXACT("Poverty") OR MAINSUBJECT.EXACT("Socioeconomic status") OR noft(social NEAR/3 (environment OR situation OR condition*)) OR noft(welfare NEAR/3 (child OR social))) AND (noft(screen* OR detect OR assess* OR tool OR question* OR ask* OR surveillance OR standardised OR standardized) OR MAINSUBJECT.EXACT("Questionnaires") OR MAINSUBJECT.EXACT("Surveys")) AND ((MAINSUBJECT.EXACT("Primary Health Care") OR MAINSUBJECT.EXACT("Health Care Services")) OR MAINSUBJECT.EXACT("Community Health") OR MAINSUBJECT.EXACT("social services") OR noft("clinical setting" OR "medical home" OR "clinical care")) AND noft(child* OR infant* OR toddler OR pediatr* OR paediatr* OR adolescen* OR baby OR babies)) AND (la.exact("ENG")

Supplementary Table 2 Prevalence of housing problems

Table 2 Prevalence of housing problems

Housing problem category	**Description of housing problem (Housing question)**	**Prevalence (%)**
General housing problems or multiple problems	Problems with housing conditions e.g. overcrowding, evictions, lead, utilities, mold, rodents among all infants in sample (Beck 2012)	3.4
	Problems with housing conditions e.g. overcrowding, evictions, lead, utilities, mold, rodents among infants with complete documentation (Beck 2012)	5.6
	Housing problems (Colvin 2016; Garg 2009)	22; 10
	Housing quality and/or availability Fiori (2019); Fiori (2020)	40
	Housing major problem: Homeless; utilities shut off; structural problems (Hassan 2013 & 2015)	34
	Housing minor problem: Need fuel/heating assistance; threat of utilities shut off in past 12 months; if renting or receiving subsidized housing, concern about transfer or eviction (Hassan 2015)	22
Housing instability/ insecurity and/or homelessness	Housing insecurity (Arbour 2021; Higginbotham 2019; Matiz 2022; Molina 2019; Oldfield 2021; Hensley 2017; Sokol 2021)	13; 18.8; 10; 23; 6.7-12; 1.8; 10
	Housing instability (Bottino 2019; De Marchis 2020; Farrell 2017; Garg 2015; Gottlieb 2018; Heller 2020; Kumar 2017; Ray 2020; Sandel 2018; Vaz 2020)	40; 52; 2.9; 43; 28.6; 4.8; 15.1-40.6; 89; 8; 5
	Homeless (Bottino 2019; Farrell 2017; Gottlieb 2016; Hardy 2021; Hassan 2013; Molina 2019; Razani 2020; Zielinski 2017)	1.8; 1.4; 29.2; 29.5; 2; 7; 25; 2
	Homeless in child's lifetime (Sandel 2018)	12
	Risk of homelessness (Hardy 2021; Spencer 2020	1.8; 5.3
	Hard to access housing (Costich 2019)	23
	Concerns about the cost or stability of your housing (Gottlieb 2014)	43.8
	Concern for stable housing (Hershey 2021; Molina 2019)	17; 14
	Concerned for eviction (Bottino 2019)	10.6

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Supplementary Table 2 Prevalence of housing problems

	Housing as unmet need (Power-Hays 2020; Messmer 2020)	11; 29
	Behind on rent in past 12 months (Sandel 2018)	27
	Would like support finding housing search resources and emergency shelters (Polk 2020)	11.4 - 24.6
	Needed housing resources at: 3 months postpartum; 6 months postpartum (Kumar 2017)	35.4; 42.3
Physical or structural housing problems	Housing condition: Potential for threat or harm; safety planning essential; unsustainable housing for any reason, significant risk, level 3; and Housing condition: Unsafe; immediate danger/risk; CPS call; needs urgent safety planning; or chronic, severe problems, severe risk, level 4 (Farrell 2017)	2.1
	Concerns about the physical condition of your housing (Gottlieb 2014)	35.9
	Structural problems (Hassan 2013)	29
	Housing quality risk (Heller 2020)	6.5
	Concerns about the condition or quality of housing (Hensley 2017; Molina 2019)	11.5; 13
	Mold on the walls of bedrooms or living areas (Pierse 2020)	88
	Unhealthy living environments (Gottlieb 2016; Razani 2020)	22.7; 17
	Problems like mold, insects, rats or mice (Vaz 2020)	8
Crowding	Need enlarged rooms at home (Badia 2014)	56
	Doubled up with another family (Bottino 2019)	1.2
Utilities and home heating	Unmet basic needs: Home heating (Garg 2015; Sokol 2021)	9; 24.2
	Inadequate heating and draughts (Pierse 2020)	81
	“Is your home usually colder than you would like?” (Pierse 2020)	85
	“During the winter months, was your house so cold that you shivered inside?” (Pierse 2020)	67
	Utilities shut off (Hassan 2013; Molina 2019)	4; 4

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Supplementary Table 2 Prevalence of housing problems

	Utilities as unmet need (Messmer 2020; Ray 2020)	21; 84
	Housing and bill insecurity (Selvaraj 2019)	15.9
	Utilities threatened or shut off (Bottino 2019)	33.2
	Not enough money for utilities (Vaz 2020)	26
Need for modifications in home	Need adapted toilet at home (Badia 2014)	61
	Need modified kitchen at home (Badia 2014)	21
	Need walking aids (Badia 2014)	39
	Need hoists at home (Badia 2014)	30
Unsafe neighbourhoods	Threats to your child's safety at school or in the neighborhood (Gottlieb 2014)	51.8
	Do you have concerns about safety in your neighbourhood? (Hensley 2017; Molina 2019)	16.1; 15
	Are you afraid you might be hurt in your apartment building or house? (Molina 2019)	5
	Personal safety ("Is there any domestic violence in the home? Do you feel safe in your neighborhood?") (Patel 2018)	9
Other	Unhealthy housing (Arbour 2021)	3.5
	Other concerns with housing (Gottlieb 2016; Razani 2020)	16.5; 13
	Housing subsidy (Patel 2018)	4
	Basic living needs (Rhodes 2012)	6

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Reporting checklist for systematic review (with or without a meta-analysis).

Based on the PRISMA guidelines.

Instructions to authors

Complete this checklist by entering the page numbers from your manuscript where readers will find each of the items listed below.

Your article may not currently address all the items on the checklist. Please modify your text to include the missing information. If you are certain that an item does not apply, please write "n/a" and provide a short explanation.

Upload your completed checklist as an extra file when you submit to a journal.

In your methods section, say that you used the PRISMAreporting guidelines, and cite them as:

Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, Shamseer L, Tetzlaff JM, Akl EA, Brennan SE, Chou R, Glanville J, Grimshaw JM, Hróbjartsson A, Lalu MM, Li T, Loder EW, Mayo-Wilson E, McDonald S, McGuinness LA, Stewart LA, Thomas J, Tricco AC, Welch VA, Whiting P, Moher D. The PRISMA 2020 statement: An updated guideline for reporting systematic reviews

		Reporting Item	Page Number
Title			
Title	#1	Identify the report as a systematic review	1
Abstract			

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Abstract	#2	Report an abstract addressing each item in the PRISMA 2020 for Abstracts checklist	1
Introduction			
Background/rationale	#3	Describe the rationale for the review in the context of existing knowledge	1
Objectives	#4	Provide an explicit statement of the objective(s) or question(s) the review addresses	3 and 4
Methods			
Eligibility criteria	#5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses	4 and 5
Information sources	#6	Specify all databases, registers, websites, organisations, reference lists, and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted	4 and 5
Search strategy	#7	Present the full search strategies for all databases, registers, and websites, including any filters and limits used	4 and 5 and Supplementary Table 1
Selection process	#8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each	4 and 5

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		record and each report retrieved, whether they worked independently, and, if applicable, details of automation tools used in the process	
Data collection process	#9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and, if applicable, details of automation tools used in the process	4 and 5
Data items	#10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (for example, for all measures, time points, analyses), and, if not, the methods used to decide which results to collect	4 and 5, and Tables 1 and 2 Pages 24-30
Study risk of bias assessment	#11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and, if applicable, details of automation tools used in the process	Pages 6-7 and Tables 3a and 3b pages 30-32

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Effect measures	#12	Specify for each outcome the effect measure(s) (such as risk ratio, mean difference) used in the synthesis or presentation of results	NA - effect measures not relevant to this study
Synthesis methods	#13a	Describe the processes used to decide which studies were eligible for each synthesis (such as tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item #5))	NA - did not complete a synthesis of quantitative data and did not have an intervention
Synthesis methods	#13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics or data conversions	NA - did not complete a synthesis of quantitative data and did not have an intervention
Synthesis methods	#13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses	NA - did not complete a synthesis of quantitative data and did

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			not have an intervention
Synthesis methods	#13d	Describe any methods used to synthesise results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used	NA - did not complete a synthesis of quantitative data and did not have an intervention
Synthesis methods	#13e	Describe any methods used to explore possible causes of heterogeneity among study results (such as subgroup analysis, meta-regression)	NA - did not complete a synthesis of quantitative data and did not have an intervention
Synthesis methods	#13f	Describe any sensitivity analyses conducted to assess robustness of the synthesised results	NA - did not complete a synthesis of quantitative data and did not have an intervention

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Reporting bias assessment	#14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases)	NA - results were qualitative and did not have missing information
Certainty assessment	#15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome	NA - confidence intervals not relevant
Data items	#10b	List and define all other variables for which data were sought (such as participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information	4 and 5, and Tables 1 and 2 Pages 24-30
Results			
Study selection	#16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram (http://www.prisma-statement.org/PRISMAStatement/FlowDiagram)	6

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Study selection	#16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded	6
Study characteristics	#17	Cite each included study and present its characteristics	6 and Table 1 page 24
Risk of bias in studies	#18	Present assessments of risk of bias for each included study	Pages 6-7 and Tables 3a and 3b pages 30-32
Results of individual studies	#19	For all outcomes, present for each study (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (such as confidence/credible interval), ideally using structured tables or plots	NA - not quantitative measures
Results of syntheses	#20a	For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies	Pages 6-7 and Tables 3a and 3b pages 30-32
Results of syntheses	#20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (such as confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect	NA - no synthesis conducted

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Results of syntheses	#20c	Present results of all investigations of possible causes of heterogeneity among study results	10
Results of syntheses	#20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesised results	NA - no synthesis conducted and no sensitivity analysis conducted due to nature of study
Risk of reporting biases in syntheses	#31	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed	NA - not applicable to this study
Certainty of evidence	#22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed	NA - study did not involve quantitative measures
Discussion			
Results in context	#23a	Provide a general interpretation of the results in the context of other evidence	Pages 10-16
Limitations of included studies	#23b	Discuss any limitations of the evidence included in the review	Pages 10-16

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Limitations of the review methods	#23c	Discuss any limitations of the review processes used	Pages 10-16
Implications	#23d	Discuss implications of the results for practice, policy, and future research	Pages 10-16
Other information			
Registration and protocol	#24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered	4
Registration and protocol	#24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared	4
Registration and protocol	#24c	Describe and explain any amendments to information provided at registration or in the protocol	NA - no amendments made
Support	#25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review	17
Competing interests	#26	Declare any competing interests of review authors	17
Availability of data, code, and other materials	#27	Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included	17

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studies; data used for all analyses; analytic code;

any other materials used in the review

Notes:

- 7: 4 and 5 and Supplementary Table 1
- 10a: 4 and 5, and Tables 1 and 2 Pages 24-30
- 11: Pages 6-7 and Tables 3a and 3b pages 30-32
- 12: NA - effect measures not relevant to this study
- 13a: NA - did not complete a synthesis of quantitative data and did not have an intervention
- 13b: NA - did not complete a synthesis of quantitative data and did not have an intervention
- 13c: NA - did not complete a synthesis of quantitative data and did not have an intervention
- 13d: NA - did not complete a synthesis of quantitative data and did not have an intervention
- 13e: NA - did not complete a synthesis of quantitative data and did not have an intervention
- 13f: NA - did not complete a synthesis of quantitative data and did not have an intervention
- 14: NA - results were qualitative and did not have missing information
- 15: NA - confidence intervals not relevant
- 17: 6 and Table 1 page 24
- 18: Pages 6-7 and Tables 3a and 3b pages 30-32
- 19: NA - not quantitative measures
- 20a: Pages 6-7 and Tables 3a and 3b pages 30-32

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- 10b: 4 and 5, and Tables 1 and 2 Pages 24-30
- 20b: NA - no synthesis conducted
- 20d: NA - no synthesis conducted and no sensitivity analysis conducted due to nature of study
- 31: NA - not applicable to this study
- 22: NA - study did not involve quantitative measures
- 23a: Pages 10-16
- 23b: Pages 10-16
- 23c: Pages 10-16
- 23d: Pages 10-16
- 24c: NA - no amendments made The PRISMA checklist is distributed under the terms of the Creative Commons Attribution License CC-BY. This checklist was completed on 08. June 2021 using https://www.goodreports.org/, a tool made by the EQUATOR Network in collaboration with Penelope.ai

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